

# From Memory to Testable Belief: A Survey of LLM Agents

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## Abstract

Language-model agents now act over many steps, and many of their failures are failures of state rather than of reasoning. They lose track of what they have learned, keep acting on contradicted evidence, or revise a belief without changing what they do. What an agent must hold between decisions is not a memory of what happened but a testable belief, a hypothesis that later evidence can support or falsify. Surveys of memory, uncertainty, and post-training each cover one part of such a belief. This is the first survey of all three parts, a state that says what is believed and how firmly, a revision that carries it across an action, and a test that scores it against the next observation, and it sorts the literature by which parts a system has and where each comes from. Memory and uncertainty research supplied the state and post-training the revision, but the test, which lets the agent's own observations correct its belief, is still supplied from outside. Only two systems train a written belief against the agent's own observations, and both use a binary belief. Belief-level metrics, aligned by construct, are rarely reported on the same systems. We outline what connecting representation, testing, revision, and credit would require, and release the taxonomy and screening record.

**AI disclosure.** The literature search, screening, and coding pipeline released with this paper is scripted and uses large-language-model agents to read and code each sampled paper under a fixed codebook (Appendix A); large-language-model assistance (Claude, Anthropic) was also used in drafting and revising the manuscript. The authors defined the framework, the inclusion criteria, and the codebook, checked the codings that support the claims in the text, and are responsible for the content.

**Table 1 | Which part of the belief each neighbouring survey supplies.**

| Survey                | Organising axis                       | S | R | T | E |
|-----------------------|---------------------------------------|---|---|---|---|
| <i>Memory</i>         |                                       |   |   |   |   |
| Zhang et al. (2024b)  | design, evaluation, use               | ■ | ■ | ■ | ■ |
| Hu et al. (2025d)     | factual / experiential / working      | ■ | ■ | ■ | ■ |
| Luo et al. (2026c)    | storage → reflection → experience     | ■ | ■ | ■ | ■ |
| <i>Reward</i>         |                                       |   |   |   |   |
| Wu (2025)             | reward model × stage                  | ■ | ■ | ■ | ■ |
| Zheng et al. (2025a)  | data → PRM → use                      | ■ | ■ | ■ | ■ |
| Zhang et al. (2025d)  | capability × application              | ■ | ■ | ■ | ■ |
| Zhang (2026)          | granularity × method                  | ■ | ■ | ■ | ■ |
| <i>Uncertainty</i>    |                                       |   |   |   |   |
| Xia et al. (2025)     | four estimation avenues               | ■ | ■ | ■ | ■ |
| Lin et al. (2026c)    | what / how / where                    | ■ | ■ | ■ | ■ |
| <i>Evaluation</i>     |                                       |   |   |   |   |
| Yehudai et al. (2026) | capability × domain                   | ■ | ■ | ■ | ■ |
| Gao et al. (2025)     | what / when / how to evolve           | ■ | ■ | ■ | ■ |
| Chen et al. (2026b)   | lifecycle × horizon carrier           | ■ | ■ | ■ | ■ |
| Xu et al. (2026d)     | forecasting architecture              | ■ | ■ | ■ | ■ |
| Dong et al. (2025b)   | financial function                    | ■ | ■ | ■ | ■ |
| <i>Belief</i>         |                                       |   |   |   |   |
| Cai et al. (2026)     | control-loop components               | ■ | ■ | ■ | ■ |
| Tang et al. (2026c)   | decision pipeline × policy            | ■ | ■ | ■ | ■ |
| <b>This survey</b>    | <b>belief state · revision · test</b> | ■ | ■ | ■ | ■ |

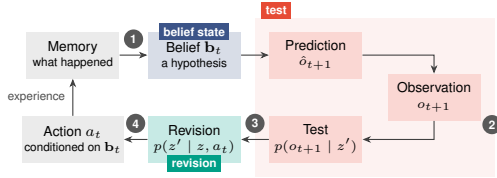
Evaluation includes horizon, self-evolution, forecasting and finance surveys, and Belief those that formalise a Bayes-filter belief for one setting. S, the belief state and its persistence; R, revision, how the state is revised; T, a test that reaches the belief as a learning signal; E, belief-level evaluation. Solid, central; tint, present; grey, absent. The ratings show how much of each part a survey covers.

## 1 Introduction

Language-model agents now act over horizons of hours or months, and the failures reported from these settings are failures of state, not of reasoning. A model that solves a problem in one pass loses much of its accuracy when the same information arrives over several turns (Laban et al., 2025), an agent given only observations wins a fraction of the episodes won by one that reads the simulator's latent state (Alaswad et al., 2026), and in markets stronger backbones do not reliably produce stronger trading agents (Xie et al., 2024; Li et al.,



**Fig. 1 | Taxonomy of belief-state work.** Work is split by the three parts of a testable belief, each branch by where the part comes from. The highlighted leaf holds two systems, both without uncertainty.



**Fig. 2 | From memory to testable belief.** Tints mark the three parts. Any evidence-based check can serve as the test, and the formulas give its Bayes-filter form (Eq. 1). Numbers mark the failure modes of Section 5.

2024a). What these failures share is that the task required the agent to hold something between decisions, a belief about what is true, how certain it is, and what would change it, and the agent never formed it, never tested it, or revised it without acting on the revision.

That something is a testable belief, and this is the shift the survey argues for. A memory records what has happened, whereas a belief is a hypothesis about what is true of the world, which a prediction exposes to evidence, a test supports or falsifies, and a revision updates (Fig. 2). The test can take many forms (Section 3). This review follows three parts, a *belief state* that says what is believed and how firmly, a *revision* that carries it across an action, and a *test* that scores it against the next observation, with the Bayes filter as their principled formalisation (Åström, 1965; Kaelbling et al., 1998; Thrun et al., 2005; Friston, 2010). A store that attaches a confidence and an update rule to its

claims already has the first two parts (Zhang et al., 2024b; Luo et al., 2026c), and what makes it a belief is the third, a test that connects the hypothesis to evidence that can support or falsify it. Memory and uncertainty research gave the agent a state with a confidence (Park et al., 2023; Shinn et al., 2023; Lin et al., 2022; Kadavath et al., 2022), and post-training taught it to revise, rewarding the response, the step, or the action (Ouyang et al., 2022; Lightman et al., 2024; Zhou et al., 2024b; Zhang, 2026). The test is still supplied from outside, by a fixed observation model, a simulator, a verifier, or a teacher.

Work on the belief itself exists, from dialogue state tracking (Lee et al., 2021) to agents that write beliefs about tasks, opponents, and markets (Kim et al., 2024; Cross et al., 2024; Yu et al., 2024) and benchmarks that score the belief rather than its outcome (Xu et al., 2026a; Deb and Krishnan, 2026; Chao et al., 2026), but it rarely cites itself (Fig. 6) and the nearest surveys organise by what is stored, stop at the action, or formalise a Bayes-filter belief for one setting at inference time only (Section 2, Table 1). This survey takes the belief state as its object and sorts the literature by which of its three parts a system has and where each comes from, whether written by the model, fixed by the designer, learned, or absent (Fig. 1), and follows the belief around the cycle of Fig. 2. Section 4 asks where the state lives and what form it takes, Section 5 how it is carried and revised and

where revision fails, and Sections 6–8 who scores the belief against evidence, whether the score becomes a learning signal and *which object receives it*, and how evaluation measures each part. Section 10 states the open problems. Against the nearest surveys, which organise by what is stored, by where the reward comes from, or by how confidence is estimated, this is the first survey to follow the belief through all three parts. It contributes:

- **The belief as the object:** a three-part definition, state, revision, and test, that joins what those surveys treat apart (Section 3).
- **A taxonomy by source:** 517 papers from 2020 to September 2026 sorted by which part a system has and where it comes from (Fig. 1), locating the missing test from the agent’s own observations.
- **A construct-level map of evaluation:** more than twenty belief-level metrics aligned by what they measure (Section 8).
- **A rerunnable screening record,** every coding checked against the full text (Appendix A).

## 2 Related Work

Table 1 places this survey among sixteen neighbouring ones by which part of the belief each supplies. Memory surveys (Zhang et al., 2024b; Hu et al., 2025d; Luo et al., 2026c) supply the belief state and its persistence, the newest with a probability or revision operator on each item, but they do not ask what evidence would retire an item, which is the test. Uncertainty surveys (Xia et al., 2025; Lin et al., 2026c) supply a confidence on the state, but attached to an output and neither carried forward nor revised. Reward and credit surveys (Wu, 2025; Zheng et al., 2025a; Zhang et al., 2025d; Zhang, 2026) supply the revision and the learning signal, following credit down to tokens, steps, and memory operations, and stop at the action. World-model and long-horizon surveys (Li et al., 2026d; Chen et al., 2026b) supply a learned revision and test whose state the agent cannot read. Evaluation, self-evolution, forecasting, and finance surveys (Yehudai et al., 2026; Gao et al., 2025; Xu et al., 2026d; Dong et al., 2025b) score the outcome and treat the belief as one component among many. Two recent surveys formalise the belief itself as a Bayes filter, for complex problem solving (Cai et al., 2026) and for proactive service agents (Tang et al., 2026c), but update it at inference time in one setting, so the test never reaches it

as a learning signal. Each strand holds one part or one setting, and none follows the three parts across agents into learning and evaluation, which is the question Section 3 turns into a definition.

## 3 Problem Formulation

A belief  $\mathbf{b}_t$  is a hypothesis about the environment that the agent carries between decisions. It yields a prediction, evidence  $o_{t+1}$  returns, a *test relation* supports or falsifies the hypothesis, and a revision produces  $\mathbf{b}_{t+1}$  for the next action (Fig. 2). The test may be a symbolic verifier, a consistency check, or a comparison with an external record. Partially observed decision-making gives this cycle its most principled realisation (Åström, 1965; Kaelbling et al., 1998). With hidden state  $z_t$  and history  $H_t$ , the belief is  $\mathbf{b}_t = p(z_t \mid H_t)$ , and a Bayes filter carries it forward as

$$\mathbf{b}_{t+1}(z') \propto p(o_{t+1} \mid z') \sum_z p(z' \mid z, a_t) \mathbf{b}_t(z) \quad (1)$$

(Thrun et al., 2005). Its three factors are the Bayesian form of the three parts the review follows. The factor  $\mathbf{b}_t$  is the *belief state*, what is believed and how firmly, the transition  $p(z' \mid z, a_t)$  is the *revision* that carries it across the action, and the likelihood  $p(o_{t+1} \mid z')$  is the *test*. Language-model agents rarely separate the filter’s prediction and correction steps, so we call revision the whole move from  $\mathbf{b}_t$  to  $\mathbf{b}_{t+1}$ , which the test feeds. A verifier or consistency check is a non-probabilistic test of the same kind and is coded as one. Agentic reinforcement learning casts the language-model agent as such a POMDP, with the context as history and any explicit state in a working memory the model writes (Zhang et al., 2025d; Sumers et al., 2023; Yao et al., 2023b), and we write  $\mathbf{b}_t$  for whatever explicit state the span carries before the action  $a_t$ .

We call such an object a *belief state* only if (i) it is a claim about the environment carried between decisions, (ii) it carries, or admits, uncertainty, and (iii) it is *testable*, committing to a claim that an observation the agent receives can support or falsify, usually a prediction of the next one. Conditions (i) and (ii) are properties of the belief state and fix the operation CONSTRUCT. Condition (iii) requires a test and fixes TEST, which makes the state a hypothesis rather than a record. REVISE is the revision, which belief-revision theory constrains to change the claim without discarding the rest (Alchourrón et al., 1985; Friston, 2010). We therefore

**Table 2 | Classes of the belief state and where each part comes from.**

| Locus     | Form                             | State | Uncert. | Rev. | Test                 | Belief | Example             |
|-----------|----------------------------------|-------|---------|------|----------------------|--------|---------------------|
| Context   | trace as state                   | M     | ■       | M    | —                    | ■      | ReAct               |
|           | written claim, calibrated        | M     | ■       | M    | D (target)           | ■      | Agent-BRACE, BOND   |
|           | written claim, testable          | M     | ■       | M    | D (oracle)           | ■      | CBM, EvoSCM         |
| Store     | trace store                      | M     | ■       | D    | —                    | ■      | MemGPT, A-MEM       |
|           | claims with probabilities        | M     | ■       | D    | D (update rule)      | ■      | Nous, BeliefMem     |
| Estimator | Bayesian filter, fixed space     | D     | ■       | D    | D (closed form)      | ■      | Belief-State Engine |
| Learned   | latent world model               | L     | ■       | L    | L (prediction error) | ■      | Dreamer, BB-WM      |
|           | written claim + parametric check | M     | ■       | M    | L (check)            | ■      | Hybrid-WM           |

Locus: in the context, in a store outside the model, in an external estimator, or in learned dynamics. M, written by the model; D, fixed by the designer; L, learned; —, absent (parts of Section 3). Uncert., the state carries uncertainty; Belief, meets conditions (i)–(iii) of Section 3. Solid, yes; tint, partly; grey, no.

place every system by which of the three parts it has and where each comes from, whether written by the model, fixed by the designer, learned from data, or absent. A trace *memory* fails (i) and (ii). A store that attaches a probability and an update rule to each item meets them but scores nothing against what returns, so it fails (iii) (Luo et al., 2026c). A *world model* is itself a revisable hypothesis, a learned revision and test whose state the agent cannot read, so it meets (iii) but seldom (i) (Li et al., 2026d).

## 4 The Belief State: Where It Lives and Who Writes It

The belief state is placed not by method family but by two attributes, its *locus* (in the context, in a store outside the model, in an external estimator, or in learned dynamics), which also says who writes it, and its *form* (a trace, a claim, or a distribution). The classes in Table 2 are locus–form pairs, so a system can occupy more than one, and the last column records which of conditions (i)–(iii) each class meets. Fig. 3 sketches the designs.

### 4.1 Context and Memory

The most common design has no representation at all. The reasoning trace is the state (Wei et al., 2022b; Yao et al., 2023b), the belief implicit and its uncertainty unstated.

Memory-augmented agents moved the state into a store retrieved into context (Packer et al., 2023; Park et al., 2023; Zhong et al., 2024; Xu et al., 2025d; Luo et al., 2026c; Jiang et al., 2026a). Two limits matter for belief. The stored unit is a trace rather than a claim with a probability, and an update is an operation on the store rather than on a hypothesis, so answer accuracy hides failures in the update and forget operations themselves (Hao et al., 2026). Structured stores narrow the first limit with schemas, graphs, bitemporal transitions, or AGM revision semantics (Luo et al., 2026d; Xu,

2026; Park, 2026), and probabilistic stores narrow the second by attaching a number to each claim (Liao et al., 2026a; Song and Xin, 2026; Yang et al., 2026c; Zhang et al., 2026j).

### 4.2 External Bayesian Filters

A different line takes the belief out of the model. The model maps evidence to a likelihood over a fixed set of latent regimes, hypotheses, or user states, and an explicit filter maintains the posterior (Chattopadhyay and Halder, 2026; Dixon, 2026b; Wu et al., 2026d; Soru, 2026; Akbar et al., 2026; Azran et al., 2026; Ansell and Toney-Wails, 2026). These beliefs are validated against a closed-form oracle and auditable by construction, at the price of a hypothesis space fixed in advance.

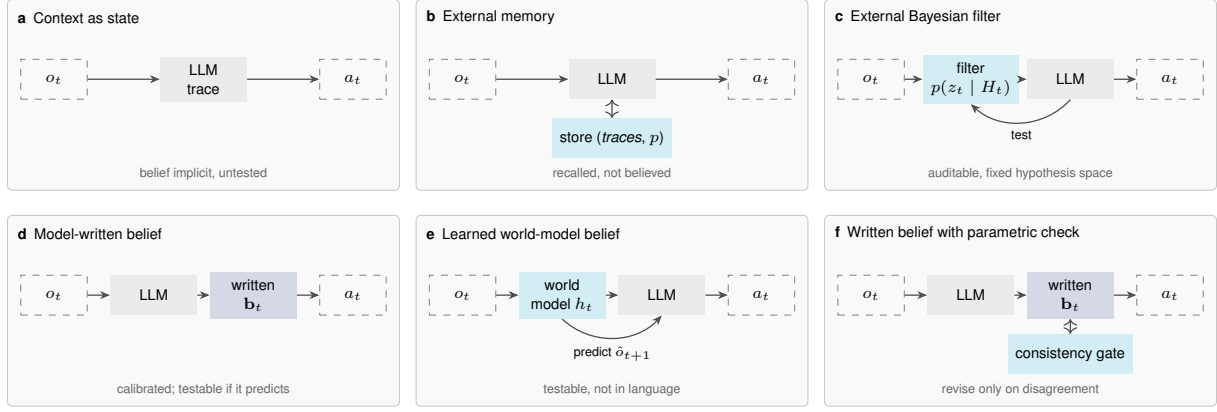
### 4.3 Model-Written Beliefs

The third line keeps the belief in context but gives it structure the model writes and reads, from slot-value dialogue states and verbalised confidence (Lee et al., 2021; Lin et al., 2022; Kadavath et al., 2022) to beliefs about tasks, opponents, and markets (Kim et al., 2024; Christakopoulou and Mourad, 2024; Cross et al., 2024; Yu et al., 2024) and states rewritten each turn or before each action (Zhou et al., 2025d; Kim et al., 2025; Kim and Hwang, 2025). The newest add certainty tags, Bayesian-teacher posteriors, symbolic verification, or populations of causal models (Singh et al., 2026b; Cui and Lin, 2026; Xu et al., 2026a; Zhao et al., 2026b).

Condition (iii) separates them. A written belief is testable only if it names an observation that can contradict it, usually a future one with a horizon, so that the observation yields a signed error. Most written beliefs are calibrated but not testable, stating confidence without naming what would falsify it, whereas Hypothetical Minds, CBM, EvoSCM, and ReBel (Tang et al., 2026b) name it.

### 4.4 Learned World Models

The fourth line learns the dynamics and exposes the resulting belief to the policy, through recurrent latent states (Ha and Schmidhuber, 2018; Hafner et al., 2019, 2023), the language model as its own world model for tree search (Hao et al., 2023; Zhao et al., 2023b; Zhou et al., 2023a), LLM-proposed POMDPs refined to a belief-based likelihood (Six et al., 2026), and a world-model belief the policy can query (Kumar et al., 2026; Li et al., 2026d; Zuo et al., 2026). World-model beliefs are testable



**Fig. 3 | Who maintains the belief.** Blue-grey marks a belief the model writes and can reason about, light blue an external estimator. Only c, d (when testable), e and f commit to a prediction that later evidence can falsify.

**Table 3 | Four failure modes of belief revision.**

| Failure mode     | Definition                             | Part at fault                   | Measured by                   | Headline result                                             |
|------------------|----------------------------------------|---------------------------------|-------------------------------|-------------------------------------------------------------|
| Failed stay      | belief moves without evidence          | test absent: update unweighted  | BeliefShift, MEDLEY-BENCH     | consensus sensitivity from near 0 to large across 35 models |
| Failed update    | belief survives contradicting evidence | carried forward, not updated    | DeltaLogic, DToM-Track, STALE | one model: 66.7% before a premise edit, 46.7% after         |
| Failed isolation | irrelevant input enters the belief     | belief state contaminated       | BeliefTrack, MemOps           | RL on the belief: 71% fewer failures than untrained         |
| Failed act       | belief revised, action unchanged       | policy not conditioned on $b_t$ | STOCKTAKE, Werewolf audit     | 34–43% of diagnosed weeks still stock out (4 models)        |

For each mode, the part at fault, the benchmarks that isolate it, and their headline results (Xu et al., 2026a; Chao et al., 2026; Myakala et al., 2026; Abtahi et al., 2026; Dhanda, 2026; Nguyen et al., 2026; Hao et al., 2026; Deb and Krishnan, 2026; Gao et al., 2026b).

by construction but, unless written back, cannot be reasoned about in language. Hybrids use the learned model as a check on a written belief, and on graph planning a parametric consistency gate cuts GPT-4o-mini’s hallucinated-state rate from 0.176 to 0.035 (Song and Cai, 2026a).

**Insight 1. Structure is not testability.** Written beliefs could name the observation that would refute them, yet few do, and what is testable by construction, filters and world models, cannot be read in language.

## 5 Revision and Its Failures

Bayesian filtering and active inference agree on how a belief should change. It predicts, evidence arrives, and the mismatch drives the update (Friston, 2010; Friston et al., 2017), routed by factor graphs to the factors that produced the prediction (Kschischang et al., 2001; Loeliger, 2004). In Equation 1 the transition carries the belief forward

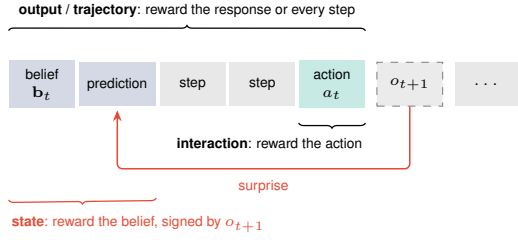
and the likelihood decides how far the observation moves it. The language-model literature imported the vocabulary but rarely the routing, appending feedback to the context or folding it into self-critique loops (Huang et al., 2022b; Madaan et al., 2023; Kim et al., 2023a).

### 5.1 Four Failure Modes

Contextual Belief Management names three ways a belief can fail on new input, a *failed stay* in which the belief moves without evidence (an update no test weighs), a *failed update* in which it survives evidence against it (carried forward without an update), and a *failed isolation* in which irrelevant input contaminates it (Xu et al., 2026a). In consolidated memory a quarter or more of the agent’s own speculations resurface as user-stated facts (Verbovoy, 2026). Failed update is also the base model’s default. Assistants revise correct answers under pushback (Sharma et al., 2023; Atwell et al., 2025), drift as context accumulates (Geng et al., 2025), and update in context like a filter that forgets exponentially (Zhang et al., 2025e), while memory systems write the misconception in (Bensal et al., 2026). Independent studies document a fourth, the *failed act*, in which the belief is revised but the policy is not conditioned on it, so that a belief-level oracle finds the diagnosis right while the action stays wrong (Deb and Krishnan, 2026; Gao et al., 2026b). Table 3 and Fig. 2 place them.

### 5.2 Mechanisms of Revision

The mechanisms that respond to failed update fall into three families by what triggers them. The first is triggered by *contradiction* among stored claims and resolves it with operator algebras, first-class



**Fig. 4 | The credited object on one span of a trajectory.** State credit reaches the written belief, signed by the observation returned after the action.

**Table 4 | Learning signals by credited object and source of the test.**

| Level       | Credited object       | Test from                    | Representative                |
|-------------|-----------------------|------------------------------|-------------------------------|
| Output      | finished response     | human preference             | RLHF, DPO                     |
| Trajectory  | sampled steps         | verifier, PRM                | PRMs, GRPO, LATS              |
| Interaction | actions, tool calls   | task reward                  | ArCHer, RAGEN, HIPER          |
|             | memory operations     | task reward, verifier        | MEM1, Memory-R1, VerMem       |
| State       | written belief        | symbolic verifier            | CBM                           |
|             | claim + certainty     | calibration target           | Agent-BRACE                   |
|             | posterior over claims | Bayesian teacher             | BOND, DEL-ToM                 |
|             | state prediction      | simulator state              | VAGEN, Dyna-Think, SPA        |
|             | belief over answers   | known answer                 | T3, IGPO, InfoPO, CORE, GAVEL |
|             | written belief        | <b>own next observation</b>  | <b>ReBel</b>                  |
|             | written belief        | <b>own past observations</b> | <b>ABBEL</b>                  |

The object that receives the gradient and where the test that signs it comes from (Section 6). Bold, the test comes from the agent’s own observations. Table 14 in the appendix adds granularity and evidence level.

update and forget operations, or supersession contracts (Wang, 2026b; Hao et al., 2026; Xu, 2026; Qin and Lu, 2026). The second is triggered *before commitment*, by self-audit, a parametric veto, semantic entropy, or resampling (Yuan et al., 2026a; Song and Cai, 2026a; Yan et al., 2026b; Wang et al., 2026b). These see no future observation and so cannot catch a belief that is consistent and wrong. The third is triggered by *surprise*. Reward-prediction error routes writes to a fast or a slow path (Song and Xin, 2026), and prediction–observation mismatches become semantic rules (Zhang et al., 2026g). FinCon revises its investment beliefs after each episode and sends each revision only to the agents that need it, routing the error to an agent though not yet to a claim (Yu et al., 2024).

**Insight 2. Revision is triggered, not routed.** Contradiction, self-audit, and surprise decide *when* a belief changes, but none sends the error to the claim that made it.

## 6 The Test as a Learning Signal

A learning signal is a test that has become a gradient. Only 296 of the 517 included papers train anything, and for the rest the belief is built and revised at inference time. Where existing surveys organise post-training by reward source and den-

sity (Wu, 2025; Zhang, 2026; Zheng et al., 2025a), we separate two axes that are usually run together (Fig. 4, Table 4), the *credited object*, the quantity that receives the gradient, and the *source of the test*, whatever signs it (humans, a verifier, the task reward, a teacher, a simulator, or the agent’s own later observation). A method is at the *state level* if the credited object is a claim about the environment, whatever the source, and the source then decides whether the sign exists outside a simulator.

### 6.1 Output and Trajectory

Supervised fine-tuning and preference optimisation credit the completed response (Ouyang et al., 2022; Wei et al., 2022a; Rafailov et al., 2023; Wang et al., 2022; Huang et al., 2022a). Trajectory-level credit began by imitating and searching over agent trajectories (Chen et al., 2023a; Zhou et al., 2023a; Putta et al., 2024; Fu et al., 2025), then credited sampled steps through PPO, GRPO, and process reward models (Schulman et al., 2017; Shao et al., 2024; Lightman et al., 2024; Zheng et al., 2025a).

### 6.2 Interaction

Agentic reinforcement learning credits actions over several turns (Zhou et al., 2024b; Wang et al., 2025f; Zhou et al., 2025c; Feng et al., 2025b; Xiong et al., 2025a), and the open question is how to make that credit denser without making it wrong, by hierarchical advantages, counterfactual estimators, potential-based shaping, or verifier-structured credit (Peng et al., 2026a; Wang et al., 2026a; Diao et al., 2026; Wang et al., 2026j; Ren et al., 2026a; Liao et al., 2026b; Liu et al., 2026b; Li et al., 2026b). The nearest this level comes to the state is a group that credits the write to a memory (Zhou et al., 2025a,d; Yu et al., 2025b; Yan et al., 2025; Sun et al., 2026b; Yan et al., 2026a). These credit the *operation*, so a well-executed update to a wrong claim is still rewarded.

### 6.3 State

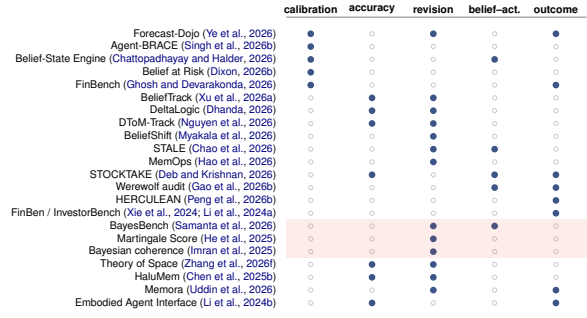
Until 2026 every such method took its test from outside the environment. CBM trains with belief-state rewards from symbolic verification and cuts belief-management failures by 71% on average over untrained models (Xu et al., 2026a), Agent-BRACE trains belief model and policy jointly with verbalised certainty as the belief’s output (Singh et al., 2026b), and BOND distills a Bayesian teacher’s posterior into an 8B student (Cui and

**Table 5 | Evidence-acquisition decisions by what they are conditioned on.**

| Acquisition conditioned on        | Systems                                                                                                                           | Belief object                                |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| task-level POMDP                  | CGDP (Kausik et al., 2026), MedExAgent (Gao et al., 2026a)                                                                        | searcher, patient state                      |
| uncertainty of an explicit belief | BayesEvolve (Wu et al., 2026d), EvoSCM (Zhao et al., 2026b), Align-While-Search (Bae et al., 2025), CA-BED (Arnould et al., 2026) | hypothesis, causal, configuration posteriors |
| belief over own resources         | Bayesian-Agent (Wu et al., 2026b), REDEREF (Hosseini et al., 2026), CMI (Srivastava, 2026b), REFLEX (Wu and Lim, 2026)            | skill, delegate, memory reliability          |
| information gain as reward        | PICA (Liu et al., 2026b), TIPS (Xie et al., 2026)                                                                                 | implicit                                     |
| written testable belief           | <i>none in corpus</i>                                                                                                             | —                                            |

Lin, 2026). VAGEN rewards a written state prediction against the simulator’s state (Wang et al., 2025d), T3 truncates rollouts by a belief deviation anchored to the true hypothesis (Zou et al., 2025), IGPO and InfoPO reward a turn by the belief update it causes toward the known answer (Wang et al., 2025a; Kong et al., 2026), DynaThink, SPA, and RLVR-World train state prediction as a loss or verifiable reward (Yu et al., 2025c; Chen et al., 2025e; Wu et al., 2025a), DEL-ToM scores each belief-update step with a process belief model (Wu et al., 2025d), and CORE and GAVEL train a probabilistic belief against gold slots or held-out placements (Zhang et al., 2026j; Wang et al., 2026e). Each needs the answer or the latent state at training time. Two now take it from the agent’s own observations: ReBel rewards each predicate of a written belief that the next observation supports (Tang et al., 2026b), and ABBEL grades a natural-language belief by how well it reconstructs the prior observation (Lidayan et al., 2025). PaW’s next-observation loss reaches the policy, not a belief (Lu et al., 2026a). None carries uncertainty, and the dense form is fragile: through group-normalised advantages it collapses every run, and as an auxiliary loss shuffled targets match it (Wang, 2026a).

**Insight 3. Credit has reached the belief, not yet its uncertainty.** The first tests signed by the agent’s own observations score binary beliefs, and the systems that train a calibrated belief with the policy, such as Agent-BRACE, still take their test from a calibration target.



**Fig. 5 | Belief-level constructs reported by each evaluation paper.** Only the Bayesian-coherence rows (shaded) share a reference posterior.

## 7 The Test as a Choice of Evidence

A testable belief tells the agent what evidence would change it, which turns retrieval, tool use, and routing into decisions (Table 5). Expected information gain needs a probabilistic test, so the systems that choose evidence by belief are those that carry one. Search and diagnosis have been framed as POMDPs whose actions include what to observe next (Kausik et al., 2026; Gao et al., 2026a), and in a lending simulator failing agents spend 62% of their observation calls on internal state, surviving ones 67% on external signals (Han et al., 2026; Zhang et al., 2026c).

Agentic retrieval let the model decide when to retrieve (Gao et al., 2024), and recent systems condition that decision on measured uncertainty about the task (Wu et al., 2026d; Zhao et al., 2026b; Xu and Fekri, 2026; Yan et al., 2026b; Bae et al., 2025; Arnould et al., 2026; Wang et al., 2026e), make information gain the training signal itself (Liu et al., 2026b; Xie et al., 2026), or keep a belief over the agent’s own skills, delegates, or memories (Wu et al., 2026b; Hosseini et al., 2026; Srivastava, 2026b; Wu and Lim, 2026).

**Insight 4. Acquisition is a belief problem.** A testable belief names the evidence that would change it, so retrieval should follow from the belief rather than precede it.

## 8 The Test as a Metric: Evaluating Belief Apart from Outcome

A correct answer can follow from a correct belief, a wrong belief that did not matter, or a belief revised too late, and outcome metrics such as task success (Shridhar et al., 2020; Liu et al., 2023b; Yao et al., 2024) cannot separate the three. Recent work added belief-level metrics, each defining its

own and each measuring one part from outside (calibration the belief state, accuracy the revision, Bayesian coherence the test, the belief–action gap whether the policy uses any). Fig. 5 aligns them by construct, and Table 15 in the appendix catalogues the benchmarks.

## 8.1 Constructs

*Calibration over a trajectory* has the most statistics and the least comparability. Checkpoint and episode calibration, posterior entropy, and proper scoring rules on time-gated forecasts express one construct four ways, none computed on another paper’s system (Lin et al., 2026c; Singh et al., 2026b; Chattopadhyay and Halder, 2026; Dixon, 2026b; Ghosh and Devarakonda, 2026; Ye et al., 2026), and the one replicated deficit flipped sign across model versions (Joshi, 2026). *Belief accuracy and revision* are measured against a known answer in a closed world, after a minimal premise edit, and across recall, inference, and change detection (Xu et al., 2026a; Dhanda, 2026; Nguyen et al., 2026), against human annotation or a hidden ledger (Myakala et al., 2026; Chao et al., 2026; Hao et al., 2026; Yang et al., 2026b; Srivastava, 2026a), a probed spatial map or a screenplay’s state (Zhang et al., 2026f; Tian et al., 2026), and theory-of-mind benchmarks (Kim et al., 2023b; Bawatneh et al., 2026; Wang et al., 2026c). *Memory validity* is scored per operation or by penalising invalidated memories (Chen et al., 2025b; Wu et al., 2024; Liu et al., 2026a; Uddin et al., 2026). *Bayesian coherence* compares a belief trajectory with an ideal updater, through total-variation distance to the exact posterior (Samanta et al., 2026), a martingale test on successive beliefs (He et al., 2025), and the correlation between the model’s update and the log-odds of the evidence (Imran et al., 2025; Yuan et al., 2025b). The *belief–action gap* is the youngest construct. STOCKTAKE scores the agent against an exact Bayes filter on the same observations (Deb and Krishnan, 2026), the Werewolf audit reports action–belief consistency (Gao et al., 2026b), and SimpleToM separates inferring a belief from applying it (Gu et al., 2024).

Every construct beyond calibration needs a latent state to compare against. Controlled environments provide it (Deb and Krishnan, 2026; Xu et al., 2026a; Alaswad et al., 2026; Wang et al., 2026i), long-horizon simulators know it but report belief through proxies (He et al., 2026b; Zhang et al., 2026c; Han et al., 2026), and historical re-

**Table 6 | Credit levels compared by where their test comes from.**

| Level       | Test from                     | Env. | No oracle | Belief | Act gap | Systems |
|-------------|-------------------------------|------|-----------|--------|---------|---------|
| Output      | humans, verifiers             |      |           |        |         |         |
| Trajectory  | verifier, PRM                 |      |           |        |         |         |
| Interaction | actions                       |      |           |        |         |         |
|             | task reward                   |      |           |        |         |         |
| State       | memory ops                    |      |           |        |         |         |
|             | task reward, teacher          |      |           |        |         |         |
|             | oracle or teacher             |      |           |        |         |         |
|             | closed world, teacher, target |      |           |        |         |         |
|             | own observation               |      |           |        |         |         |
|             | returned observation          |      |           |        |         |         |

Env., the test comes from the environment; No oracle, no external source of truth is needed; Belief, the written belief receives the gradient; Act gap, belief and action error are separated; Systems, solid many, tint a few, grey none. Otherwise solid, yes; tint, partly; grey, no.

play provides neither (Peng et al., 2026b).

### Insight 5. Belief cannot be scored by outcome.

More than twenty metrics share almost no system, and only a known latent state can score belief, so controlled environments, not replay, will carry progress.

## 9 Discussion

**From memory to testable belief: the test is scarce.** The evidence supports the survey’s central claim that agents need to move from memory, which records what happened, to belief, a hypothesis their own observations can support or falsify. Memory has become a belief state and post-training a revision, but the step from record to hypothesis is the test, and Table 4 shows the state level taking it from a source the environment does not supply in every case but two, ReBel, which settles a predicate belief against the next observation, and ABBEL, which reconstructs the previous one. Both are binary and isolated. The step of Fig. 2 that makes a belief a hypothesis is no longer empty, but it is thin. The cycle is broader than its Bayesian form and applies to agentic systems generally.

**Auditability and expressiveness pull apart.** Filters are validated in closed form but cannot introduce a factor they were not given, written beliefs can name a new factor but are rarely testable, and world models are testable but mute. Hybrids that write the belief and check it parametrically are the convergence point (Song and Cai, 2026a; Zhang et al., 2026g; Kumar et al., 2026).

**The belief–action gap is a training problem.** Wherever belief and action are scored on the same episodes, prompting and oracles improve the belief while action–belief consistency barely moves (Deb and Krishnan, 2026; Gao et al., 2026b), and the system that moves both trains belief and policy jointly (Singh et al., 2026b). Closing the gap requires the state-level and interaction-level sig-

nals of Table 6 in one objective, not a better belief alone.

## 10 Open Problems

**A calibrated belief with an environment-signed test.** ReBel signs a written belief with the next observation, but a binary one, and Agent-BRACE carries certainty, but signed by a calibration target. Joining the two without the collapse dense prediction rewards show under group normalisation is the most direct open problem. ReBel’s predicates already expose the unit a proper scoring rule needs, so the step is to attach a probability to each and score it against the observation that returns.

**Latent beliefs without ground truth.** Historical data has no oracle or annotation, and its candidate substitutes, filter-recovered latents and representation drift, are unvalidated (Dixon, 2026b; Xue, 2026); a controlled environment paired with a replay of the same instrument would test whether belief-level gains transfer.

**Belief under governance.** Regulated domains add auditability, which only fixed-space filters provide, and LLM-proposed spaces are the candidate bridge (Six et al., 2026; Soru, 2026; Chen et al., 2026c; Dixon, 2026c).

## 11 Conclusion

A memory records what has happened, whereas a belief is a hypothesis about what is true, open to testing and revision by future evidence. This survey asked of every system which of the belief’s three parts it has and where each comes from. The field has given the agent a state, from the trace to stores, filters, and world models, a revision, with credit made denser at every level below the belief, and more than twenty ways to measure belief without a shared one. The test that turns a record into a hypothesis is still supplied from outside: testable representations are not the ones the agent can reason about, revision does not route the error to the claim that produced it, and the two methods that train a written belief on the agent’s own observations use a binary one. The next step is a written belief that carries uncertainty, is scored by the agent’s own matured predictions, receives the error at the claim that made it, and is evaluated where the latent state is known. The taxonomy, screening record, and benchmark catalogue are re-

leased to make that step measurable.<sup>1</sup>

## Limitations

This survey covers work from January 2020 to 30 September 2026 that the keyword, citation-tree, and update searches of Appendix A reach in OpenAlex and arXiv. Each system is assigned one source per part of the belief, although some combine several. Reported numbers are those stated by the cited papers; we did not re-run any method.

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## Declaration of interests

Jimin Huang and Xueqing Peng are affiliated with The Fin AI, which produced several of the finance benchmarks discussed here, among them FinBen and InvestorBench; these were screened, included, and described under the same protocol as every other paper (Appendix A). One benchmark discussed in Section 8 and Table 15, Herculean (Peng et al., 2026b), is co-authored by the authors of this survey. The authors declare no other financial or personal relationships with people or organisations that could be perceived to have influenced the work reported in this paper. The funder had no role in the design of the review, the screening protocol, the analysis, or the decision to publish.

## References

- Farhad Abtahi, Abdolamir Karbalaie, Eduardo Illueca-Fernandez, and Fernando Seoane. 2026. **MEDLEY-BENCH: Benchmarking behavioural metacognition and belief revision under social pressure in large language models.** *Preprint*, arXiv:2604.16009.
- Ashutosh Adhikari, Xingdi Yuan, Marc-Alexandre Côté, Mikuláš Zelinka, Marc-Antoine Rondeau, Romain Laroche, Pascal Poupart, Jian Tang, Adam Trischler, and William L. Hamilton. 2020. **Learning dynamic belief graphs to generalize on text-based games.** In *Proceedings of NeurIPS 2020*.
- Naman Agarwal, Siddhartha R. Dalal, and Vishal Misra. 2025. **The Bayesian geometry of Transformer attention.** *arXiv preprint arXiv:2512.22471*.

<sup>1</sup><https://github.com/jiminHuang/belief-state-survey>

- Milad Aghajohari, Kamran Chitsaz, Amirhossein Kazemnejad, Sarath Chandar, Alessandro Sordoni, Aaron Courville, and Siva Reddy. 2025. [The markovian thinker: Architecture-agnostic linear scaling of reasoning](#). *arXiv preprint arXiv:2510.06557*.
- Nesreen K. Ahmed and Nima Nafisi. 2026. [Agent-ToM: Learning to monitor autonomous LLM agents via theory-of-mind reasoning](#). *Preprint*, arXiv:2605.24216.
- Shimaa Ahmed, Yiwei Cai, Mohsen Minaei, and Rahul Rachuri. 2026. [Can LLM agents price competitively? a dynamic multi-attribute auction benchmark for agentic commerce](#). *arXiv preprint arXiv:2608.00102*.
- Michael Ahn, Anthony Brohan, Noah Brown, Yevgen Chebotar, Omar Cortes, Byron David, Chelsea Finn, Chuyuan Fu, Keerthana Gopalakrishnan, Karol Hausman, Alex Herzog, Daniel Ho, Jasmine Hsu, Julian Ibarz, Brian Ichter, Alex Irpan, Eric Jang, Rosario Jauregui Ruano, Kyle Jeffrey, and 26 others. 2022. Do as i can, not as i say: Grounding language in robotic affordances. In *CoRL 2022*. ArXiv:2204.01691.
- Qingyao Ai, Yichen Tang, Changyue Wang, Jianming Long, Weihang Su, and Yiqun Liu. 2025. [MemoryBench: A benchmark for memory and continual learning in LLM systems](#). *arXiv preprint arXiv:2510.17281*.
- Hafsa Akbar, Daniel Platnick, Marjan Alirezaie, Hossein Rahnama, and Alex 'Sandy' Pentland. 2026. [Bayesian belief layer for controllable opinion dynamics in LLM agents](#). *arXiv preprint arXiv:2609.21997*.
- Feisal Alaswad, Batoul Aljaddouh, Maher Alrahal, Poovammal E, and Talal Bonny. 2026. [Why we need world models for AGI: Where LLMs fail and how world models may outperform](#). *Preprint*, arXiv:2605.23972.
- Carlos E. Alchourrón, Peter Gärdenfors, and David Makinson. 1985. On the logic of theory change: Partial meet contraction and revision functions. *Journal of Symbolic Logic*, 50(2):510–530.
- Amez Amanj Ali and Kuo-Kun Tseng. 2026. [Reward-driven LLM agent workflows: Synthesizing POMDP routing and self-correction for autonomous decision-making](#). *Preprint*, arXiv:2607.17038.
- Danial Amin. 2026. [Bayesian orchestration of Multi-LLM agents for cost-aware sequential decision-making](#). *arXiv preprint arXiv:2601.01522*.
- Jacob Andreas. 2022. Language models as agent models. In *EMNLP 2022 Findings*. ArXiv:2212.01681.
- Huan ang Gao, Zikang Zhang, Tianwei Luo, Kaisen Yang, Xinzhe Juan, Jiahao Qiu, Tianxing Chen, Bingxiang He, Hao Zhao, Hao Zhou, Shilong Liu, and Mengdi Wang. 2025. [CubeBench: Diagnosing interactive, long-horizon spatial reasoning under partial observations](#). *arXiv preprint arXiv:2512.23328*.
- Rebecca Ansell and Autumn Toney-Wails. 2026. [Clueing up LLMs with tool-augmented deductive reasoning](#). *arXiv preprint arXiv:2609.18736*.
- Daniel Arnould, Rashad Aziz, Zixuan Kang, Tanav Chagal, Kevin Zhu, Sunishchal Dev, Gabriel Grand, and Shreyas Sunil Kulkarni. 2026. [CA-BED: Conversation-aware Bayesian experimental design](#). *arXiv preprint arXiv:2606.01182*.
- Akari Asai, Zeqiu Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. 2023. [Self-RAG: Learning to retrieve, generate, and critique through self-reflection](#). In *Proceedings of ICLR 2023*.
- Karl Johan Åström. 1965. Optimal control of Markov processes with incomplete state information. *Journal of Mathematical Analysis and Applications*, 10(1):174–205.
- Katherine Atwell, Pedram Heydari, Anthony Sicilia, and Malihe Alikhani. 2025. [BASIL: Bayesian assessment of sycophancy in LLMs](#). In *Proceedings of Conference on Fairness, Accountability and Transparency 2025*.
- Guy Azran, Michael Navat, and Sarah Keren. 2026. [Bridging learned visual perception and symbolic belief-space planning](#). In *Proceedings of NeuS 2026*.
- Joris Baan, Wilker Aziz, Barbara Plank, and Raquel Fernández. 2026. [Clarify, abstain or answer? strategising in conversation with belief-augmented generation](#). *arXiv preprint arXiv:2605.25831*.
- Axel Backlund and Lukas Petersson. 2025. [Vending-bench: A benchmark for long-term coherence of autonomous agents](#). *arXiv preprint arXiv:2502.15840*.
- Seohui Bae, Jeonghye Kim, Youngchul Sung, and Woohyung Lim. 2025. [Align while search: Belief-guided exploratory inference for world-grounded embodied agents](#). *arXiv preprint arXiv:2512.24461*.
- Vaidehi Bagaria, Bijo Sebastian, and Nirav Kumar Patel. 2026. [Recursive belief vision language action models](#). *Preprint*, arXiv:2602.20659.
- Hao Bai, Yifei Zhou, Mert Cemri, Jiayi Pan, Alane Suhr, Sergey Levine, and Aviral Kumar. 2024. [Di-giRL: Training in-the-wild device-control agents with autonomous reinforcement learning](#). In *Proceedings of NeurIPS 2024*.
- Jiaxin Bai, Yue Guo, Yifei Dong, Jiaxuan Xiong, Tianshi Zheng, Yixia Li, Tianqing Fang, Yufei Li, Yisen Gao, Haoyu Huang, Zhongwei Xie, Hong Ting Tsang, Zihao Wang, Lihui Liu, Jeff Z. Pan, and Yangqiu Song. 2026. [PatchWorld: Gradient-free optimization of executable world models for agent environments](#). *arXiv preprint arXiv:2605.30880*.

- Victor Barres, Honghua Dong, Soham Ray, Xujie Si, and Karthik Narasimhan. 2025. [tau2-bench: Evaluating conversational agents in a dual-control environment](#). *Preprint*, arXiv:2506.07982.
- Semanti Basu et al. 2026. Augmenting large language models with psychologically grounded models of causal reasoning for planning under uncertainty. In *Frontiers in Artificial Intelligence*.
- Adam Bawatneh, Sagar Sapkota, Amrit Singh Bedi, Santu Karmaker, and Mubarak Shah. 2026. [OmniToM: Benchmarking theory of mind in LLMs via explicit belief modeling](#). *Preprint*, arXiv:2605.26322.
- Shelly Bensal, Axel Magnuson, Aparna Balagopalan, and Daniel M. Bikel. 2026. [Recalling too well: Sycophancy evaluation and mitigation in memory-augmented models](#). *arXiv preprint arXiv:2606.10949*.
- Shmuel Berman and Jia Deng. 2026. [Good memory has ECC: Evaluating the memory of vision-language models beyond accuracy](#). *arXiv preprint arXiv:2609.00103*.
- Gregor Betz et al. 2023. Probabilistic coherence, logical consistency, and Bayesian learning: Neural language models as epistemic agents. In *PLoS ONE*.
- William E Bishop, Alice Li, Christopher Rawles, and Oriana Riva. 2024. [Latent state estimation helps UI agents to reason](#). *arXiv preprint arXiv:2405.11120*.
- Igor Bogdanov, Chung-Horng Lung, Thomas Kunz, Jie Gao, Adrian Taylor, and Marzia Zaman. 2026. [Context, reasoning, and hierarchy: A cost-performance study of compound LLM agent design in an adversarial POMDP](#). *Preprint*, arXiv:2605.16205.
- Siddharth Boppana, Annabel Ma, Max Loeffler, Raphael Sarfati, Eric Bigelow, Atticus Geiger, Owen Lewis, and Jack Merullo. 2026. [Reasoning theater: Disentangling model beliefs from Chain-of-Thought](#). *arXiv preprint arXiv:2603.05488*.
- Ailiya Borjigin, Igor Stadnyk, Ben Bilski, Maksym Chikita, Dmytro Kyrylenko, Sofia Pidrunkina, and Julia Stadnyk. 2026. [Absorbing complexity: An interaction-native knowledge harness for financial LLM agents](#). *Preprint*, arXiv:2606.01886.
- Jiazhang Cai, Tao Wang, Ruidong Zhang, Siyuan Li, Terry Ma, Luyang Fang, Haoran Lu, Huimin Cheng, Yingchuan Zhang, Shushan Wu, Rui Xie, Lin Tang, Chao Huang, Rongjie Liu, Ziyu Liu, Meizhi Yu, Yongkai Chen, Yifan Zhou, Zeliang Sun, and 19 others. 2026. [Complex problem solving in large language models: A statistical control survey and diagnostic framework](#). *arXiv preprint arXiv:2609.20973*.
- Hyunjoo Chae, Namyoun Kim, Kai Tzu iunn Ong, Minju Gwak, Gwanwoo Song, Jihoon Kim, Sunghwan Kim, Dongha Lee, and Jinyoung Yeo. 2024. [Web agents with world models: Learning and leveraging environment dynamics in web navigation](#). In *Proceedings of ICLR 2025*.
- Chunkit Chan, Cheng Jiayang, Yauwai Yim, Zheyue Deng, Wei Fan, Haoran Li, Xin Liu, Hongming Zhang, Weiqi Wang, and Yangqiu Song. 2024. [NegotiationToM: A benchmark for stress-testing machine theory of mind on negotiation surrounding](#). In *Proceedings of EMNLP 2024*.
- Edward Y. Chang. 2026. [Trivium: Temporal regret as a first-class objective for causal-memory controllers](#). *arXiv preprint arXiv:2606.04421*.
- Edward Y. Chang and Emily J. Chang. 2026. [TRACE: An operational reasoning schema for auditable agentic commitments](#). *arXiv preprint arXiv:2607.12480*.
- Hanxiang Chao, Yihan Bai, Rui Sheng, Tianle Li, and Yushi Sun. 2026. [STALE: Can LLM agents know when their memories are no longer valid?](#) *Preprint*, arXiv:2605.06527.
- Arnab Chattopadhyay and Debdipta Halder. 2026. [Belief-state engine: Augmenting LLMs for principled planning under partial observability](#). *Preprint*, arXiv:2609.10036.
- Baian Chen, Chang Shu, Ehsan Shareghi, Nigel Collier, Karthik Narasimhan, and Shunyu Yao. 2023a. [Fire-Act: Toward language agent fine-tuning](#). *Preprint*, arXiv:2310.05915.
- Cong Chen, Kaixiang Ji, Hao Zhong, Muzhi Zhu, Anzhou Li, Guo Gan, Ziyuan Huang, Cheng Zou, Jiajia Liu, Jingdong Chen, Hao Chen, and Chunhua Shen. 2025a. [GUI-Shepherd: Reliable process reward and verification for long-sequence GUI tasks](#). *arXiv preprint arXiv:2509.23738*.
- Ding Chen, Simin Niu, Kehang Li, Peng Liu, Xiangping Zheng, Bo Tang, Xinchu Li, Feiyu Xiong, and Zhiyu Li. 2025b. [HaluMem: Evaluating hallucinations in memory systems of agents](#). *arXiv preprint arXiv:2511.03506*.
- Guo Chen, Lidong Lu, Yicheng Liu, Liangrui Dong, Lidong Zou, Jixin Lv, Zhenquan Li, Xinyi Mao, Baoqi Pei, Shihao Wang, Zhiqi Li, Karan Sapra, Fuxiao Liu, Yin-Dong Zheng, Yifei Huang, Limin Wang, Zhiding Yu, Andrew Tao, Guilin Liu, and Tong Lu. 2026a. [Towards multimodal lifelong understanding: A dataset and agentic baseline](#). *arXiv preprint arXiv:2603.05484*.
- Kevin Chen, Marco Cusumano-Towner, Brody Huval, Aleksei Petrenko, Jackson Hamburger, and Vladlen Koltun. 2025c. [Reinforcement learning for long-horizon interactive LLM agents](#). *Preprint*, arXiv:2502.01600.
- Lili Chen, Kevin Lu, Aravind Rajeswaran, Kimin Lee, Aditya Grover, Michael Laskin, Pieter Abbeel, Aravind Srinivas, and Igor Mordatch. 2021. [Decision Transformer: Reinforcement learning via sequence modeling](#). In *NeurIPS 2021*. ArXiv:2106.01345.

- Mingguang Chen, Licheng Wang, and Bo Qu. 2026b. [The horizon gap: Planning, memory, execution, training, and evaluation for long-horizon LLM agents](#). *Preprint*, arXiv:2608.06663.
- Mingyang Chen, Linzhuang Sun, Tianpeng Li, Haoze Sun, Yijie Zhou, Chenzheng Zhu, Haofen Wang, Jeff Z. Pan, Wen Zhang, Huajun Chen, Fan Yang, Zenan Zhou, and Weipeng Chen. 2025d. [Re-Search: Learning to reason with search for LLMs via reinforcement learning](#). In *NeurIPS 2025*. ArXiv:2503.19470.
- Shiqi Chen, Tongyao Zhu, Zian Wang, Jinghan Zhang, Kangrui Wang, Ruochen Zhou, Siyang Gao, Teng Xiao, Yee Whye Teh, Junxian He, and Manling Li. 2025e. [Why do LLM agents fail in exploring new environments? a world-modeling perspective](#). *arXiv preprint arXiv:2510.15047*.
- Xinyun Chen, Maxwell Lin, and Denny Zhou. 2023b. [Teaching large language models to self-debug](#). In *ICLR 2024*. ArXiv:2304.05128.
- Yuhan Chen, Zhihua Tian, Mahavir Dabas, Charith Peris, Rahul Gupta, Ming Jin, Feiyang Kang, Siyuan Zhang, Nan Wang, and Ruoxi Jia. 2026c. [Belief-calibrated optimization: An explicit world model for agentic optimization](#). *Preprint*, arXiv:2609.01861.
- Zehui Chen, Kuikun Liu, Qiuchen Wang, Wenwei Zhang, Jiangning Liu, Dahua Lin, Kai Chen, and Feng Zhao. 2024a. [Agent-FLAN: Designing data and methods of effective agent tuning for large language models](#). In *Proceedings of ACL 2024*.
- Zhuang Chen, Jincenzi Wu, Jinfeng Zhou, Bosi Wen, Guanqun Bi, Gongyao Jiang, Yaru Cao, Mengting Hu, Yunghwei Lai, Zexuan Xiong, and Minlie Huang. 2024b. [ToMBench: Benchmarking theory of mind in large language models](#). In *Proceedings of ACL 2024*.
- Mingyue Cheng, Shuo Yu, Daoyu Wang, Qingchuan Li, Xiaoyu Tao, Jie Ouyang, Yucong Luo, Yitong Zhou, Qi Liu, and Enhong Chen. 2025. [Agent-rl: A unified and modular framework for agentic reinforcement learning](#). *Preprint*, arXiv:2511.14460.
- Zihao Cheng, Yixia Li, Shengbin Yue, and Yuheng Lu. 2026. [Theory of agent: The science of internalization and externalization for LLM-based agents](#). Preprints.org.
- Prateek Chhikara, Dev Khant, Saket Aryan, Taranjeet Singh, and Deshraj Yadav. 2025. [Mem0: Building production-ready AI agents with scalable long-term memory](#). *Preprint*, arXiv:2504.19413.
- Hyeong Kyu Choi, Xiaojin Zhu, and Sharon Li. 2025. [Debate or vote: Which yields better decisions in multi-agent large language models?](#) In *Proceedings of NeurIPS 2025*.
- Konstantina Christakopoulou and Shibl Mourad. 2024. [Agents thinking fast and slow: A talker-reasoner architecture](#). *Preprint*, arXiv:2410.08328.
- Sebastian Cochinescu. 2026. [Truth for believable AI: Expressed doubt, provenance, and belief revision as an engineerable stance](#). *arXiv preprint arXiv:2609.26035*.
- Cédric Colas, Tracey Mills, Ben Prystawski, Michael Henry Tessler, Noah Goodman, Jacob Andreas, and Joshua Tenenbaum. 2025. [Language and experience: A computational model of social learning in complex tasks](#). *arXiv preprint arXiv:2509.00074*.
- Logan Cross, Violet Xiang, Agam Bhatia, Daniel LK Yamins, and Nick Haber. 2024. [Hypothetical minds: Scaffolding theory of mind for multi-agent tasks with large language models](#). *Preprint*, arXiv:2407.07086.
- Yanwei Cui, Xing Zhang, Yulong Zhang, Li Shao, Xiaofeng Shi, Guanghui Wang, and Peiyang He. 2026. [Closing the feedback loop: From experience extraction to insight governance in verbal reinforcement learning](#). *arXiv preprint arXiv:2606.17591*.
- Zongqi Cui and Baihan Lin. 2026. [Distilling Bayesian belief states into language models for auditable negotiation](#). *Preprint*, arXiv:2605.04507.
- Nicola Dainese, Matteo Merler, Minttu Alakuijala, and Pekka Marttinen. 2024. [Generating code world models with large language models guided by Monte Carlo tree search](#). In *Proceedings of NeurIPS 2024*.
- Sahith Dambekodi, Spencer Frazier, Prithviraj Ammanabrolu, and Mark O. Riedl. 2020. [Playing text-based games with common sense](#). *arXiv preprint arXiv:2012.02757*.
- Sagar Deb and Ashwanth Krishnan. 2026. [STOCK-TAKE: Measuring the gap between perception and action in LLM agents with a fair oracle](#). *Preprint*, arXiv:2607.13618.
- Darshan Deshpande. 2026. [Masked diffusion language models are strong and steerable text-based world models for agentic RL](#). *arXiv preprint arXiv:2607.16204*.
- Amit Dhanda. 2026. [DeltaLogic: Minimal premise edits reveal belief-revision failures in logical reasoning models](#). *Preprint*, arXiv:2604.02733.
- Juncheng Diao, Zhicong Lu, Peiguang Li, Yongwei Zhou, Changyuan Tian, Qingbin Li, Rongxiang Weng, Jingang Wang, and Xunliang Cai. 2026. [HIPIF: Hierarchical planning and information folding for long-horizon LLM agent learning](#). *Preprint*, arXiv:2606.10507.
- Jie Ding, Rui Sun, Xinyuan Zhang, Zeyu Zhang, and Xin Liu. 2026a. [APEX: Distillation of agent procedural experience for adaptive deep research question answering](#). *arXiv preprint arXiv:2609.02253*.

- Tianyu Ding, Jianhong Xin, and Juan Pablo De la Cruz Weinstein. 2026b. [Keep policy gradient in charge: Sibling-guided credit distillation for long-horizon tool-use agents](#). *arXiv preprint arXiv:2606.12634*.
- Matthew Francis Dixon. 2026a. [Adaptive AI delegation under uncertainty: A Bayesian governance policy for sequential decision authority](#). *Preprint*, arXiv:2606.29406.
- Matthew Francis Dixon. 2026b. [Belief at risk: Quantifying agentic AI model risk with LLM-inferred Bayesian state filters](#). *Preprint*, arXiv:2606.15473.
- Matthew Francis Dixon. 2026c. [Model validation of agentic AI systems: A POMDP-based framework for belief-state, forecast, and policy validation](#). *Preprint*, arXiv:2606.17383.
- Guanting Dong, Hangyu Mao, Kai Ma, Licheng Bao, Yifei Chen, Zhongyuan Wang, Zhongxia Chen, Jiazhen Du, Huiyang Wang, Fuzheng Zhang, Guorui Zhou, Yutao Zhu, Ji-Rong Wen, and Zhicheng Dou. 2025a. [Agentic reinforced policy optimization](#). *Preprint*, arXiv:2507.19849.
- Yifei Dong, Fengyi Wu, Kunlin Zhang, Yilong Dai, Sanjian Zhang, Wanghao Ye, Sihan Chen, and Zhi-Qi Cheng. 2025b. Large language model agents in finance: A survey bridging research, practice, and real-world deployment. In *Findings of the Association for Computational Linguistics: EMNLP 2025*.
- Yifei Dong, Minggen Zheng, Linqun Wu, Jeff Z. Pan, and Jiaxin Bai. 2026. [Mind-studio: Executable world models with lookahead evaluation for partially observable games](#). *arXiv preprint arXiv:2606.16070*.
- Jialu Du, Guiyang Hou, Yihui Fu, Chen Wu, Wenqi Zhang, Yongliang Shen, and Weiming Lu. 2025. [Active confusion expression in large language models: Leveraging world models toward better social reasoning](#). *arXiv preprint arXiv:2510.07974*.
- Feiyu Duan, Xuanjing Huang, and Zhongyu Wei. 2026. [LifeSim: Long-horizon user life simulator for personalized assistant evaluation](#). In *Proceedings of ACL 2026*.
- Xianzhe Fan, Xuhui Zhou, Chuanyang Jin, Kolby Nottingham, Hao Zhu, and Maarten Sap. 2025. [SoMi-ToM: Evaluating multi-perspective theory of mind in embodied social interactions](#). In *Proceedings of NeurIPS 2025*.
- Haishuo Fang, Xiaodan Zhu, and Iryna Gurevych. 2024. [Preemptive detection and correction of misaligned actions in LLM agents](#). In *Proceedings of EMNLP 2024*.
- Jiazhan Feng, Shijue Huang, Xingwei Qu, Ge Zhang, Yujia Qin, Baoquan Zhong, Chengquan Jiang, Jinxin Chi, and Wanjuan Zhong. 2025a. [ReTool: Reinforcement learning for strategic tool use in LLMs](#). *arXiv preprint arXiv:2504.11536*.
- Lang Feng, Zhenghai Xue, Tingcong Liu, and Bo An. 2025b. [Group-in-group policy optimization for LLM agent training](#). In *NeurIPS 2025*. ArXiv:2505.10978.
- Tao Feng, Xinke Jiang, and Chao Wu. 2026. [KbSD: Knowledge boundary aware self-distillation for behavioral calibration in agentic search](#). *arXiv preprint arXiv:2606.29863*.
- Karl Friston. 2010. The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 11(2):127–138.
- Karl Friston, Thomas FitzGerald, Francesco Rigoli, Philipp Schwartenbeck, and Giovanni Pezzulo. 2017. Active inference: a process theory. *Neural Computation*, 29(1):1–49.
- Dayuan Fu, Keqing He, Yejie Wang, Wentao Hong, Zhuoma Gongque, Weihao Zeng, Wei Wang, Jingang Wang, Xunliang Cai, and Weiran Xu. 2025. [AgentRefine: Enhancing agent generalization through refinement tuning](#). In *ICLR 2025*. ArXiv:2501.01702.
- Huan-ang Gao, Jiayi Geng, Wenyue Hua, Mengkang Hu, Xinzhe Juan, Hongzhang Liu, Shilong Liu, Jiahao Qiu, Xuan Qi, Yiran Wu, Hongru Wang, Han Xiao, Yuhang Zhou, Shaokun Zhang, Jiayi Zhang, Jinyu Xiang, Yixiong Fang, Qiwen Zhao, Dongrui Liu, and 8 others. 2025. [A survey of self-evolving agents: What, when, how, and where to evolve on the path to artificial super intelligence](#). *arXiv preprint arXiv:2507.21046*.
- Yicheng Gao, Xiaolin Zhou, Yahan Li, Yue Zhao, and Ruishan Liu. 2026a. [MedExAgent: Training LLM agents to ask, examine, and diagnose in noisy clinical environments](#). *Preprint*, arXiv:2605.07058.
- Yuan Gao, Jiangyi Yang, Yao Zhao, and Yichi Zhang. 2026b. [Auditing belief-conditioned LLM agents in hidden-information social deduction games](#). *Preprint*, arXiv:2607.10814.
- Yunfan Gao, Yun Xiong, Xinyu Gao, Kangxiang Jia, Jinliu Pan, Yuxi Bi, Yi Dai, Jiawei Sun, Meng Wang, and Haofen Wang. 2024. [Retrieval-augmented generation for large language models: A survey](#). *arXiv preprint arXiv:2312.10997*.
- Jiayi Geng, Howard Chen, Ryan Liu, Manoel Horta Ribeiro, Robb Willer, Graham Neubig, and Thomas L. Griffiths. 2025. [Accumulating context changes the beliefs of language models](#). *arXiv preprint arXiv:2511.01805*.
- Luay Gharzeddine and Samer Saab. 2026. [Complete cyclic subtask graphs for tool-using LLM agents: Flexibility, cost, and bottlenecks in long-horizon workflows](#). *Preprint*, arXiv:2604.22820.
- Rishab Ghosh and Vinay Devarakonda. 2026. [Fin-Bench: Time-gated calibration and uncertainty benchmarking for agentic financial forecasting](#). *Preprint*, arXiv:2607.16229.

- Sayontan Ghosh, Mahnaz Koupaee, Yash Kumar Lal, Pegah Alipoormolabashi, Mohammad Saqib Hasan, Jun Seok Kang, and Niranjana Balasubramanian. 2025. [DIAMONDS: A dataset for Dynamic Information And Mental modeling Of Numeric Discussions](#). *arXiv preprint arXiv:2505.12651*.
- Tommaso Giovannelli and Griffin D. Kent. 2026. [CHAL: Council of hierarchical agentic language](#). *Preprint*, arXiv:2605.12718.
- Nanxu Gong, Zixin Chen, Haotian Li, Zishu Zhao, Jianxun Lian, Huamin Qu, Yanjie Fu, and Xing Xie. 2026. [Does theory of mind improvement really benefit Human-AI interactions? empirical findings from interactive evaluations](#). In *Proceedings of ACL 2026*.
- Mikhail Gritsaenko. 2026. [Executable epistemic contracts in deterministic agent simulation: The ENIGMA architecture](#). Zenodo preprint.
- Yuling Gu, Oyvind Tafjord, Hyunwoo Kim, Jared Moore, Ronan Le Bras, Peter Clark, and Yejin Choi. 2024. [SimpleToM: Exposing the gap between explicit ToM inference and implicit ToM application in LLMs](#). In *Proceedings of ICLR 2026*.
- Jiacheng Guo, Ling Yang, Peter Chen, Qixin Xiao, Yinjie Wang, Xinzhe Juan, Jiahao Qiu, Ke Shen, and Mengdi Wang. 2025. [GenEnv: Difficulty-aligned co-evolution between LLM agents and environment simulators](#). *arXiv preprint arXiv:2512.19682*.
- Tengda Guo, Jie Leng, Hanlei Li, Yaoyuan Liang, Qingyue Zhang, Dian Yang, Mingyu Zhang, Yuhua Fu, and Shao-Lun Huang. 2026a. [CAVE: A structured credit assignment approach for fragmented visual evidence reasoning](#). *arXiv preprint arXiv:2605.16416*.
- Weiyang Guo, Zesheng Shi, Longhui Zhang, Zeen Zhu, Min Zhang, and Jing Li. 2026b. [Agent reinforcement learning via pivotal-aware self-feedback retry](#). *arXiv preprint arXiv:2607.03702*.
- Yucan Guo, Xiaohan Wang, Miao Su, Saiping Guan, Zhongni Hou, Jiajun Chai, Wei Lin, Guojun Yin, Xiaolong Jin, Jiafeng Guo, and Xueqi Cheng. 2026c. [HiDiffTIR: Hierarchical difficulty-aware policy optimization for multi-turn tool-integrated reasoning](#). *Preprint*, arXiv:2608.21863.
- David Ha and Jürgen Schmidhuber. 2018. World models. In *Advances in Neural Information Processing Systems*.
- Danijar Hafner, Timothy Lillicrap, Ian Fischer, Ruben Villegas, David Ha, Honglak Lee, and James Davidson. 2019. Learning latent dynamics for planning from pixels. In *International Conference on Machine Learning*.
- Danijar Hafner, Jurgis Pasukonis, Jimmy Ba, and Timothy Lillicrap. 2023. Mastering diverse domains through world models. *arXiv preprint arXiv:2301.04104*.
- Haoliang Han. 2026. [Memory depth, not memory access: Selective parametric consolidation for long-running language agents](#). *arXiv preprint arXiv:2606.26806*.
- Yi Han, Yan Wang, Lingfei Qian, Haohang Li, Yupeng Cao, Yueru He, Xueqing Peng, Nanhan Shen, Yitao Xu, Yankai Chen, Dongji Feng, Jimin Huang, Xue Liu, Jian-Yun Nie, and Sophia Ananiadou. 2026. [Can LLM agents be CFOs? benchmarking long-horizon resource allocation in an uncertain enterprise environment](#). *Preprint*, arXiv:2603.23638.
- Hakeem Hannon, Andrew Zhao, Mihir Narayan, Sharvin Goyal, and Ivaxi Sheth. 2026. [Mitigating over-personalization in LLMs via structured memory](#). *arXiv preprint arXiv:2608.08300*.
- Shibo Hao, Yi Gu, Haodi Ma, Joshua Jiahua Hong, Zhen Wang, Daisy Zhe Wang, and Zhiting Hu. 2023. Reasoning with language model is planning with world model. In *EMNLP 2023*. ArXiv:2305.14992.
- Xixuan Hao, Zeyu Zhang, Zehao Lin, Yihang Sun, Ziliang Guo, Xichong Zhang, Yuxuan Liang, Feiyu Xiong, and Zhiyu Li. 2026. [MemOps: Benchmarking lifecycle memory operations in long-horizon conversations](#). *Preprint*, arXiv:2607.12893.
- Peter Hase, Mona Diab, Asli Celikyilmaz, Xian Li, Zornitsa Kozareva, Veselin Stoyanov, Mohit Bansal, and Srinivasan Iyer. 2021. [Do language models have beliefs? methods for detecting, updating, and visualizing model beliefs](#). *arXiv preprint arXiv:2111.13654*.
- Peter Hase, Thomas Hofweber, Xiang Zhou, Elias Stengel-Eskin, and Mohit Bansal. 2024. [Fundamental problems with model editing: How should rational belief revision work in LLMs?](#) In *Transactions on Machine Learning Research*.
- Jianliang He, Siyu Chen, Fengzhuo Zhang, and Zhuoran Yang. 2024. [From words to actions: Unveiling the theoretical underpinnings of LLM-Driven autonomous systems](#). In *Proceedings of ICML 2024*.
- Kailai He, Zhihao Wu, Linhai Zhang, Runcong Zhao, Yulan He, and Jiazheng Li. 2026a. [CoLearn: An agentic tutor that learns its learner in a Human-AI co-learning loop](#). *arXiv preprint arXiv:2609.21154*.
- Muyu He, Adit Jain, Anand Kumar, Vincent Tu, Soumyadeep Bakshi, Sachin Patro, and Nazneen Rajani. 2026b. [YC-bench: Benchmarking AI agents for long-term planning and consistent execution](#). *Preprint*, arXiv:2604.01212.
- Shuo He, Lang Feng, Qi Wei, Xin Cheng, Lei Feng, and Bo An. 2026c. [Hierarchy-of-Groups policy optimization for long-horizon agentic tasks](#). *arXiv preprint arXiv:2602.22817*.
- Yinghui He, Yufan Wu, Yilin Jia, Rada Mihalcea, Yulong Chen, and Naihao Deng. 2023. [HI-TOM: A benchmark for evaluating higher-order theory of](#)

- mind reasoning in large language models. In *Findings of EMNLP 2023*.
- Zhida He, Xiaoyu Wen, Han Qi, Ziyuan Zhou, Peng Yu, Xingcheng Xu, Dongrui Liu, Xia Hu, Chaochao Lu, and Qiaosheng Zhang. 2026d. [Not all turns matter: Credit assignment for multi-turn jailbreaking](#). *Preprint*, arXiv:2605.08778.
- Zhonghao He, Tianyi Qiu, Hirokazu Shirado, and Maarten Sap. 2025. [Martingale score: An unsupervised metric for Bayesian rationality in LLM reasoning](#). In *Proceedings of NeurIPS 2025*.
- Thomas Hofweber, Peter Hase, Elias Stengel-Eskin, and Mohit Bansal. 2024. [Are language models rational? the case of coherence norms and belief revision](#). *arXiv preprint arXiv:2406.03442*.
- Jiseung Hong, Grace Byun, Seungone Kim, Kai Shu, and Jinho D. Choi. 2025. [Measuring sycophancy of language models in multi-turn dialogues](#). In *Proceedings of EMNLP 2025*.
- S M Asif Hossain, Ruksat Khan Shayoni, and Md Kishor Morol. 2026. [CAPTURE: Disentangling preference drift from memory poisoning in personalized LLM agents](#). *Preprint*, arXiv:2609.02265.
- Mohammad Parsa Hosseini, Ankit Shah, Saiyra Qureshi, Alex Huang, Connie Miao, and Wei Wei. 2026. [Training-free agentic AI: Probabilistic control and coordination in multi-agent LLM systems](#). *Preprint*, arXiv:2603.13256.
- Dengzhe Hou, Lingyu Jiang, Deng Li, Zirui Li, Fangzhou Lin, and Kazunori D Yamada. 2026. [WMF-AM: Probing LLM working memory via depth-parameterized cumulative state tracking](#). *arXiv preprint arXiv:2603.27343*.
- Guiyang Hou, Wenqi Zhang, Yongliang Shen, Linjuan Wu, and Weiming Lu. 2024. [TimeToM: Temporal space is the key to unlocking the door of large language models’ Theory-of-Mind](#). In *Proceedings of ACL 2024*.
- Jinli Hu, Ross M. Clarke, Yichuan Zhang, and José Miguel Hernández-Lobato. 2026a. [Infinite-parameter LLMs: Generating and adapting weights from live data](#). *arXiv preprint arXiv:2609.18842*.
- Mengkang Hu, Tianxing Chen, Qiguang Chen, Yao Mu, Wenqi Shao, and Ping Luo. 2024. [HiAgent: Hierarchical working memory management for solving long-horizon agent tasks with large language model](#). In *Proceedings of ACL 2024*.
- Mengkang Hu, Tianxing Chen, Yude Zou, Yuheng Lei, Qiguang Chen, Ming Li, Yao Mu, Hongyuan Zhang, Wenqi Shao, and Ping Luo. 2025a. [Text2World: Benchmarking large language models for symbolic world model generation](#). In *Proceedings of ACL 2025*.
- Mengkang Hu, Bowei Xia, Yuran Wu, Ailing Yu, Yude Zou, Qiguang Chen, Shijian Wang, Jiarui Jin, Kexin Li, Wenxiang Jiao, Yuan Lu, and Ping Luo. 2025b. [Agent2World: Learning to generate symbolic world models via adaptive multi-agent feedback](#). *arXiv preprint arXiv:2512.22336*.
- Yuanzhe Hu, Yu Wang, and Julian McAuley. 2025c. [Evaluating memory in LLM agents via incremental multi-turn interactions](#). *Preprint*, arXiv:2507.05257.
- Yuyang Hu, Shichun Liu, Yanwei Yue, Guibin Zhang, Boyang Liu, Fangyi Zhu, Jiahang Lin, Honglin Guo, Shihan Dou, Zhiheng Xi, et al. 2025d. [Memory in the age of AI agents](#). *arXiv preprint arXiv:2512.13564*.
- Zhiting Hu and Tianmin Shu. 2023. [Language models, agent models, and world models: The LAW for machine reasoning and planning](#). *arXiv preprint arXiv:2312.05230*.
- Zhiyuan Hu, Yunhai Hu, Juncheng Liu, Shuyue Stella Li, Yucheng Wang, Zhen Xu, See-Kiong Ng, Anh Tuan Luu, Xinxing Xu, Bryan Hooi, Cynthia Breazeal, and Hae Won Park. 2026b. [Collaborative multi-agent test-time reinforcement learning for reasoning](#). *Preprint*, arXiv:2601.09667.
- Zican Hu, Xuyang Hu, Yiming Liu, Zuwei Long, Wei Liu, Yunzhuo Hao, Jiawei Gu, Linjie Li, Yu Cheng, Zhenhong Sun, Weibo Gu, Xing Sun, and Zhi Wang. 2026c. [Bridging interleaved multi-modal reasoning as a unified decision process](#). *arXiv preprint arXiv:2607.03748*.
- Jiaxin Huang, Shixiang Shane Gu, Le Hou, Yuexin Wu, Xuezhi Wang, Hongkun Yu, and Jiawei Han. 2022a. [Large language models can self-improve](#). In *EMNLP 2023*. ArXiv:2210.11610.
- Wenlong Huang, Fei Xia, Ted Xiao, Harris Chan, Jacky Liang, Pete Florence, Andy Zeng, Jonathan Tompson, Igor Mordatch, Yevgen Chebotar, Pierre Sermanet, Noah Brown, Tomas Jackson, Linda Luu, Sergey Levine, Karol Hausman, and Brian Ichter. 2022b. [Inner monologue: Embodied reasoning through planning with language models](#). In *CoRL 2022*. ArXiv:2207.05608.
- EunJeong Hwang, Yuwei Yin, Giuseppe Carenini, Peter West, and Vered Shwartz. 2025. [Infusing theory of mind into socially intelligent LLM agents](#). In *Proceedings of ACL 2025*.
- Yechan Hwang, Sangjun Bae, Jeongmo Kim, Sangwoo Bang, and Seungyul Han. 2026. [MARBO: Relational belief grounding for LLM agents in social deduction games](#). In *Proceedings of EMNLP 2026*.
- Jonas Hübner, Frederike Lübeck, Lejs Behric, Anton Baumann, Marco Bagatella, Daniel Marta, Ido Hakimi, Idan Shenfeld, Thomas Kleine Büning, Carlos Guestrin, and Andreas Krause. 2026. [Reinforcement learning via self-distillation](#). *arXiv preprint arXiv:2601.20802*.

- Sohaib Imran, Ihor Kendiukhov, Matthew Broerman, Aditya Thomas, Riccardo Campanella, Rob Lamb, and Peter M. Atkinson. 2025. [Are LLM belief updates consistent with Bayes’ theorem?](#) *arXiv preprint arXiv:2507.17951*.
- Hyejun Jeong, Amir Houmansadr, Shlomo Zilberstein, and Eugene Bagdasarian. 2026. [Understanding persuasion in long-running agents.](#) *arXiv preprint arXiv:2602.00851*.
- Alexandru-Iulius Jerpelea. 2026. [Planting a latent variable in natural-looking text: a more realistic test of belief states in LLMs and their link to concept geometry.](#) *arXiv preprint arXiv:2608.26887*.
- Saurabh Jha, Rohan Arora, Bhavya, Noah Zheutlin, Paulina Toro Isaza, Laura Schwartz, Yu Deng, Daby Sow, Ruchi Mahindru, and Ruchir Puri. 2026. [Think locally, explain globally: Graph-guided LLM investigations via local reasoning and belief propagation.](#) *arXiv preprint arXiv:2601.17915*.
- Haoxuan Jia, Yang Liu, Yingguang Yang, Yancheng Chen, Chongyang Zhang, Hao Zheng, Qian Li, Yulin Huang, Jianshen Zhang, Yongzhi Qi, Shang Luo, Kefu Xu, Hao Peng, Junyu Lu, Du Cheng, Philip S. Yu, and Bin Chong. 2026. [Hindsight Memory-PRM: Supervising memory management with auditable hindsight credit.](#) *arXiv preprint arXiv:2608.29605*.
- Dongming Jiang, Yi Li, and Bingzhe Li. 2026a. [Jevmem: System-one-controlled agentic memory for efficient AI agents.](#) *Preprint*, arXiv:2609.23986.
- Dongming Jiang, Yi Li, Guanpeng Li, Qiannan Li, and Bingzhe Li. 2026b. [HAGE: Harnessing agentic memory via RL-Driven weighted graph evolution.](#) *arXiv preprint arXiv:2605.09942*.
- Haitao Jiang, Lin Ge, Hengrui Cai, and Rui Song. 2026c. [PABU: Progress-aware belief update for efficient LLM agents.](#) *arXiv preprint arXiv:2602.09138*.
- Yuxuan Jiang and Francis Ferraro. 2026. [SCRIBE: Structured mid-level supervision for tool-using language models.](#) *arXiv preprint arXiv:2601.03555*.
- Bowen Jin, Jinsung Yoon, Priyanka Kargupta, Serkan O. Arik, and Jiawei Han. 2025a. [An empirical study on reinforcement learning for reasoning-search interleaved LLM agents.](#) *arXiv preprint arXiv:2505.15117*.
- Bowen Jin, Hansi Zeng, Zhenrui Yue, Jinsung Yoon, Serkan Arik, Dong Wang, Hamed Zamani, and Jiawei Han. 2025b. [Search-rl: Training LLMs to reason and leverage search engines with reinforcement learning.](#) *Preprint*, arXiv:2503.09516.
- Chuanyang Jin, Yutong Wu, Jing Cao, Jiannan Xiang, Yen-Ling Kuo, Zhiting Hu, Tomer Ullman, Antonio Torralba, Joshua B. Tenenbaum, and Tianmin Shu. 2024. [MMToM-QA: Multimodal theory of mind question answering.](#) In *Proceedings of ACL 2024*.
- Bhushan Kashinath Joshi. 2026. [Replication without persistence in hosted LLMs: Measurement sensitivity in action-time belief evaluation.](#) *arXiv preprint arXiv:2609.22478*.
- Chani Jung, Dongkwan Kim, Jiho Jin, Jiseon Kim, Yeon Seonwoo, Yejin Choi, Alice Oh, and Hyunwoo Kim. 2024. [Perceptions to beliefs: Exploring precursory inferences for theory of mind in large language models.](#) In *Proceedings of EMNLP 2024*.
- Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, Scott Johnston, Sheer El-Showk, Andy Jones, Nelson Elhage, Tristan Hume, Anna Chen, Yuntao Bai, Sam Bowman, Stanislav Fort, and 17 others. 2022. [Language models \(mostly\) know what they know.](#) *Preprint*, arXiv:2207.05221.
- Leslie Pack Kaelbling, Michael L. Littman, and Anthony R. Cassandra. 1998. Planning and acting in partially observable stochastic domains. *Artificial Intelligence*, 101(1-2):99–134.
- Adam Kamel, Tanish Rastogi, Michael Ma, Kailash Ranganathan, and Kevin Zhu. 2025. [Emergent world beliefs: Exploring Transformers in stochastic games.](#) *arXiv preprint arXiv:2512.23722*.
- Jiazheng Kang, Mingming Ji, Zhe Zhao, and Ting Bai. 2025a. [Memory OS of AI agent.](#) In *Proceedings of EMNLP 2025*.
- Minki Kang, Wei-Ning Chen, Dongge Han, Huseyin A. Inan, Lukas Wutschitz, Yanzhi Chen, Robert Sim, and Saravan Rajmohan. 2025b. [ACON: Optimizing context compression for long-horizon LLM agents.](#) *arXiv preprint arXiv:2510.00615*.
- Mark Kashirskiy and Ilya Makarov. 2026. [SuS: Strategy-aware surprise for intrinsic exploration.](#) *Preprint*, arXiv:2601.10349.
- Nora Kassner, Oyvind Tafjord, Hinrich Schütze, and Peter Clark. 2021. [BeliefBank: Adding memory to a pre-trained language model for a systematic notion of belief.](#) In *Proceedings of EMNLP 2021*.
- Chinmaya Kausik, Adith Swaminathan, and Nathan Kallus. 2026. [The context gathering decision process: A POMDP framework for agentic search.](#) *Preprint*, arXiv:2605.07042.
- Kim et al. 2024. [QuBE: Question-based belief enhancement for agentic LLM reasoning.](#) In *EMNLP 2024*.
- Geunwoo Kim, Pierre Baldi, and Stephen McAleer. 2023a. Language models can solve computer tasks. In *NeurIPS 2023*. ArXiv:2303.17491.
- Hyunwoo Kim, Melanie Sclar, Xuhui Zhou, Ronan Le Bras, Gunhee Kim, Yejin Choi, and Maarten Sap. 2023b. [FANToM: A benchmark for stress-testing machine theory of mind in interactions.](#) In *Proceedings of EMNLP 2023*.

- Jeonghye Kim, Sojeong Rhee, Minbeom Kim, Dohyung Kim, Sangmook Lee, Youngchul Sung, and Kyomin Jung. 2025. [ReflAct: World-grounded decision making in LLM agents via goal-state reflection](#). In *Proceedings of EMNLP 2025*.
- Minsoo Kim and Seung-won Hwang. 2025. [CoEx – co-evolving world-model and exploration](#). In *Proceedings of EMNLP 2025*.
- Jing Yu Koh, Robert Lo, Lawrence Jang, Vikram Duvvur, Ming Chong Lim, Po-Yu Huang, Graham Neubig, Shuyan Zhou, Ruslan Salakhutdinov, and Daniel Fried. 2024. VisualWebArena: Evaluating multimodal agents on realistic visual web tasks. In *ACL 2024*. ArXiv:2401.13649.
- Fanqi Kong, Jiayi Zhang, Mingyi Deng, Chenglin Wu, Yuyu Luo, and Bang Liu. 2026. [InfoPO: Information-driven policy optimization for user-centric agents](#). *arXiv preprint arXiv:2603.00656*.
- Michal Kosinski. 2023. Evaluating large language models in theory of mind tasks. In *PNAS 2024*. ArXiv:2302.02083.
- Frank R. Kschischang, Brendan J. Frey, and Hans-Andrea Loeliger. 2001. Factor graphs and the sum-product algorithm. *IEEE Transactions on Information Theory*, 47(2):498–519.
- Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. 2023. Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation. In *ICLR 2023*. ArXiv:2302.09664.
- Aviral Kumar, Vincent Zhuang, Rishabh Agarwal, Yi Su, John D Co-Reyes, Avi Singh, Kate Baumli, Shariq Iqbal, Colton Bishop, Rebecca Roelofs, Lei M Zhang, Kay McKinney, Disha Shrivastava, Cosmin Paduraru, George Tucker, Doina Precup, Feryal Behbahani, and Aleksandra Faust. 2024. [Training language models to self-correct via reinforcement learning](#). In *Proceedings of ICLR 2025*.
- Shubham Kumar, Harshit Kumar, Narendra Ahuja, and Saurabh Jha. 2026. [Towards a belief-based world model for LLM agents](#). *Preprint*, arXiv:2609.00455.
- Brandon Kynoch, Hugo Latapie, and Dwane van der Sluis. 2023. [RecallM: An adaptable memory mechanism with temporal understanding for large language models](#). *arXiv preprint arXiv:2307.02738*.
- Philippe Laban, Hiroaki Hayashi, Yingbo Zhou, and Jennifer Neville. 2025. LLMs get lost in multi-turn conversation. *arXiv preprint arXiv:2505.06120*.
- Chris Latimer, Nicoló Boschi, Andrew Neeser, Chris Bartholomew, Gaurav Srivastava, Xuan Wang, and Naren Ramakrishnan. 2025. [Hindsight is 20/20: Building agent memory that retains, recalls, and reflects](#). *arXiv preprint arXiv:2512.12818*.
- Chris Latimer et al. 2026. Hindsight: Structured agent memory that retains, recalls, and reflects. In *Annual Meeting of the Association for Computational Linguistics*.
- Chia-Hsuan Lee, Hao Cheng, and Mari Ostendorf. 2021. Dialogue state tracking with a language model using schema-driven prompting. In *EMNLP 2021*. ArXiv:2109.07506.
- Hyunji Lee, Justin Chih-Yao Chen, Joykirat Singh, Zaid Khan, Elias Stengel-Eskin, and Mohit Bansal. 2026a. [MINTEval: Evaluating memory under multi-target interference in long-horizon agent systems](#). *arXiv preprint arXiv:2605.18565*.
- Kuang-Huei Lee, Xinyun Chen, Hiroki Furuta, John Canny, and Ian Fischer. 2024. [A human-inspired reading agent with gist memory of very long contexts](#). In *Proceedings of ICML 2024*.
- Yoonho Lee, Roshen Nair, Qizheng Zhang, Kangwook Lee, Omar Khattab, and Chelsea Finn. 2026b. [Meta-harness: End-to-End optimization of model harnesses](#). *arXiv preprint arXiv:2603.28052*.
- Wolfgang Lehrach, Daniel Hennes, Miguel Lazaro-Gredilla, Xinghua Lou, Carter Wendelken, Zun Li, Antoine Dedieu, Jordi Grau-Moya, Marc Lanctot, Atil Iscen, John Schultz, Marcus Chiam, Ian Gemp, Piotr Zielinski, Satinder Singh, and Kevin P. Murphy. 2025. [Code world models for general game playing](#). *arXiv preprint arXiv:2510.04542*.
- Matthew Levinson. 2026. [Finding belief geometries with sparse autoencoders](#). *arXiv preprint arXiv:2604.02685*.
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. [Retrieval-augmented generation for knowledge-intensive NLP tasks](#). In *Proceedings of NeurIPS 2020*.
- Chenliang Li, Adel Elmahdy, Alex Boyd, Zhongruo Wang, Siliang Zeng, Alfredo Garcia, Parminder Bhatia, Taha Kass-Hout, Cao Xiao, and Mingyi Hong. 2025a. [Stabilizing off-policy training for long-horizon LLM agent via turn-level importance sampling and clipping-triggered normalization](#). *arXiv preprint arXiv:2511.20718*.
- Haohang Li, Yupeng Cao, Yangyang Yu, Shashidhar Reddy Javaji, Zhiyang Deng, Yueru He, et al. 2024a. [InvestorBench: A benchmark for financial decision-making tasks with LLM-based agents](#). *arXiv preprint arXiv:2412.18174*.
- Huaoli, Yu Quan Chong, Simon Stepputtis, Joseph Campbell, Dana Hughes, Michael Lewis, and Katia Sycara. 2023a. [Theory of mind for multi-agent collaboration via large language models](#). In *Proceedings of EMNLP 2023*.

- Jian Li, Yizhang Jin, Dongqi Liu, Hang Ding, Jiafu Wu, Dongsheng Chen, Yunhang Shen, Yulei Qin, Ying Tai, Chengjie Wang, Xiaotong Yuan, and Yabiao Wang. 2026a. [SE-Search: Self-evolving search agent via memory and dense reward](#). *arXiv preprint arXiv:2603.03293*.
- Jiazheng Li, Yawei Wang, David Yan, Yijun Tian, Zhichao Xu, Huan Song, Panpan Xu, and Lin Lee Cheong. 2025b. [SALT: Step-level advantage assignment for long-horizon agents via trajectory graph](#). In *Proceedings of EACL 2026*.
- Junbo Li, Peng Zhou, Rui Meng, Meet P. Vadera, Li-hong Li, and Yang Li. 2025c. [Turn-PPO: Turn-level advantage estimation with PPO for improved multi-turn RL in agentic LLMs](#). In *Proceedings of EACL 2026*.
- Manling Li, Shiyu Zhao, Qineng Wang, Kangrui Wang, Yu Zhou, Sanjana Srivastava, Cem Gokmen, Tony Lee, Li Erran Li, Ruohan Zhang, Weiyu Liu, Percy Liang, Li Fei-Fei, Jiayuan Mao, and Jiajun Wu. 2024b. [Embodied agent interface: Benchmarking LLMs for embodied decision making](#). In *Proceedings of NeurIPS 2024*.
- Mengfan Li, Xuanhua Shi, and Yang Deng. 2025d. [RecToM: A benchmark for evaluating machine theory of mind in LLM-based conversational recommender systems](#). In *Proceedings of AAAI 2026*.
- Pengcheng Li, Zhengyang Zhang, Dongxu Zhang, Sui Huang, and Shaohua Ma. 2026b. [VICT: Verifier-instrumented credit tracing for long-horizon LLM agent reinforcement learning](#). *Preprint*, arXiv:2608.28128.
- Xiaoxi Li, Wenxiang Jiao, Jiarui Jin, Guanting Dong, Jiajie Jin, Yinuo Wang, Hao Wang, Yutao Zhu, Ji-Rong Wen, Yuan Lu, and Zhicheng Dou. 2025e. [DeepAgent: A general reasoning agent with scalable toolsets](#). In *Proceedings of WWW 2026*.
- Xiaozhe Li, Tianyi Lyu, Yang Li, Yichuan Ma, Peiji Li, Linyang Li, Qipeng Guo, Dahua Lin, and Kai Chen. 2026c. [What and when to distill: Selective hindsight distillation for multi-turn agents](#). *Preprint*, arXiv:2605.19447.
- Yang Li, Yangyang Yu, Haohang Li, Zhi Chen, and Khaldoun Khashanah. 2023b. [TradingGPT: Multi-agent system with layered memory and distinct characters for enhanced financial trading performance](#). *Preprint*, arXiv:2309.03736.
- Yixia Li, Hongru Wang, Peng Lai, Zhiwen Ruan, He Zhu, Youxin Zhu, Ganlong Zhao, Minda Hu, Yun Chen, Sibe Yang, Peng Li, Jeff Z. Pan, Jia Pan, Guanhua Chen, Yang Liu, and Guanbin Li. 2026d. [Bridging the agent-world gap: Text world models for LLM-based agents](#). *Preprint*, arXiv:2606.09032.
- Yixia Li, Hongru Wang, Jiahao Qiu, Zhenfei Yin, Dongdong Zhang, Cheng Qian, Zeping Li, Pony Ma, Guanhua Chen, and Heng Ji. 2025f. [From word to world: Can large language models be implicit text-based world models?](#) In *Proceedings of ACL 2026*.
- Yuanfan Li, Qi Zhou, Wenjing Duan, and Lu Chen. 2026e. [When denser credit is not enough: Evidence-calibrated policy optimization for long-horizon LLM agent training](#). *Preprint*, arXiv:2606.05885.
- Yuante Li, Yicheng Tao, Kate Zhang, Taozhi Wang, Gefei Gu, and Yaxin Zhou. 2026f. [Diverse evidence, better forecasts: Multi-agent deliberation under information asymmetry](#). *arXiv preprint arXiv:2607.01661*.
- Yugu Li, Zehong Cao, Jianglin Qiao, and Siyi Hu. 2026g. [RTPO: Reverse-turn policy optimization for stabilizing agentic RL training](#). *arXiv preprint arXiv:2608.18682*.
- Zhiyu Li, Chenyang Xi, Chunyu Li, Ding Chen, Boyu Chen, Shichao Song, Simin Niu, Hanyu Wang, Jiawei Yang, Chen Tang, Qingchen Yu, Jiahao Zhao, Yezhaohui Wang, Peng Liu, Zehao Lin, Pengyuan Wang, Jiahao Huo, Tianyi Chen, Kai Chen, and 20 others. 2025g. [MemOS: A memory OS for AI system](#). *arXiv preprint arXiv:2507.03724*.
- Zhuofeng Li, Haoxiang Zhang, Seungju Han, Sheng Liu, Jianwen Xie, Yu Zhang, Yejin Choi, James Zou, and Pan Lu. 2025h. [In-the-Flow agentic system optimization for effective planning and tool use](#). *arXiv preprint arXiv:2510.05592*.
- Qiao Liang, Yuke Zhu, Chao Ge, Lei Yang, Ying Shen, Bo Zheng, and Sheng Guo. 2026. [Learning from the irrecoverable: Error-localized policy optimization for tool-integrated LLM reasoning](#). In *Proceedings of ACL 2026*.
- Junfeng Liao, Qizhou Wang, Jianing Zhu, Bo Du, Rui Yan, and Xiuying Chen. 2026a. [Belief memory: Agent memory under partial observability](#). *Preprint*, arXiv:2605.05583.
- Sicong Liao, Zhi Chen, and Yaohua Tang. 2026b. [TCPO: Turn-level credit policy optimization](#). *Preprint*, arXiv:2608.01667.
- Mircea Lic, Ojas Shirekar, Baptiste Colle, and Chirag Raman. 2024. [MindForge: Empowering embodied agents with theory of mind for lifelong cultural learning](#). In *Proceedings of NeurIPS 2025*.
- Aly Lidayan, Jakob Bjorner, Satvik Golechha, Kartik Goyal, and Alane Suhr. 2025. [ABBEL: Learning natural-language belief states for memory-efficient interaction](#). *arXiv preprint arXiv:2512.20111*.
- Hunter Lightman, Vineet Kosaraju, Yuri Burda, Harrison Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. 2024. [Let’s verify step by step](#). In *International Conference on Learning Representations*.

- Hongxiang Lin, Zhirui Kuai, Erpeng Xue, and Lei Wang. 2026a. [SKILLC: Learning autonomous skill internalization in LLM agents via contrastive credit assignment](#). *Preprint*, arXiv:2605.27899.
- Jianzhe Lin, Xiaolin Li, Yunda Liu, Fei Wang, and Jubin Chheda. 2026b. [When LLM agents fail to read the room: ReAdapt for relational social reasoning](#). *arXiv preprint arXiv:2609.25284*.
- Moule Lin, Qizhen Lan, Shuhao Guan, Weipeng Jing, Jiexin Fan, David Gregg, and Goetz Botterweck. 2026c. [Uncertainty quantification for LLM agents: A taxonomy, an evaluation protocol, and an empirical study](#). *Preprint*, arXiv:2609.07395.
- Stephanie Lin, Jacob Hilton, and Owain Evans. 2022. Teaching models to express their uncertainty in words. In *TMLR 2022*. ArXiv:2205.14334.
- Bo Liu, Yuqian Jiang, Xiaohan Zhang, Qiang Liu, Shiqi Zhang, Joydeep Biswas, and Peter Stone. 2023a. [LLM+P: Empowering large language models with optimal planning proficiency](#). *Preprint*, arXiv:2304.11477.
- Chengzhi Liu, Yuzhe Yang, Sophia Xiao Pu, Yepeng Liu, Lin Long, Yichen Guo, Nuo Chen, Zhaotian Weng, Elena Kochkina, Simerjot Kaur, Charese Smiley, Xiaomo Liu, James Zou, Sheng Liu, Yuheng Bu, Songyou Peng, and Xin Eric Wang. 2026a. [World-MemArena: Evaluating multimodal agent memory through action-world interaction](#). *arXiv preprint arXiv:2605.29341*.
- Dongyi Liu, Yifan Niu, Qinwen Wang, Han Xiao, and Jia Li. 2026b. [PiCA: Pivot-based credit assignment for search agentic reinforcement learning](#). *Preprint*, arXiv:2605.09287.
- Hanbing Liu, Chunhao Tian, Nan An, Ziyuan Wang, Pinyan Lu, Changyuan Yu, and Qi Qi. 2026c. [Budget-constrained agentic large language models: Intention-based planning for costly tool use](#). *arXiv preprint arXiv:2602.11541*.
- Jiaqi Liu, Yaofeng Su, Peng Xia, Siwei Han, Zeyu Zheng, Cihang Xie, Mingyu Ding, and Huaxiu Yao. 2026d. [SimpleMem: Efficient lifelong memory for LLM agents](#). *arXiv preprint arXiv:2601.02553*.
- Xiao Liu, Hao Yu, Hanchen Zhang, Yifan Xu, Xuanyu Lei, Hanyu Lai, Yu Gu, Hangliang Ding, Kaiwen Men, Kejuan Yang, Shudan Zhang, Xiang Deng, Aohan Zeng, Zhengxiao Du, Chenhui Zhang, Sheng Shen, Tianjun Zhang, Yu Su, Huan Sun, and 3 others. 2023b. AgentBench: Evaluating LLMs as agents. In *ICLR 2024*. ArXiv:2308.03688.
- Xiaoqian Liu, Ke Wang, Yuchuan Wu, Fei Huang, Yongbin Li, Junge Zhang, and Jianbin Jiao. 2025. [Agentic reinforcement learning with implicit step rewards](#). *arXiv preprint arXiv:2509.19199*.
- Yicheng Liu, Bolin Zhang, Weiran Liu, Yakun Zhang, Yangqin Jiang, Zhiying Tu, and Dianhui Chu. 2026e. [MADE: Belief-driven dual-agent coordination for autonomous model deployment](#). *Preprint*, arXiv:2608.01189.
- Yu Liu, Wenwen Li, Yifan Dou, and Guangnan Ye. 2026f. [Recursive reasoning or statistical extrapolation? in-context learning in multi-agent interdependent decision-making](#). *arXiv preprint arXiv:2609.18591*.
- Yu-Ze Liu et al. 2026g. SHADOW: Dynamic-aware credit assignment against long-horizon tasks. In *AAAI 2026*.
- Zhihan Liu, Hao Hu, Shenao Zhang, Hongyi Guo, Shuqi Ke, Boyi Liu, and Zhaoran Wang. 2023c. [Reason for future, act for now: A principled framework for autonomous LLM agents with provable sample efficiency](#). *arXiv preprint arXiv:2309.17382*.
- Zijian Liu et al. 2026h. PRISM: Preference-guided semantic reasoning with vision-language models for object goal navigation. In *International Conference on Multimedia Retrieval*.
- Ziyan Liu, Zhezheng Hao, Yeqiu Chen, Hong Wang, Jingren Hou, Ruiyi Ding, Yongkang Yang, Wence Ji, Wei Xia, and Feng Liu. 2026i. [Meta-cognitive memory policy optimization for long-horizon LLM agents](#). *arXiv preprint arXiv:2605.30159*.
- Hans-Andrea Loeliger. 2004. An introduction to factor graphs. *IEEE Signal Processing Magazine*, 21(1):28–41.
- Lin Long, Yichen He, Wentao Ye, Yiyuan Pan, Yuan Lin, Hang Li, Junbo Zhao, and Wei Li. 2025. [Seeing, listening, remembering, and reasoning: A multimodal agent with long-term memory](#). *arXiv preprint arXiv:2508.09736*.
- Xinghua Lou, Miguel Lázaro-Gredilla, Antoine Dedieu, Carter Wendelken, Wolfgang Lehrach, and Kevin P. Murphy. 2026. [AutoHarness: improving LLM agents by automatically synthesizing a code harness](#). *arXiv preprint arXiv:2603.03329*.
- Fanbin Lu, Zhisheng Zhong, Shu Liu, Chi-Wing Fu, and Jiaya Jia. 2025. [ARPO:End-to-End policy optimization for GUI agents with experience replay](#). *arXiv preprint arXiv:2505.16282*.
- Ning Lu, Baijiong Lin, Shengcai Liu, Jiahao Wu, Haoze Lv, Yanbin Wei, Lingting Zhu, Shengju Qian, Xin Wang, Ying-Cong Chen, Qi Wang, and Ke Tang. 2026a. [Policy and world modeling co-training for language agents](#). In *Proceedings of EMNLP 2026*.
- Zhengxi Lu, Fei Tang, Guangyi Liu, Kaitao Song, Xu Tan, Jin Ma, Wenqi Zhang, Weiming Lu, Jun Xiao, Yueting Zhuang, and Yongliang Shen. 2026b. [UI-Copilot: Advancing long-horizon GUI automation via tool-integrated policy optimization](#). In *Proceedings of ACL 2026*.

- Zhicong Lu, Zichuan Lin, Wei Jia, Changyuan Tian, Deheng Ye, Peiguang Li, Li Jin, Nayu Liu, Guan-guan Xu, and Wei Feng. 2026c. [HISR: Hindsight information modulated segmental process rewards for multi-turn agentic reinforcement learning](#). *arXiv preprint arXiv:2603.18683*.
- Haotian Luo, Huaisong Zhang, Xuelin Zhang, Haoyu Wang, Zeyu Qin, Wenjie Lu, Guozheng Ma, Haiying He, Yingsha Xie, Qiyang Zhou, Zixuan Hu, Hongze Mi, Yibo Wang, Naiqiang Tan, Hong Chen, Yi R. Fung, Chun Yuan, and Li Shen. 2025a. [UltraHorizon: Benchmarking agent capabilities in ultra long-horizon scenarios](#). *arXiv preprint arXiv:2509.21766*.
- Jia Luo. 2026. [From topology to trajectory: LLM-driven world models for supply chain resilience](#). *Preprint*, arXiv:2604.11041.
- Jiabao Luo et al. 2026a. ReflectiChain: Mitigating semantic-execution drift in long-horizon LLM agents via retrospective reflection and double-loop policy adaptation. In *Electronics*.
- Jiani Luo, Xiaoyan Zhao, Yang Zhang, Shuyi Miao, Bingbing Xu, Stefan Konigorski, and Tat-Seng Chua. 2026b. [Know you before you speak: User-state modeling for LLM personalization in multi-turn conversation](#). *arXiv preprint arXiv:2605.24647*.
- Jinghao Luo, Yuchen Tian, Chuxue Cao, Ziyang Luo, Hongzhan Lin, Kaixin Li, Chuyi Kong, Ruichao Yang, and Jing Ma. 2026c. [From storage to experience: A survey on the evolution of LLM agent memory mechanisms](#). *Preprint*, arXiv:2605.06716.
- Xufang Luo, Yuge Zhang, Zhiyuan He, Zilong Wang, Siyun Zhao, Dongsheng Li, Luna K. Qiu, and Yuqing Yang. 2025b. [Agent lightning: Train ANY AI agents with reinforcement learning](#). *arXiv preprint arXiv:2508.03680*.
- Yu Luo, Rongchen Gao, Lu Teng, Xidao Wen, Jiamin Jiang, Qingliang Zhang, Yongqian Sun, Shenglin Zhang, Jiasong Feng, Tong Liu, Wenjie Zhang, and Dan Pei. 2026d. [Graph of states: Solving abductive tasks with large language models](#). *Preprint*, arXiv:2603.21250.
- Weitao Ma, Xiaocheng Feng, Lei Huang, Xiachong Feng, Zhanyu Ma, Jun Xu, Jiuchong Gao, Jinghua Hao, Renqing He, and Bing Qin. 2026a. [Fine-mem: Fine-grained feedback alignment for long-horizon memory management](#). *arXiv preprint arXiv:2601.08435*.
- Yijun Ma, Zehong Wang, Yiyang Li, Ziming Li, Xiaoguang Guo, Weixiang Sun, Chuxu Zhang, and Yanfang Ye. 2026b. [ProPlay: Procedural world models for self-evolving LLM agents](#). *Preprint*, arXiv:2606.12780.
- Ziyu Ma, Hailang Huang, Shun Zou, Yong Wang, Shidong Yang, Yiming Hu, Fei Wei, and Xiangxiang Chu. 2026c. [LongHorizon-Harness: Advancing long-horizon agents for real-world tasks](#). *arXiv preprint arXiv:2608.01964*.
- Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegrefe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Shashank Gupta, Bodhisattwa Prasad Majumder, Katherine Hermann, Sean Welleck, Amir Yazdanbakhsh, and Peter Clark. 2023. Self-refine: Iterative refinement with self-feedback. In *NeurIPS 2023*. ArXiv:2303.17651.
- Adyasha Maharana, Dong-Ho Lee, Sergey Tulyakov, Mohit Bansal, Francesco Barbieri, and Yuwei Fang. 2024. Evaluating very long-term conversational memory of LLM agents. In *ACL 2024*. ArXiv:2402.17753.
- Marina Mao, Alexandr Liu, Pengbo Li, Siheng Li, Bo Zhou, and Xiang Wang. 2026a. [Tree-based credit assignment for multi-agent memory system](#). *arXiv preprint arXiv:2605.04811*.
- Wenyu Mao, Haoyang Liu, Haosong Tan, Yaorui Shi, Jiancan Wu, An Zhang, and Xiang Wang. 2026b. [Joint optimization of multi-agent memory system](#). *arXiv preprint arXiv:2603.12631*.
- Kohsei Matsutani, Gouki Minegishi, Core Francisco Park, Takeshi Kojima, Yusuke Iwasawa, and Yutaka Matsuo. 2026. [Few-shot demonstrations elicit the use of in-context world representations in LLMs](#). *arXiv preprint arXiv:2609.24352*.
- Kai Mei, Jiang Guo, Shuaichen Chang, Mingwen Dong, Dongkyu Lee, Xing Niu, and Jiarong Jiang. 2025. [R-WoM: Retrieval-augmented world model for computer-use agents](#). *arXiv preprint arXiv:2510.11892*.
- Tianyi Men, Zhuoran Jin, Kang Liu, and Jun Zhao. 2026. [The physics of multi-turn long-horizon planning: From pre-training to post-training via single- and multi-teacher on-policy agentic distillation](#). *arXiv preprint arXiv:2607.24720*.
- Renwei Meng. 2026. [Policy-conditioned counterfactual credit for verifiable reinforcement learning of long-horizon language agents](#). *arXiv preprint arXiv:2606.05263*.
- Yuqiao Meng, Sakshi Sunil Narvekar, Luoxi Tang, Rupali Rajendra Vaje, Yingxue Zhang, Muchao Ye, and Zhaohan Xi. 2026. [EquiMem: Calibrating shared memory in multi-agent debate via game-theoretic equilibrium](#). *arXiv preprint arXiv:2605.09278*.
- Nikola Milosevic. 2026. [Episodic-semantic memory architecture for long-horizon scientific agents](#). *arXiv preprint arXiv:2605.17625*.

- Bhavana Dalvi Mishra, Oyvind Tafjord, and Peter Clark. 2022. [Towards teachable reasoning systems: Using a dynamic memory of user feedback for continual system improvement](#). In *Proceedings of EMNLP 2022*.
- Zhanfeng Mo, Xingxuan Li, Yuntao Chen, and Lidong Bing. 2025. [Multi-agent tool-integrated policy optimization](#). *arXiv preprint arXiv:2510.04678*.
- Ali Modarressi, Ayyoob Imani, Mohsen Fayyaz, and Hinrich Schütze. 2023. [RET-LLM: Towards a general read-write memory for large language models](#). *arXiv preprint arXiv:2305.14322*.
- Wachiravit Modecrua, Krittanon Kaewtawee, Krittin Pachtrachai, and Touchapon Kraisingkorn. 2026. [Multi-turn reinforcement learning for tool-calling agents with iterative reward calibration](#). *Preprint*, arXiv:2604.02869.
- Jared Moore, Ned Cooper, Rasmus Overmark, Beba Cibralic, Nick Haber, and Cameron R. Jones. 2025. [Do large language models have a planning theory of mind? evidence from MindGames: a multi-step persuasion task](#). In *Proceedings of COLM 2025*.
- Kevin Murphy. 2026. [Agentic forecasting using sequential Bayesian updating of linguistic beliefs](#). *Preprint*, arXiv:2604.18576.
- Praveen Kumar Myakala, Manan Agrawal, and Rahul Manche. 2026. [BeliefShift: Benchmarking temporal belief consistency and opinion drift in LLM agents](#). *Preprint*, arXiv:2603.23848.
- Reiichiro Nakano, Jacob Hilton, Suchir Balaji, Jeff Wu, Long Ouyang, Christina Kim, Christopher Hesse, Shantanu Jain, Vineet Kosaraju, William Saunders, Xu Jiang, Karl Cobbe, Tyna Eloundou, Gretchen Krueger, Kevin Button, Matthew Knight, Benjamin Chess, and John Schulman. 2021. [WebGPT: Browser-assisted question-answering with human feedback](#). *arXiv preprint arXiv:2112.09332*.
- Kazuki Nakayashiki. 2026. [When stale constraints go unchecked: Budgeted verification failures in inherited agent memory](#). *Preprint*, arXiv:2608.25553.
- Siddharth Nayak, Adelmo Morrison Orozco, Marina Ten Have, Vittal Thirumalai, Jackson Zhang, Darren Chen, Aditya Kapoor, Eric Robinson, Karthik Gopalakrishnan, James Harrison, Brian Ichter, Anuj Mahajan, and Hamsa Balakrishnan. 2024. [LLaMAR: Long-horizon planning for multi-agent robots in partially observable environments](#). *arXiv preprint arXiv:2407.10031*.
- Thuy Ngoc Nguyen, Duy Nhat Phan, and Cleotilde Gonzalez. 2026. [Dynamic theory of mind as a temporal memory problem: Evidence from large language models](#). *Preprint*, arXiv:2603.14646.
- Chang Nie, Jiaju Wei, Junlan Feng, Chaoyou Fu, and Caifeng Shan. 2026. [Light-omni: Reflex over reasoning in agentic video understanding with long-term memory](#). *arXiv preprint arXiv:2607.05511*.
- Dujun Nie, Xianda Guo, Yiqun Duan, Ruijun Zhang, and Long Chen. 2025. [WMNav: Integrating vision-language models into world models for object goal navigation](#). In *IEEE/RJS International Conference on Intelligent Robots and Systems 2025*.
- Xuying Ning, Dongqi Fu, Tianxin Wei, Hanqing Zeng, Yuanchen Bei, Bingxuan Li, Zihao Li, Qifan Wang, Xiang Shen, Yifan Wu, Jiayi Liu, Hong Li, Yinglong Xia, Xiangjun Fan, Hanghang Tong, and Jingrui He. 2026. [EvoHarness-RL: Learning self-evolving runtime harness for long-horizon LLM agents](#). *Preprint*, arXiv:2608.05446.
- Haochen Niu, Lantao Zhang, Xingwu Ji, Rendong Ying, Peilin Liu, and Fei Wen. 2026. [SSMG-Nav: Enhancing lifelong object navigation with semantic skeleton memory graph](#). *arXiv preprint arXiv:2603.01813*.
- Kolby Nottingham, Prithviraj Ammanabrolu, Alane Suhr, Yejin Choi, Hannaneh Hajishirzi, Sameer Singh, and Roy Fox. 2023. [Do embodied agents dream of pixelated sheep: Embodied decision making using language guided world modelling](#). In *Proceedings of ICML 2023*.
- Long Ouyang, Jeff Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, et al. 2022. [Training language models to follow instructions with human feedback](#). In *Advances in Neural Information Processing Systems*.
- Siru Ouyang, Jun Yan, Yanfei Chen, Rujun Han, Zifeng Wang, Bhavana Dalvi Mishra, Rui Meng, Chun-Liang Li, Yizhu Jiao, Kaiwen Zha, Maohao Shen, Vishy Tirumalashetty, George Lee, Jiawei Han, Tomas Pfister, and Chen-Yu Lee. 2026. [SkillIOS: Learning skill curation for self-evolving agents](#). *arXiv preprint arXiv:2605.06614*.
- Siru Ouyang, Jun Yan, I-Hung Hsu, Yanfei Chen, Ke Jiang, Zifeng Wang, Rujun Han, Long T. Le, Samira Daruki, Xiangru Tang, Vishy Tirumalashetty, George Lee, Mahsan Rofouei, Hangfei Lin, Jiawei Han, Chen-Yu Lee, and Tomas Pfister. 2025. [ReasoningBank: Scaling agent self-evolving with reasoning memory](#). *arXiv preprint arXiv:2509.25140*.
- Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G. Patil, Ion Stoica, and Joseph E. Gonzalez. 2023. [MemGPT: Towards LLMs as operating systems](#). *arXiv preprint arXiv:2310.08560*.
- Dheeraj Mohandas Pai and Lu Xian. 2026. [Long-horizon state tracking in LLMs: Executing MD5 through a deep sequence of dependent tool calls](#). *Preprint*, arXiv:2609.00012.
- Fabian Paischer, Thomas Adler, Vihang Patil, Angela Bitto-Nemling, Markus Holzleitner, Sebastian Lehner, Hamid Eghbal-zadeh, and Sepp Hochreiter. 2022. [History compression via language models in reinforcement learning](#). In *Proceedings of ICML 2022*.

- Yijun Pan, Yukun Lian, Kunyu Shi, Junbo Li, Hongwei Xue, Sicong Xie, Guannan Zhang, and Xiaoying Xing. 2026a. [Business arena: Benchmarking LLM agents in a realistic marketplace](#). *Preprint*, arXiv:2608.08621.
- Zhuoshi Pan, Qizhi Pei, Junru Lu, Honglin Lin, H. Vicky Zhao, Di Yin, and Xing Sun. 2026b. [ContextPilot: Teaching agents for proactive context management via fine-grained RL](#). In *Proceedings of EMNLP 2026*.
- Theodore Papamarkou, Pierre Alquier, Matthias Bauer, Wray Buntine, Andrew Davison, Gintare Karolina Dziugaite, Maurizio Filippone, Andrew Y. K. Foong, Vincent Fortuin, Dimitris Fouskakis, Jes Frellsen, Eyke Hüllermeier, Theofanis Karaletsos, Mohammad Emtiyaz Khan, Nikita Kotelevskii, Salem Lahlou, Yingzhen Li, Fang Liu, Clare Lyle, and 11 others. 2026. [Position: agentic AI orchestration should be Bayes-consistent](#). *arXiv preprint arXiv:2605.00742*.
- Joon Sung Park, Joseph O’Brien, Carrie Jun Cai, Meredith Ringel Morris, Percy Liang, and Michael S. Bernstein. 2023. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the ACM Symposium on User Interface Software and Technology*.
- Young Bin Park. 2026. [Graph-native cognitive memory for AI agents: Formal belief revision semantics for versioned memory architectures](#). *arXiv preprint arXiv:2603.17244*.
- Jiangweizhi Peng, Yuanxin Liu, Ruida Zhou, Charles Fleming, Zhaoran Wang, Alfredo Garcia, and Mingyi Hong. 2026a. HiPER: Hierarchical reinforcement learning with explicit credit assignment for large language model agents. *arXiv preprint arXiv:2602.16165*.
- Xueqing Peng, Zhuohan Xie, Yupeng Cao, Haohang Li, Lingfei Qian, Yan Wang, Vincent Jim Zhang, Huan He, Xuguang Ai, Linhai Ma, Ruoyu Xiang, Yueru He, Yi Han, Shuyao Wang, Yuqing Guo, Mingyang Jiang, Yilun Zhao, Youzhong Dong, Xiaoyu Wang, and 44 others. 2026b. [Herculean: An agentic benchmark for financial intelligence](#). *Preprint*, arXiv:2605.14355.
- Daniel Platnick, Mohamed E. Bengueddache, Marjan Alirezaie, Dava J. Newman, Alex “Sandy” Pentland, and Hossein Rahnama. 2025. [ID-RAG: Identity retrieval-augmented generation for long-horizon persona coherence in generative agents](#). *arXiv preprint arXiv:2509.25299*.
- Nikhil Prakash, Natalie Shapira, Arnab Sen Sharma, Christoph Riedl, Yonatan Belinkov, Tamar Rott Shalom, David Bau, and Atticus Geiger. 2025. [Language models use lookbacks to track beliefs](#). *arXiv preprint arXiv:2505.14685*.
- Pradyumna Shyama Prasad and Minh Nhat Nguyen. 2025. [When two LLMs debate, both think they’ll win](#). *arXiv preprint arXiv:2505.19184*.
- Liming Pu, Xiaoxia Li, Yifu Liu, Teng Cao, and Bin Yang. 2026. [Explore more, drift less: Outcome-only reinforcement learning can suffice for long-horizon interactive agents](#). *Preprint*, arXiv:2609.01245.
- Pranav Putta, Edmund Mills, Naman Garg, Sumeet Motwani, Chelsea Finn, Divyansh Garg, and Rafael Rafailov. 2024. [Agent q: Advanced reasoning and learning for autonomous AI agents](#). *Preprint*, arXiv:2408.07199.
- Qiuyi Qi, Tian Liang, Mutian Bao, Jinjian Zhang, Dongnan Liu, Wei Zhou, Linjian Mo, Ming Kong, Jie Liu, Feng Zhang, and Qiang Zhu. 2026. [STAPO: Selective trajectory-aware policy optimization for LLM agent training](#). *Preprint*, arXiv:2607.04963.
- Zehan Qi, Xiao Liu, Iat Long Iong, Hanyu Lai, Xueqiao Sun, Wenyi Zhao, Yu Yang, Xinyue Yang, Jidai Sun, Shuntian Yao, Tianjie Zhang, Wei Xu, Jie Tang, and Yuxiao Dong. 2024. WebRL: Training LLM web agents via self-evolving online curriculum reinforcement learning. In *ICLR 2025*. ArXiv:2411.02337.
- Cheng Qian, Emre Can Acikgoz, Qi He, Hongru Wang, Xiusi Chen, Dilek Hakkani-Tür, Gokhan Tur, and Heng Ji. 2025. [ToolRL: Reward is all tool learning needs](#). In *Proceedings of NeurIPS 2025*.
- Cheng Qian, Emre Can Acikgoz, Bingxuan Li, Xiusi Chen, Yuji Zhang, Bingxiang He, Qinyu Luo, Dilek Hakkani-Tür, Gokhan Tur, Yunzhu Li, and Heng Ji. 2026. [Current agents fail to leverage world model as tool for foresight](#). In *Proceedings of ACL 2026*.
- Bowen Qin and Yao Lu. 2026. [MoM: Memory of memory](#). *arXiv preprint arXiv:2609.25054*.
- Linlu Qiu, Fei Sha, Kelsey Allen, Yoon Kim, Tal Linzen, and Sjoerd van Steenkiste. 2026a. [Bayesian teaching enables probabilistic reasoning in large language models](#). *Nature Communications*, 17:1238.
- Yifu Qiu, Zheng Zhao, Waylon Li, Yftah Ziser, Anna Korhonen, Shay B. Cohen, and Edoardo M. Ponti. 2026b. [Self-improving world modelling with latent actions](#). *arXiv preprint arXiv:2602.06130*.
- Changle Qu, Sunhao Dai, Hengyi Cai, Jun Xu, Shuaiqiang Wang, and Dawei Yin. 2026. [MatchTIR: Fine-grained supervision for tool-integrated reasoning via bipartite matching](#). In *Proceedings of ACL 2026*.
- Rafael Rafailov, Joey Hejna, Ryan Park, and Chelsea Finn. 2024. [From  \$r\$  to  \$q^\*\$ : Your language model is secretly a Q-Function](#). *arXiv preprint arXiv:2404.12358*.
- Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D. Manning, Stefano Ermon, and Chelsea Finn. 2023. Direct preference optimization: Your language model is secretly a reward model. In *Advances in Neural Information Processing Systems*.

- Babak Rahmani. 2026. [Debugging code world models](#). *arXiv preprint arXiv:2602.07672*.
- Ashish Rana, Chia-Chien Hung, Qumeng Sun, Julian Martin Kunkel, and Carolin Lawrence. 2026. [Oblivion: Self-adaptive agentic memory control through decay-driven activation](#). *arXiv preprint arXiv:2604.00131*.
- Preston Rasmussen, Pavlo Paliychuk, Travis Beauvais, Jack Ryan, and Daniel Chalef. 2025. [Zep: A temporal knowledge graph architecture for agent memory](#). *arXiv preprint arXiv:2501.13956*.
- Param Raval, Rohit Shenoy, and Archana Vaidheeswaran. 2026. [Triggers and diagnostics for LLM-Based interpretability failures in active inference agents](#). *arXiv preprint arXiv:2609.23215*.
- Bo Ren, Yirong Mao, Yi Yang, and Wenhui Que. 2026a. [IAPO: Influence-aware policy optimization for credit assignment in multi-turn service agents](#). *Preprint*, arXiv:2608.24588.
- Wencan Ren et al. 2026b. Causal-dependency state space prompting: bounded-memory long-horizon reasoning for streaming IT incidents. In *Engineering Research Express*.
- Tessa Rusch et al. 2020. Theory of mind and decision science: Towards a typology of tasks and computational models. In *Neuropsychologia*.
- Martiño Ríos-García, Nawaf Alampara, Chandan Gupta, Indrajeet Mandal, Sajid Mannan, Ali Asghar Aghajani, N. M. Anoop Krishnan, and Kevin Maik Jablonka. 2026. [AI scientists produce results without reasoning scientifically](#). *arXiv preprint arXiv:2604.18805*.
- Ankur Samanta, Akshayaa Magesh, Tal Lancewicki, Ayush Jain, Youliang Yu, Paul Sajda, Kaveh Hassani, Aditya Modi, Daniel R. Jiang, and Yonathan Efroni. 2026. [BayesBench: Evaluating LLM belief trajectories under multi-turn evidence accumulation](#). *arXiv preprint arXiv:2606.30850*.
- John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. 2017. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*.
- Melanie Sclar, Sachin Kumar, Peter West, Alane Suhr, Yejin Choi, and Yulia Tsvetkov. 2023. [Minding language models’ \(lack of\) theory of mind: A Plug-and-Play multi-character belief tracker](#). In *Proceedings of ACL 2023*.
- SeungWon Seo, SooBin Lim, SeongRae Noh, Haneul Kim, and HyeongYeop Kang. 2026. [From assumptions to actions: Turning LLM reasoning into uncertainty-aware planning for embodied agents](#). *Preprint*, arXiv:2602.04326.
- Adam S. Shai, Sarah E. Marzen, Lucas Teixeira, Alexander Gietelink Oldenziel, and Paul M. Riechers. 2024. [Transformers represent belief state geometry in their residual stream](#). In *Proceedings of NeurIPS 2024*.
- Lianlei Shan, Shixian Luo, Zezhou Zhu, Yu Yuan, and Yong Wu. 2025. [Cognitive memory in large language models](#). *arXiv preprint arXiv:2504.02441*.
- Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, et al. 2024. DeepSeek-Math: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*.
- Natalie Shapira, Mosh Levy, Seyed Hossein Alavi, Xuhui Zhou, Yejin Choi, Yoav Goldberg, Maarten Sap, and Vered Shwartz. 2023. [Clever hans or neural theory of mind? stress testing social reasoning in large language models](#). In *Proceedings of EACL 2023*.
- Abhishek Sharma. 2026. [FinalityBench: An effect-level benchmark for agent decisions under delayed and conflicting financial finality](#). *arXiv preprint arXiv:2609.04706*.
- Mrinank Sharma, Meg Tong, Tomasz Korbak, David Duvenaud, Amanda Askell, Samuel R. Bowman, Newton Cheng, Esin Durmus, Zac Hatfield-Dodds, Scott R. Johnston, Shauna Kravec, Timothy Maxwell, Sam McCandlish, Kamal Ndousse, Oliver Rausch, Nicholas Schiefer, Da Yan, Miranda Zhang, and Ethan Perez. 2023. [Towards understanding sycophancy in language models](#). In *Proceedings of ICLR 2023*.
- Junhong Shen, Hao Bai, Lunjun Zhang, Yifei Zhou, Amrith Setlur, Shengbang Tong, Diego Caples, Nan Jiang, Tong Zhang, Ameet Talwalkar, and Aviral Kumar. 2025. [Thinking vs. doing: Agents that reason by scaling test-time interaction](#). *Preprint*, arXiv:2506.07976.
- Haojun Shi, Suyu Ye, Xinyu Fang, Chuanyang Jin, Leyla Isik, Yen-Ling Kuo, and Tianmin Shu. 2024. [MuMA-ToM: Multi-modal multi-agent theory of mind](#). In *Proceedings of AAAI 2024*.
- Qiming Shi, Zhaolu Kang, Yunfan Zhou, Di Weng, and Yingcai Wu. 2026. [SPADER: Step-wise peer advantage with diversity-aware exploration rewards for multi-answer question answering](#). *arXiv preprint arXiv:2606.00593*.
- Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. 2023. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems*.
- Kazutoshi Shinoda, Nobukatsu Hojo, Kyosuke Nishida, Saki Mizuno, Keita Suzuki, Ryo Masumura, Hiroaki Sugiyama, and Kuniko Saito. 2025. [ToMATO: Verbalizing the mental states of role-playing LLMs for](#)

- benchmarking theory of mind. In *Proceedings of AAAI 2025*.
- Mohit Shridhar, Xingdi Yuan, Yonatan Bisk, Adam Trischler, and Matthew Hausknecht. 2020. ALF-World: Aligning text and embodied environments for interactive learning. In *ICLR 2021*. ArXiv:2010.03768.
- Bao Shu, Yan Cai, Jianjian Sun, Chunrui Han, En Yu, Liang Zhao, Jingcheng Hu, Yinmin Zhang, Hao-ran Lv, Yuang Peng, Zheng Ge, Xiangyu Zhang, Daxin Jiang, and Xiangyu Yue. 2025. Thinking by doing: Building efficient world model reasoning in LLMs via multi-turn interaction. *arXiv preprint arXiv:2511.23476*.
- Anthony Sicilia, Hyunwoo Kim, Khyathi Raghavi Chandu, Malihe Alikhani, and Jack Hessel. 2024. Deal, or no deal (or who knows)? forecasting uncertainty in conversations using large language models. In *Proceedings of ACL 2024*.
- Inderjeet Singh, Vikas Pahuja, Aishvariya Priya Rathina Sabapathy, Chiara Picardi, Amit Giloni, Roman Vainshtein, Andrés Murillo, Hisashi Kojima, Motoyoshi Sekiya, Yuki Unno, and Junichi Suga. 2026a. Adversarial intent is a latent variable: Stateful trust inference for securing multimodal agentic RAG. *arXiv preprint arXiv:2602.21447*.
- Joykirat Singh, Zaid Khan, Archiki Prasad, Justin Chih-Yao Chen, Akshay Nambi, Hyunji Lee, Elias Stengel-Eskin, and Mohit Bansal. 2026b. Agent-BRACE: Decoupling beliefs from actions in long-horizon tasks via verbalized state uncertainty. *Preprint*, arXiv:2605.11436.
- Pranav Singh. 2026. When does belief-based agent memory help? reliability-conditional updating and provenance-capped poisoning defense. *Preprint*, arXiv:2606.22030.
- Valentin Six, Frederik Panse, and Mathis Fajeau. 2026. Learning POMDP world models from observations with language-model priors. *Preprint*, arXiv:2605.13740.
- Charlie Snell, Ilya Kostrikov, Yi Su, Mengjiao Yang, and Sergey Levine. 2022. Offline RL for natural language generation with implicit language q learning. In *Proceedings of ICLR 2022*.
- Aagam Sogani, Botao Rui, Swetha Vaidyanathan, Rishi Agarwal, Minghao Yan, and Shivaram Venkataraman. 2026. When web agents finish but still fail: Reproducible triggers and trace diagnostics for parallel web exploration. *arXiv preprint arXiv:2606.20724*.
- Tin Stribor Sohn, Maximilian Dillitzer, Jason J. Corso, and Eric Sax. 2025. R4: Retrieval-augmented reasoning for vision-language models in 4d spatio-temporal space. *arXiv preprint arXiv:2512.15940*.
- Huatong Song, Jinhao Jiang, Yingqian Min, Jie Chen, Zhipeng Chen, Wayne Xin Zhao, Lei Fang, and Jirong Wen. 2025. R1-Searcher: Incentivizing the search capability in LLMs via reinforcement learning. *arXiv preprint arXiv:2503.05592*.
- Xinyuan Song and Zekun Cai. 2026a. Agent vs. parametric world models: Hybrid planning for reliable language agents. *Preprint*, arXiv:2606.27806.
- Xinyuan Song and Zekun Cai. 2026b. Repair the amplifier, not the symptom: Stable world-model correction for agent rollouts. *Preprint*, arXiv:2607.01767.
- Yuru Song and Qi Xin. 2026. D-MEM: Dopamine-gated agentic memory via reward prediction error routing. *Preprint*, arXiv:2603.14597.
- Tommaso Soru. 2026. Semantic Bayesian world models. *Preprint*, arXiv:2609.03834.
- Adil Soubki, John Murzaku, Arash Yousefi Jordehi, Peter Zeng, Magdalena Markowska, Seyed Abolghasem Mirroshandel, and Owen Rambow. 2024. Views are my own, but also yours: Benchmarking theory of mind using common ground. In *Proceedings of ACL 2024*.
- Mehul Srivastava. 2026a. Constructing a benchmark when every component is a language model. Zenodo preprint.
- Saksham Sahai Srivastava. 2026b. Causal intervention-based memory selection for long-horizon LLM agents. *Preprint*, arXiv:2605.17641.
- Kaya Stechly, Karthik Valmeekam, and Subbarao Kambhampati. 2024. On the self-verification limitations of large language models on reasoning and planning tasks. In *Proceedings of ICLR 2024*.
- Sukesh Subaharan. 2026. Controlling long-horizon behavior in language model agents with explicit state dynamics. *Preprint*, arXiv:2601.16087.
- Theodore R. Sumers, Shunyu Yao, Karthik Narasimhan, and Thomas L. Griffiths. 2023. Cognitive architectures for language agents. In *TMLR 2024*. ArXiv:2309.02427.
- Hao Sun, Zile Qiao, Jiayan Guo, Xuanbo Fan, Yingyan Hou, Yong Jiang, Pengjun Xie, Yan Zhang, Fei Huang, and Jingren Zhou. 2025a. ZeroSearch: Incentivize the search capability of LLMs without searching. *arXiv preprint arXiv:2505.04588*.
- Haofei Sun and Lin He. 2026. When memory updates but behavior does not: Repairing implicit state dependencies in personalized agent responses. *Preprint*, arXiv:2608.01619.
- Lingfeng Sun, Devesh K. Jha, Chiori Hori, Siddarth Jain, Radu Corcodel, Xinghao Zhu, Masayoshi Tomizuka, and Diego Romeres. 2023. Interactive planning using large language models for partially observable robotics tasks. *arXiv preprint arXiv:2312.06876*.

- Shuang Sun, Guoxin Chen, Fanzhe Meng, Jia Deng, Huatong Song, Jinhao Jiang, Wayne Xin Zhao, Hongteng Xu, and Ji-Rong Wen. 2026a. [Agent-editing world model: Rethinking world modeling for LLM agents](#). *arXiv preprint arXiv:2609.28416*.
- Weiwei Sun, Miao Lu, Zhan Ling, Kang Liu, Xuesong Yao, Yiming Yang, and Jiecao Chen. 2025b. [Scaling long-horizon LLM agent via context-folding](#). *arXiv preprint arXiv:2510.11967*.
- Xiaolong Sun, Qichao Wang, Hangyu Li, and Liang Chen. 2026b. [Verifiable memory: Learning unified memory management with local and global verifiers for large language model agents](#). *arXiv preprint arXiv:2608.03137*.
- Mirac Suzgun, Mert Yuksekgonul, Federico Bianchi, Dan Jurafsky, and James Zou. 2025. [Dynamic cheat-sheet: Test-time learning with adaptive memory](#). In *Proceedings of EACL 2025*.
- Andrew Szot, Max Schwarzer, Harsh Agrawal, Bogdan Mazouze, Walter Talbott, Katherine Metcalf, Natalie Mackraz, Devon Hjelm, and Alexander Toshev. 2023. [Large language models as generalizable policies for embodied tasks](#). In *Proceedings of ICLR 2023*.
- Haoran Tan, Zeyu Zhang, Chen Ma, Xu Chen, Quanyu Dai, and Zhenhua Dong. 2025a. [MemBench: Towards more comprehensive evaluation on the memory of LLM-based agents](#). In *Proceedings of ACL 2025*.
- Hui-Ze Tan, Xiao-Wen Yang, Hao Chen, Jie-Jing Shao, Yi Wen, Yuteng Shen, Weihong Luo, Xiku Du, Lan-Zhe Guo, and Yu-Feng Li. 2026. [Hindsight credit assignment for long-horizon LLM agents](#). *Preprint*, arXiv:2603.08754.
- Weiting Tan, Xinghua Qu, Ming Tu, Meng Ge, Andy T. Liu, Philipp Koehn, and Lu Lu. 2025b. [Process-supervised reinforcement learning for interactive multimodal tool-use agents](#). *arXiv preprint arXiv:2509.14480*.
- Bo Tang, Yang Zhang, Guomian Zhuang, Wenqiang Wei, Gaoyang Zheng, Lindong Xie, Yanchao Tan, Feiyu Xiong, Qingyu Yang, Edward Chung, and Zhiyu li. 2026a. [From memory to skills: Evidence-grounded co-evolution governance for long-horizon LLM agents](#). *arXiv preprint arXiv:2607.16621*.
- Wenjie Tang, Minne Li, Sijie Huang, Liquan Xiao, and Yuan Zhou. 2026b. [Rewarding beliefs, not actions: Consistency-guided credit assignment for long-horizon agents](#). *arXiv preprint arXiv:2605.20061*.
- Yan Tang, Tingyu Cao, Yuanbo Tang, Huaze Tang, and Keer Hu. 2026c. [Proactive service agents: A unified decision framework, methods, and evaluation](#). *arXiv preprint arXiv:2609.03727*.
- Xinrui Tao et al. 2025. [A large language model-enabled framework for simulating multi-agent cooperative game](#). In *BigData Congress [Services Society]*.
- Jayden Teoh, Manan Tomar, Kwangjun Ahn, Edward S. Hu, Tim Pearce, Pratyusha Sharma, Akshay Krishnamurthy, Riashat Islam, Alex Lamb, and John Langford. 2025. [Next-latent prediction Transformers learn compact world models](#). *arXiv preprint arXiv:2511.05963*.
- Sebastian Thrun, Wolfram Burgard, and Dieter Fox. 2005. *Probabilistic Robotics*. MIT Press.
- Qiuyu Tian, Yiding Li, Yingce Xia, Fengyi Chen, Youyong Kong, Fan Guo, Yuyao Li, Jinjing Shen, Zhijing Xie, Yiyun Luo, Xin Zhang, and Zequn Liu. 2026. [STAGE: A full-screenplay benchmark for reasoning over evolving stories](#). *arXiv preprint arXiv:2601.08510*.
- Harsh Trivedi, Tushar Khot, Mareike Hartmann, Ruskin Manku, Vinty Dong, Edward Li, Shashank Gupta, Ashish Sabharwal, and Niranjan Balasubramanian. 2024. [AppWorld: A controllable world of apps and people for benchmarking interactive coding agents](#). In *Proceedings of ACL 2024*.
- Md Nayem Uddin, Kumar Shubham, Eduardo Blanco, Chitta Baral, and Gengyu Wang. 2026. [From recall to forgetting: Benchmarking long-term memory for personalized agents](#). In *Proceedings of ACL 2026*.
- Karthik Valmeekam, Matthew Marquez, Alberto Olmo, Sarath Sreedharan, and Subbarao Kambhampati. 2022. [PlanBench: An extensible benchmark for evaluating large language models on planning and reasoning about change](#). In *NeurIPS 2023 Datasets and Benchmarks*. ArXiv:2206.10498.
- Ivan Verbovoy. 2026. [Rank-bounded memory: Self-poisoning and attribution laundering in LLM agents](#). Zenodo preprint.
- Daoyu Wang, Qingchuan Li, Mingyue Cheng, Jie Ouyang, Shuo Yu, Chunli Liu, Shijin Wang, Qi Liu, and Enhong Chen. 2026a. [CAPO: Critic-guided action-aligned policy optimization for advancing LLM agent capabilities](#). *Preprint*, arXiv:2604.18401.
- Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi Fan, and Anima Anandkumar. 2023a. [Voyager: An open-ended embodied agent with large language models](#). In *TMLR 2024*. ArXiv:2305.16291.
- Guoqing Wang, Sunhao Dai, Guangze Ye, Zeyu Gan, Wei Yao, Yong Deng, Xiaofeng Wu, and Zhenzhe Ying. 2025a. [Information gain-based policy optimization: A simple and effective approach for multi-turn search agents](#). *arXiv preprint arXiv:2510.14967*.

- Haixin Wang, Hejie Cui, Chenwei Zhang, Xin Liu, Shuowei Jin, Shijie Geng, Xinyang Zhang, Nasser Zalmout, Zhenyu Shi, and Yizhou Sun. 2026b. [T<sup>2</sup>po: Uncertainty-guided exploration control for stable multi-turn agentic reinforcement learning](#). *Preprint*, arXiv:2605.02178.
- Hanlin Wang, Chak Tou Leong, Jiashuo Wang, Jian Wang, and Wenjie Li. 2025b. [SPA-RL: Reinforcing LLM agents via stepwise progress attribution](#). *arXiv preprint arXiv:2505.20732*.
- Jiawei Wang, Jiakai Liu, Yuqian Fu, Yingru Li, Xintao Wang, Yuan Lin, Yu Yue, Lin Zhang, Yang Wang, and Ke Wang. 2025c. [Harnessing uncertainty: Entropy-modulated policy gradients for long-horizon LLM agents](#). *arXiv preprint arXiv:2509.09265*.
- Jun Wang. 2025. [Memento 2: Learning by stateful reflective memory](#). *arXiv preprint arXiv:2512.22716*.
- Kangrui Wang, Pingyue Zhang, Zihan Wang, Yaning Gao, Linjie Li, Qineng Wang, Hanyang Chen, Chi Wan, Yiping Lu, Zhengyuan Yang, Lijuan Wang, Ranjay Krishna, Jiajun Wu, Li Fei-Fei, Yejin Choi, and Manling Li. 2025d. [VAGEN: Reinforcing world model reasoning for multi-turn VLM agents](#). *arXiv preprint arXiv:2510.16907*.
- Kevin Wang, Anna Thöni, Benjamin Kempinski, et al. 2026c. [MINDGAMES: A live arena for evaluating social and strategic reasoning in multi-agent LLMs](#). *Preprint*, arXiv:2605.29512.
- Li Wang, Yandong Wang, Xin Yu, Kui Zhang, Tianhao Peng, and Wenjun Wu. 2026d. [TAMTRL: Teacher-aligned reward reshaping for multi-turn reinforcement learning in long-context compression](#). *Preprint*, arXiv:2603.21663.
- Peiyi Wang, Lei Li, Zhihong Shao, R. X. Xu, Damai Dai, Yifei Li, Deli Chen, Y. Wu, and Zhifang Sui. 2023b. [Math-shepherd: Verify and reinforce LLMs step-by-step without human annotations](#). In *Proceedings of ACL 2023*.
- Prince Zizhuang Wang and Shuli Jiang. 2026. [SLEA-RL: Step-level experience augmented reinforcement learning for multi-turn agentic training](#). *Preprint*, arXiv:2603.18079.
- Qineng Wang, Wenlong Huang, Yu Zhou, Hang Yin, Tianwei Bao, Jianwen Lyu, Weiyu Liu, Ruohan Zhang, Jiajun Wu, Li Fei-Fei, and Manling Li. 2025e. [ENACT: Evaluating embodied cognition with world modeling of egocentric interaction](#). *arXiv preprint arXiv:2511.20937*.
- Ruiyang Wang, Hao-Lun Hsu, Swarajh Mehta, Jiwoo Kim, Zhihao Dou, and Miroslav Pajic. 2026e. [GAVEL: Graph world models for verified and efficient long-horizon LLM task planning](#). *arXiv preprint arXiv:2609.19315*.
- Ruiyi Wang, Haofei Yu, Wenxin Zhang, Zhengyang Qi, Maarten Sap, Graham Neubig, Yonatan Bisk, and Hao Zhu. 2024a. [SOTOPIA-pi: Interactive learning of socially intelligent language agents](#). *Preprint*, arXiv:2403.08715.
- Taiyi Wang, Zhihao Wu, Jianheng Liu, Jianye Hao, Jun Wang, and Kun Shao. 2024b. [DistRL: An asynchronous distributed reinforcement learning framework for on-device control agents](#). In *Proceedings of ICLR 2024*.
- Xingjun Wang, Gongsheng Li, Qi Fan, Yunlin Mao, Luyan Su, and Yingda Chen. 2026f. [LENS: In-context search via latent evidence exploration over dynamic raw documents](#). *arXiv preprint arXiv:2608.16185*.
- Xinyu Wang, Hanwei Wu, Jingwei Song, Shuyuan Zhang, Jiayi Zhang, Fanqi Kong, Tung Sum Thomas Kwok, Xiao-Wen Chang, Yuyu Luo, Chenglin Wu, and Bang Liu. 2026g. [Co-evolution of policy and internal reward for language agents](#). *Preprint*, arXiv:2604.03098.
- Xucong Wang, Ziyu Ma, Yong Wang, Yuxiang Ji, Shidong Yang, Guanhua Chen, Pengkun Wang, and Xiangxiang Chu. 2026h. [APPO: Agentic procedural policy optimization](#). *arXiv preprint arXiv:2606.12384*.
- Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. 2022. Self-consistency improves chain of thought reasoning in language models. In *ICLR 2023*. ArXiv:2203.11171.
- Yan Wang, Yi Han, Lingfei Qian, Yueru He, Xueqing Peng, Dongji Feng, Zhuohan Xie, Vincent Jim Zhang, Yuqing Guo, Fengran Mo, Jimin Huang, Yankai Chen, and Jian-Yun Nie. 2026i. [Conv-FinRe: A conversational and longitudinal benchmark for utility-grounded financial recommendation](#). In *Proceedings of the 49th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 3448–3455. ACM.
- Yu Wang. 2026a. [The dark room in the reward channel: Dense prediction rewards collapse GRPO-Trained LLM agents – and the channel, not the content, decides what works](#). *arXiv preprint arXiv:2607.21273*.
- Yu Wang, Yifan Gao, Xiusi Chen, Haoming Jiang, Shiyang Li, Jingfeng Yang, Qingyu Yin, Zheng Li, Xian Li, Bing Yin, Jingbo Shang, and Julian McAuley. 2024c. [MEMORYLLM: Towards self-updatable large language models](#). In *Proceedings of ICML 2024*.
- Yunan Wang, Minghui Song, Zihan Zhang, Shaohan Huang, Haizhen Huang, Furu Wei, Weiwei Deng, Feng Sun, and Qi Zhang. 2026j. [Group-graph policy optimization for long-horizon agentic reinforcement learning](#). *Preprint*, arXiv:2606.22995.

- Zechuan Wang, Siyuan Lu, Hongxuan Zhang, Linjian Mo, Chenyi Zhuang, and Leilei Gan. 2026k. [Teach the magnitude, not the direction: Verifier-bounded credit assignment for multi-turn multi-step LLM agents](#). *Preprint*, arXiv:2608.13179.
- Zihan Wang, Kangrui Wang, Qineng Wang, Pingyue Zhang, Linjie Li, Zhengyuan Yang, Xing Jin, Kefan Yu, Minh Nhat Nguyen, Licheng Liu, Eli Gotlieb, Yiping Lu, Kyunghyun Cho, Jiajun Wu, Li Fei-Fei, Lijuan Wang, Yejin Choi, and Manling Li. 2025f. [RAGEN: Understanding self-evolution in LLM agents via multi-turn reinforcement learning](#). *Preprint*, arXiv:2504.20073.
- Zihao Wang, Shaofei Cai, Guanzhou Chen, Anji Liu, Xiaojian Ma, and Yitao Liang. 2023c. Describe, explain, plan and select: Interactive planning with large language models enables open-world multi-task agents. In *NeurIPS 2023*. ArXiv:2302.01560.
- Ziliang Wang, Xuhui Zheng, Kang An, Cijun Ouyang, Jialu Cai, Yuhang Wang, and Yichao Wu. 2025g. [StepSearch: Igniting LLMs search ability via step-wise proximal policy optimization](#). In *Proceedings of EMNLP 2025*.
- Ziming Wang. 2026b. [TOKI: A bitemporal operator algebra for contradiction resolution in LLM-agent persistent memory](#). *Preprint*, arXiv:2606.06240.
- Zixuan Wang, Yufan Zhou, Jinzhou Tang, Xinle Yu, Chengjun Wu, Lyumanshan Ye, Zhaoxiang Feng, Letian Peng, Adyasha Patra, Fan Bai, Enze Ma, Zhengding Hu, Jianyang Gu, Zhao Wang, Yufei Ding, Jingbo Shang, Tianmin Shu, Zhiting Hu, and Zhen Wang. 2026l. [Mind2Dialogue: Training human-aware language models by simulating user mental states](#). *arXiv preprint arXiv:2609.15972*.
- Zora Zhiruo Wang, Jiayuan Mao, Daniel Fried, and Graham Neubig. 2024d. [Agent workflow memory](#). *Preprint*, arXiv:2409.07429.
- Jason Wei, Maarten Bosma, Vincent Zhao, Kelvin Guu, Adams Wei Yu, Brian Lester, Nan Du, Andrew M. Dai, and Quoc V. Le. 2022a. Finetuned language models are zero-shot learners. In *International Conference on Learning Representations*.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. 2022b. Chain-of-thought prompting elicits reasoning in large language models. In *NeurIPS 2022*. ArXiv:2201.11903.
- Quan Wei, Siliang Zeng, Chenliang Li, Zhongruo Wang, William Brown, Oana Frunza, Wei Deng, Anderson Schneider, Yuriy Nevmyvaka, Yang Katie Zhao, Alfredo Garcia, and Mingyi Hong. 2025a. [Reinforcing multi-turn reasoning in LLM agents via fine-grained reward structure and credit assignment](#). *arXiv preprint arXiv:2505.11821*.
- Tianxin Wei, Ting-Wei Li, Zhining Liu, Xuying Ning, Ze Yang, Jiaru Zou, Zhichen Zeng, Ruizhong Qiu, Xiao Lin, Dongqi Fu, Zihao Li, Mengting Ai, Duo Zhou, Wenxuan Bao, Yunzhe Li, Gaotang Li, Cheng Qian, Yu Wang, Xiangru Tang, and 10 others. 2026. [Agentic reasoning for large language models](#). *Preprint*, arXiv:2601.12538.
- Tianxin Wei, Noveen Sachdeva, Benjamin Coleman, Zhankui He, Yuanchen Bei, Xuying Ning, Mengting Ai, Yunzhe Li, Jingrui He, Ed H. Chi, Chi Wang, Shuo Chen, Fernando Pereira, Wang-Cheng Kang, and Derek Zhiyuan Cheng. 2025b. [Evo-memory: Benchmarking LLM agent test-time learning with self-evolving memory](#). *arXiv preprint arXiv:2511.20857*.
- Tong Wei, Yijun Yang, Changhao Zhang, Junliang Xing, Yuanchun Shi, Zongqing Lu, and Deheng Ye. 2025c. [GTR-Turbo: Merged checkpoint is secretly a free teacher for agentic VLM training](#). *arXiv preprint arXiv:2512.13043*.
- Zhepei Wei, Wenlin Yao, Yao Liu, Weizhi Zhang, Qin Lu, Liang Qiu, Changlong Yu, Puyang Xu, Chao Zhang, Bing Yin, Hyokun Yun, and Lihong Li. 2025d. [WebAgent-R1: Training web agents via End-to-End multi-turn reinforcement learning](#). In *Proceedings of EMNLP 2025*.
- Muning Wen et al. 2024. Reinforcing LLM agents via policy optimization with action decomposition. In *NeurIPS 2024*.
- Alex Wilf, Sihyun Shawn Lee, Paul Pu Liang, and Louis-Philippe Morency. 2023. [Think twice: Perspective-taking improves large language models' Theory-of-Mind capabilities](#). In *Proceedings of ACL 2023*.
- Bryan Wilie, Samuel Cahyawijaya, Etsuko Ishii, Junxian He, and Pascale Fung. 2024. [Belief revision: The adaptability of large language models reasoning](#). In *Proceedings of EMNLP 2024*.
- Grant Wilson. 2026. [A controlled replication of a self-model loop for language-model agents: Controls, confounds, and finite-horizon path dependence](#). Zenodo preprint.
- Di Wu, Hongwei Wang, Wenhao Yu, Yuwei Zhang, Kai-Wei Chang, and Dong Yu. 2024. [Long-MemEval: Benchmarking chat assistants on long-term interactive memory](#). In *Proceedings of ICLR 2025*.
- Jialong Wu, Shaofeng Yin, Ningya Feng, and Mingsheng Long. 2025a. [RLVR-World: Training world models with reinforcement learning](#). In *Proceedings of NeurIPS 2025*.
- Shirley Wu, Evelyn Choi, Arpandeeep Khatua, Zhanhan Wang, Joy He-Yueya, Tharindu Cyril Weerasooriya, Wei Wei, Diyi Yang, Jure Leskovec, and James Zou. 2026a. [HumanLM: Simulating users with state alignment beats response imitation](#). *arXiv preprint arXiv:2603.03303*.

- Shirley Wu, Michel Galley, Baolin Peng, Hao Cheng, Gavin Li, Yao Dou, Weixin Cai, James Zou, Jure Leskovec, and Jianfeng Gao. 2025b. [CollabLLM: From passive responders to active collaborators](#). In *Proceedings of ICML 2025*.
- Tiantong Wu and Wei Yang Bryan Lim. 2026. [RE-FLEX with Jev for efficient selective control in LLM agents](#). *Preprint*, arXiv:2609.26532.
- Xiaobao Wu. 2025. Sailing by the stars: A survey on reward models and learning strategies for learning from rewards. *arXiv preprint arXiv:2505.02686*. Findings of EMNLP 2025.
- Xiaojun Wu, Cehao Yang, Honghao Liu, Xueyuan Lin, Wenjie Zhang, Zhichao Shi, Xuhui Jiang, Chengjin Xu, Jia Li, and Jian Guo. 2026b. [Bayesian-agent: Posterior-guided skill evolution across LLM agent harnesses](#). *Preprint*, arXiv:2606.08348.
- Xixi Wu, Kuan Li, Yida Zhao, Liwen Zhang, Litu Ou, Huifeng Yin, Zhongwang Zhang, Xinmiao Yu, Dingchu Zhang, Yong Jiang, Pengjun Xie, Fei Huang, Minhao Cheng, Shuai Wang, Hong Cheng, and Jingren Zhou. 2025c. [ReSum: Unlocking long-horizon search intelligence via context summarization](#). *arXiv preprint arXiv:2509.13313*.
- Xixi Wu, Qianguo Sun, Ruiyang Zhang, Chao Song, Junlong Wu, Yiyang Qi, and Hong Cheng. 2026c. [Demystifying reinforcement learning for long-horizon tool-using agents: A comprehensive recipe](#). *arXiv preprint arXiv:2603.21972*.
- Xuening Wu, Shan Yu, Qianya Xu, and Shenqin Yin. 2026d. [BayesEvolve: Explicit belief states for autonomous scientific discovery](#). *Preprint*, arXiv:2606.30335.
- Yue Wu, Xuan Tang, Tom M. Mitchell, and Yuanzhi Li. 2023. SmartPlay: A benchmark for LLMs as intelligent agents. In *ICLR 2024*. ArXiv:2310.01557.
- Yuheng Wu, Jianwen Xie, Denghui Zhang, and Zhaozhuo Xu. 2025d. [DEL-ToM: Inference-time scaling for Theory-of-Mind reasoning via dynamic epistemic logic](#). In *Proceedings of EMNLP 2025*.
- Zhiheng Xi, Yiwen Ding, Wenxiang Chen, Boyang Hong, Honglin Guo, Junzhe Wang, Dingwen Yang, Chenyang Liao, Xin Guo, Wei He, Songyang Gao, Lu Chen, Rui Zheng, Yicheng Zou, Tao Gui, Qi Zhang, Xipeng Qiu, Xuanjing Huang, Zuxuan Wu, and Yu-Gang Jiang. 2024. [AgentGym: Evolving large language model-based agents across diverse environments](#). *Preprint*, arXiv:2406.04151.
- Zhiheng Xi, Chenyang Liao, Guanyu Li, Yajie Yang, Wenxiang Chen, Zhihao Zhang, Binghai Wang, Senjie Jin, Yuhao Zhou, Jian Guan, Wei Wu, Tao Ji, Tao Gui, Qi Zhang, and Xuanjing Huang. 2025. [Agent-PRM: Process reward models for LLM agents via step-wise promise and progress](#). In *Proceedings of WWW 2025*.
- Peng Xia, Jianwen Chen, Hanyang Wang, Jiaqi Liu, Kaide Zeng, Yu Wang, Siwei Han, Yiyang Zhou, Xujiao Zhao, Haifeng Chen, Zeyu Zheng, Cihang Xie, and Huaxiu Yao. 2026. [SkillRL: Evolving agents via recursive skill-augmented reinforcement learning](#). *arXiv preprint arXiv:2602.08234*.
- Zhiqiu Xia, Jinxuan Xu, Yuqian Zhang, and Hang Liu. 2025. A survey of uncertainty estimation methods on large language models. In *Findings of the Association for Computational Linguistics: ACL 2025*.
- Jianqiang Xiao, Xiang Deng, Yuexuan Sun, Yanjin Wu, Wenbiao Yan, and Liqiang Nie. 2026a. [Dual-layer semantic-spatial belief mapping for aerial object goal navigation](#). *arXiv preprint arXiv:2609.08164*.
- Yijia Xiao, Edward Sun, Di Luo, and Wei Wang. 2024. [TradingAgents: Multi-agents LLM financial trading framework](#). *Preprint*, arXiv:2412.20138.
- Zikai Xiao, Jianhong Tu, Chuhan Zou, Yuxin Zuo, Zhi Li, Peng Wang, Bowen Yu, Fei Huang, Junyang Lin, and Zuozhu Liu. 2026b. [WebWorld: A large-scale world model for web agent training](#). *arXiv preprint arXiv:2602.14721*.
- Qianqian Xie, Weiguang Han, Zhengyu Chen, Ruoyu Xiang, Xiao Zhang, Yueru He, et al. 2024. FinBen: A holistic financial benchmark for large language models. In *Advances in Neural Information Processing Systems Datasets and Benchmarks Track*.
- Yutao Xie, Nathaniel Thomas, Nicklas Hansen, Yang Fu, Li Erran Li, and Xiaolong Wang. 2026. [TIPS: Turn-level information-potential reward shaping for search-augmented LLMs](#). *Preprint*, arXiv:2603.22293.
- Guojun Xiong, Zhiyang Deng, Keyi Wang, Yupeng Cao, Haohang Li, Yangyang Yu, Xueqing Peng, Mingquan Lin, Kaleb E Smith, Xiao-Yang Liu, Jimin Huang, Sophia Ananiadou, and Qianqian Xie. 2025a. [FLAG-TRADER: Fusion LLM-agent with gradient-based reinforcement learning for financial trading](#). In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 13921–13934, Vienna, Austria. Association for Computational Linguistics.
- Zidi Xiong, Yuping Lin, Wenya Xie, Pengfei He, Zirui Liu, Jiliang Tang, Himabindu Lakkaraju, and Zhen Xiang. 2025b. [How memory management impacts LLM agents: An empirical study of experience-following behavior](#). In *Proceedings of ACL 2025*.
- Duo Xu and Faramarz Fekri. 2026. [NeSyFS: A neuro-symbolic fast-slow thinking framework for LLM agent under partial observability](#). *Preprint*, arXiv:2607.28942.
- Feng Xu, Guangyao Zhai, Xin Kong, Tingzhong Fu, Daniel F. N. Gordon, Xueli An, and Benjamin Busam. 2025a. [STARE-VLA: Progressive stage-aware reinforcement for fine-tuning vision-language-action models](#). *arXiv preprint arXiv:2512.05107*.

- Guodong Xu. 2026. [Governed persistent memory: Source-bound state semantics and fail-closed release for long-horizon agents](#). *Preprint*, arXiv:2608.12476.
- Hainiu Xu, Siya Qi, Jiazheng Li, Yuxiang Zhou, Jinhua Du, Caroline Catmur, and Yulan He. 2025b. [EnigmaToM: Improve LLMs’ Theory-of-Mind reasoning capabilities with neural knowledge base of entity states](#). In *Proceedings of ACL 2025*.
- Haoming Xu, Weihong Xu, Zongrui Li, Mengru Wang, Yunzhi Yao, Chiyu Wu, Jin Shang, Yu Gong, and Shumin Deng. 2026a. [When should models change their minds? contextual belief management in large language models](#). *Preprint*, arXiv:2605.30219.
- Nuoyue Xu, Jiang Liu, Wenxuan Huang, Xiang Zhang, Juntao Cao, and Jiaqi Wei. 2026b. [Agent memory with episodic retrieval for financial decision-making](#). In *Findings of the Association for Computational Linguistics: AACL-IJCNLP 2026*.
- Peiran Xu, Zhuohao Li, Xiaoying Xing, Guannan Zhang, Debiao Li, and Kunyu Shi. 2025c. [Hybrid reward normalization for process-supervised non-verifiable agentic tasks](#). *arXiv preprint arXiv:2509.25598*.
- Wujiang Xu, Zujie Liang, Kai Mei, Hang Gao, Juntao Tan, and Yongfeng Zhang. 2025d. [A-MEM: Agentic memory for LLM agents](#). *arXiv preprint arXiv:2502.12110*.
- Wujiang Xu, Zujie Liang, Kai Mei, Hang Gao, Juntao Tan, and Yongfeng Zhang. 2025e. [A-MEM: Agentic memory for LLM agents](#). In *NeurIPS 2025*. ArXiv:2502.12110.
- Wujiang Xu, Yu Wang, Kai Mei, Kaiqu Liang, Zhen-ting Wang, Mingyu Jin, Han Zhang, Shi-Xiong Zhang, Wenye Hua, Sambit Sahu, and Dimitris N. Metaxas. 2026c. [MemGym: a long-horizon memory environment for LLM agents](#). *Preprint*, arXiv:2605.20833.
- Xiaogang Xu, Jiaqi Tang, Jianmin Chen, Yingying Yan, Zhenchao Tang, Xiangxin Zhou, Xiaobin Hu, Wei Wei, Jinfeng Wu, Qifeng Chen, Lu Zhou, Jiafei Wu, Zhe Liu, Jianwei Yin, and Weimin Zheng. 2026d. [LLM-based agents for forecasting and prediction: Methods, training, evaluation, and applications](#). *Preprint*, arXiv:2608.23058.
- Zishan Xu, Zhiyuan Yao, Yuxin Chen, Yifu Guo, Zhengxi Lu, Yuquan Lu, Jinyang Huang, Yan Xu, Yasheng Wang, Weinan Zhang, Xingshan Zeng, and Weiwen Liu. 2026e. [EnvACE: Internalizing environment dynamics via world rehearsal for agentic reinforcement learning](#). *arXiv preprint arXiv:2608.06197*.
- Weicheng Xue. 2026. [Representation signatures and risk-feedback alignment in LLM trading agents](#). *Preprint*, arXiv:2605.28850.
- Neeraj Yadav. 2026. [Temporal validity in retrieval memory: Eliminating stale-fact errors for AI agents over evolving knowledge](#). *arXiv preprint arXiv:2606.26511*.
- Sikuan Yan, Ahmed Bahloul, Ercong Nie, Susanna Schwarzmann, Riccardo Trivisonno, Volker Tresp, and Yunpu Ma. 2026a. [Memory-r2: Fair credit assignment for long-horizon memory-augmented LLM agents](#). *Preprint*, arXiv:2605.21768.
- Sikuan Yan, Xiufeng Yang, Zuchao Huang, Ercong Nie, Zifeng Ding, Zonggen Li, Xiaowen Ma, Jinhe Bi, Kristian Kersting, Jeff Z. Pan, Hinrich Schütze, Volker Tresp, and Yunpu Ma. 2025. [Memory-R1: Enhancing large language model agents to manage and utilize memories via reinforcement learning](#). In *Proceedings of ACL 2025*.
- Yandong Yan, Junwei Peng, Shijie Li, Chenxi Li, Yifei Shang, Can Deng, Ruiting Dai, Yongqiang Zhao, Jiaqi Zhu, and Yu Huang. 2026b. [DenoiseFlow: Uncertainty-aware denoising for reliable LLM agentic workflows](#). *Preprint*, arXiv:2603.00532.
- Bo Yang, Jiaxian Guo, Yusuke Iwasawa, and Yutaka Matsuo. 2025a. [Large language models as theory of mind aware generative agents with counterfactual reflection](#). *arXiv preprint arXiv:2501.15355*.
- Bohao Yang, Hainiu Xu, Jinhua Du, Ze Li, Yulan He, and Chenghua Lin. 2025b. [EvolvTrip: Enhancing literary character understanding with temporal Theory-of-Mind graphs](#). *arXiv preprint arXiv:2506.13641*.
- Chang Yang, Chuang Zhou, Yilin Xiao, Su Dong, Luyao Zhuang, Yujing Zhang, Zhu Wang, Zijin Hong, Zheng Yuan, Zhishang Xiang, Shengyuan Chen, Huachi Zhou, Qinggang Zhang, Ninghao Liu, Jinsong Su, Xinrun Wang, Yi Chang, and Xiao Huang. 2026a. [Graph-based agent memory: Taxonomy, techniques, and applications](#). *arXiv preprint arXiv:2602.05665*.
- Jingbo Yang, Kwei-Herng Lai, Xiaowen Wang, Shiyu Chang, Yaar Harari, and Evgeniy Gabrilovich. 2026b. [GroupMemBench: Benchmarking LLM agent memory in multi-party conversations](#). *Preprint*, arXiv:2605.14498.
- Joshua C. Yang, Maurice Flechtner, Damian Dailisan, and Michiel A. Bakker. 2026c. [Belief engine: Configurable and inspectable stance dynamics in multi-agent LLM deliberation](#). *Preprint*, arXiv:2605.15343.
- Min Yang, Yichen Pan, Jinghua Piao, Dandan Song, Yongshun Gong, and Yong Li. 2026d. [EnterpriseBench: Benchmarking LLM agents on enterprise-level strategic reasoning and decision-making](#). *arXiv preprint arXiv:2609.37658*.
- Sherry Yang. 2026. [World models as an intermediary between agents and the real world](#). *arXiv preprint arXiv:2602.00785*.

- Shuo Yang, Soyeon Caren Han, Xueqi Ma, Yan Li, Mohammad Reza Ghasemi Madani, and Eduard Hovy. 2026e. [EvoTool: Self-evolving tool-use policy optimization in LLM agents via blame-aware mutation and diversity-aware selection](#). In *Proceedings of ACL 2026*.
- Yu-Yu Yang, Ti-Rong Wu, Hung Guei, Hsing-Yu Chen, and I-Chen Wu. 2026f. [Do LLMs trust the accuser or the accusation? measuring belief shifts in Werewolf](#). *arXiv preprint arXiv:2609.12446*.
- Shunyu Yao, Howard Chen, John Yang, and Karthik Narasimhan. 2022. [WebShop: Towards scalable real-world web interaction with grounded language agents](#). In *Proceedings of NeurIPS 2022*.
- Shunyu Yao, Noah Shinn, Pedram Razavi, and Karthik Narasimhan. 2024. [tau-bench: A benchmark for tool-agent-user interaction in real-world domains](#). *Preprint*, arXiv:2406.12045.
- Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Thomas L. Griffiths, Yuan Cao, and Karthik Narasimhan. 2023a. Tree of thoughts: Deliberate problem solving with large language models. In *NeurIPS 2023*. ArXiv:2305.10601.
- Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023b. [ReAct: Synergizing reasoning and acting in language models](#). In *International Conference on Learning Representations*.
- Li Qin Ye, Haorui Wang, Fardin Ahmed, Rongzhi Zhang, Yuan He, Ziyuan Lin, Yanbin Yin, Jing Peng, Michael Galarnyk, Sudheer Chava, and Chao Zhang. 2026. [Forecast-dojos: Replayable environments for benchmarking and training LLM forecasting agents](#). *arXiv preprint arXiv:2609.28876*.
- Asaf Yehudai, Lilach Eden, Alan Li, Guy Uziel, Yilun Zhao, Roy Bar-Haim, Arman Cohan, and Michal Shmueli-Scheuer. 2026. Survey on evaluation of LLM-based agents. In *Findings of the Association for Computational Linguistics: ACL 2026*. ArXiv:2503.16416.
- Xie Yi, Zhanke Zhou, Chentao Cao, Qiyu Niu, Tongliang Liu, and Bo Han. 2025. [From debate to equilibrium: Belief-driven multi-agent LLM reasoning via Bayesian Nash equilibrium](#). In *Proceedings of ICML 2025*.
- Lance Ying, Tan Zhi-Xuan, Lionel Wong, Vikash Mansinghka, and Joshua B. Tenenbaum. 2024. [Understanding epistemic language with a language-augmented Bayesian theory of mind](#). *Transactions of the Association for Computational Linguistics*.
- Chenglin Yu, Yuchen Wang, Songmiao Wang, Hongxia Yang, and Ming Li. 2026a. [InfiAgent: An infinite-horizon framework for general-purpose autonomous agents](#). *arXiv preprint arXiv:2601.03204*.
- Fangxu Yu, Lai Jiang, Shenyi Huang, Zhen Wu, and Xinyu Dai. 2025a. [PersuasiveToM: A benchmark for evaluating machine theory of mind in persuasive dialogues](#). *arXiv preprint arXiv:2502.21017*.
- Hongli Yu, Tinghong Chen, Jiangtao Feng, Jiangjie Chen, Weinan Dai, Qiying Yu, Ya-Qin Zhang, Wei-Ying Ma, Jingjing Liu, Mingxuan Wang, and Hao Zhou. 2025b. [MemAgent: Reshaping long-context LLM with multi-conv RL-based memory agent](#). In *ICLR 2026*. ArXiv:2507.02259.
- Xiao Yu, Baolin Peng, Ruize Xu, Michel Galley, Hao Cheng, Suman Nath, Jianfeng Gao, and Zhou Yu. 2025c. [Dyna-think: Synergizing reasoning, acting, and world model simulation in AI agents](#). *arXiv preprint arXiv:2506.00320*.
- Xiaohang Yu and William Knottenbelt. 2026. [LO-CARD: An agentic framework for blockchain forensics](#). In *IEEE International Conference on Blockchain 2026*.
- Yangyang Yu, Haohang Li, Zhi Chen, Yuechen Jiang, Yang Li, Denghui Zhang, Rong Liu, Jordan W. Suchow, and Khalidoun Khashanah. 2023. [FinMem: A performance-enhanced LLM trading agent with layered memory and character design](#). In *AAAI Spring Symposium 2024*. ArXiv:2311.13743.
- Yangyang Yu, Zhiyuan Yao, Haohang Li, Zhiyong Deng, Yupeng Cao, Zhi Chen, Jordan W. Suchow, Rong Liu, Zhenyu Cui, Zhaozhuo Xu, Denghui Zhang, Koduvayur Subbalakshmi, Guojun Xiong, Yueru He, Jimin Huang, Dong Li, and Qianqian Xie. 2024. [FinCon: A synthesized LLM multi-agent system with conceptual verbal reinforcement for enhanced financial decision making](#). In *NeurIPS 2024*. ArXiv:2407.06567.
- Yi Yu, Liuyi Yao, Yuexiang Xie, Qingquan Tan, Jiaqi Feng, Yaliang Li, and Libing Wu. 2026b. [Agentic memory: Learning unified long-term and short-term memory management for large language model agents](#). In *Proceedings of ACL 2026*.
- Zhaochen Yu, Ling Yang, Jiaru Zou, Shuicheng Yan, and Mengdi Wang. 2025d. [Demystifying reinforcement learning in agentic reasoning](#). *arXiv preprint arXiv:2510.11701*.
- Huining Yuan, Zelai Xu, Zheyue Tan, Xiangmin Yi, Mo Guang, Kaiwen Long, Haojia Hui, Boxun Li, Xinlei Chen, Bo Zhao, Xiao-Ping Zhang, Chao Yu, and Yu Wang. 2025a. [MARSHAL: Incentivizing multi-agent reasoning via self-play with strategic LLMs](#). *arXiv preprint arXiv:2510.15414*.
- Wenhao Yuan, Chenchen Lin, Jian Chen, Jinfeng Xu, Xuehe Wang, and Edith Cheuk Han Ngai. 2026a. [Verify before you commit: Towards faithful reasoning in LLM agents via self-auditing](#). *Preprint*, arXiv:2604.08401.

- Zhangdie Yuan, Zifeng Ding, and Andreas Vlachos. 2025b. [Do language models update their forecasts with new information?](#) *arXiv preprint arXiv:2509.23936*.
- Zhenhang Yuan, Shenghai Yuan, and Lihua Xie. 2026b. [RPMS: Enhancing LLM-Based embodied planning through rule-augmented memory synergy.](#) *arXiv preprint arXiv:2603.17831*.
- Aohan Zeng, Mingdao Liu, Rui Lu, Bowen Wang, Xiao Liu, Yuxiao Dong, and Jie Tang. 2023. AgentTuning: Enabling generalized agent abilities for LLMs. In *ACL 2024 Findings*. ArXiv:2310.12823.
- Yaopei Zeng, Congchao Wang, JianHang Chen, Nan Wang, Yurui Chang, and Lu Lin. 2026. [Critic experience bank: Self-evolving step-level confidence estimation for LLM agents.](#) *arXiv preprint arXiv:2607.12397*.
- Bolin Zhang et al. 2025a. Ask and retrieve knowledge: Towards proactive asking with imperfect information in medical multi-turn dialogues. In *Annual International ACM SIGIR Conference on Research and Development in Information Retrieval*.
- Ceyao Zhang, Kaijie Yang, Siyi Hu, Zihao Wang, Guanghe Li, Yihang Sun, Cheng Zhang, Zhaowei Zhang, Anji Liu, Song-Chun Zhu, Xiaojun Chang, Junge Zhang, Feng Yin, Yitao Liang, and Yaodong Yang. 2023a. [ProAgent: Building proactive cooperative agents with large language models.](#) In *Proceedings of AAAI 2024*.
- Chenchen Zhang. 2026. [From reasoning to agentic: Credit assignment in reinforcement learning for large language models.](#) *Preprint*, arXiv:2604.09459.
- Chunhui Zhang, Zhongyu Ouyang, Kwonjoon Lee, Nakul Agarwal, Sean Dae Houlihan, Soroush Vosoughi, and Shao-Yuan Lo. 2025b. [Overcoming multi-step complexity in multimodal Theory-of-Mind reasoning: A scalable Bayesian planner.](#) In *Proceedings of ICML 2025*.
- Dianxing Zhang, Wendong Li, Kani Song, Jiaye Lu, Gang Li, Liuchun Yang, and Sheng Li. 2025c. [Memory in large language models: Mechanisms, evaluation and evolution.](#) *arXiv preprint arXiv:2509.18868*.
- Erhan Zhang, Yiqun Chen, Zechun Niu, Wei Yang, Xiaochi Wei, Yan Gao, Yi Wu, Yao Hu, and Jiaxin Mao. 2026a. [OASES: Outcome-aligned search-evaluation co-training for agentic search.](#) *arXiv preprint arXiv:2604.03675*.
- Guibin Zhang, Hejia Geng, Xiaohang Yu, Zhenfei Yin, Zaibin Zhang, Zelin Tan, Heng Zhou, Zhongzhi Li, Xiangyuan Xue, Yijiang Li, Yifan Zhou, Yang Chen, Chen Zhang, Yutao Fan, Zihu Wang, Songtao Huang, Francisco Piedrahita-Velez, Yue Liao, Hongru Wang, and 6 others. 2025d. The landscape of agentic reinforcement learning for LLMs: A survey. In *TMLR*. ArXiv:2509.02547.
- Haozhen Zhang, Quanyu Long, Jianzhu Bao, Tao Feng, Weizhi Zhang, Haodong Yue, and Wenya Wang. 2026b. [MemSkill: Learning and evolving memory skills for self-evolving agents.](#) *arXiv preprint arXiv:2602.02474*.
- Hongxin Zhang, Weihua Du, Jiaming Shan, Qinhong Zhou, Yilun Du, Joshua B. Tenenbaum, Tianmin Shu, and Chuang Gan. 2023b. [Building cooperative embodied agents modularly with large language models.](#) In *Proceedings of ICLR 2023*.
- Jensen Zhang, Jing Yang, and Keze Wang. 2025e. [Large language models as discounted Bayesian filters.](#) *arXiv preprint arXiv:2512.18489*.
- Kaiwei Zhang, Qi Jia, Zijian Chen, Wei Sun, Xi-angyang Zhu, Chunyi Li, Dandan Zhu, and Guangtao Zhai. 2025f. [Sycophancy under pressure: Evaluating and mitigating sycophantic bias via adversarial dialogues in scientific QA.](#) *arXiv preprint arXiv:2508.13743*.
- Linghua Zhang, Jun Wang, Jingtong Wu, and Zhisong Zhang. 2026c. [RetailBench: Evaluating long-horizon autonomous decision-making and strategy stability of LLM agents in realistic retail environments.](#) *Preprint*, arXiv:2603.16453.
- Naifan Zhang, Ruihan Sun, Jinwei Su, Hengjie Yang, Zhengyuan Pan, Zhaohan Chen, and Xiaofan Zhang. 2026d. [MICA: Multi-granularity intertemporal credit assignment for long-horizon emotional support dialogues.](#) *Preprint*, arXiv:2603.06194.
- Ningning Zhang, Xingxing Yang, Zhizhong Tan, Weiping Deng, and Wenyong Wang. 2026e. [HiMem: Hierarchical long-term memory for LLM long-horizon agents.](#) *arXiv preprint arXiv:2601.06377*.
- Pingyue Zhang, Zihan Huang, Yue Wang, Jieyu Zhang, Letian Xue, Zihan Wang, Qineng Wang, Keshigeyan Chandrasegaran, Ruohan Zhang, Yejin Choi, Ranjay Krishna, Jiajun Wu, Li Fei-Fei, and Manling Li. 2026f. [Theory of space: Can foundation models construct spatial beliefs through active exploration?](#) *arXiv preprint arXiv:2602.07055*.
- Qizheng Zhang, Changran Hu, Shubhangi Upasani, Boyuan Ma, Fenglu Hong, Vamsidhar Kamanuru, Jay Rainton, Chen Wu, Mengmeng Ji, Hanchen Li, Urmish Thakker, James Zou, and Kunle Olukotun. 2025g. [Agentic context engineering: Evolving contexts for self-improving language models.](#) *arXiv preprint arXiv:2510.04618*.
- Ruoxuan Zhang, Qiyun Zheng, Zhiyu Zhou, Ziqi Liao, Siyu Wu, Jian-Yu Jiang-Lin, Bin Wen, Hongxia Xie, Jianlong Fu, and Wen-Huang Cheng. 2025h. [MindPower: Enabling Theory-of-Mind reasoning in VLM-based embodied agents.](#) In *Proceedings of CVPR 2026*.
- Shenao Zhang, Yaqing Wang, Yinxiao Liu, Tianqi Liu, Peter Grabowski, Eugene Ie, Zhaoran Wang, and Yunxuan Li. 2025i. [Beyond markovian: Reflective](#)

- exploration via bayes-adaptive RL for LLM reasoning. *arXiv preprint arXiv:2505.20561*.
- Wenlin Zhang, Xiangyang Li, Kuicai Dong, Yichao Wang, Pengyue Jia, Xiaopeng Li, Yingyi Zhang, Derong Xu, Zhaocheng Du, Huifeng Guo, Ruiming Tang, and Xiangyu Zhao. 2025j. [Process vs. outcome reward: Which is better for agentic RAG reinforcement learning](#). In *Proceedings of NeurIPS 2025*.
- Wentao Zhang, Lingxuan Zhao, Haochong Xia, Shuo Sun, Jiaze Sun, Molei Qin, Xinyi Li, Yuqing Zhao, Yilei Zhao, Xinyu Cai, Longtao Zheng, Xinrun Wang, and Bo An. 2024a. A multi-modal foundation agent for financial trading: Tool-augmented, diversified, and generalist. In *KDD 2024*. ArXiv:2402.18485.
- Xinliang Frederick Zhang, Nick Beauchamp, and Lu Wang. 2025k. [PRIME: Large language model personalization with cognitive dual-memory and personalized thought process](#). In *Proceedings of EMNLP 2025*.
- Xuan Zhang, Wenxuan Zhang, See-Kiong Ng, and Yang Deng. 2026g. [Self-evolving world models for LLM agent planning](#). *Preprint*, arXiv:2606.30639.
- Xuan Zhang, Zhijian Zhou, Lingfeng Qiao, Yulei Qin, Ke Li, Xing Sun, Xiaoyu Tan, Chao Qu, and Yuan Qi. 2026h. [Internalizing the future: A unified agentic training paradigm for world model planning](#). *Preprint*, arXiv:2606.27483.
- Yaocheng Zhang, Haohuan Huang, Zijun Song, Yuanheng Zhu, Qichao Zhang, Zijie Zhao, and Dongbin Zhao. 2025l. [CriticSearch: Fine-grained credit assignment for search agents via a retrospective critic](#). In *Proceedings of ACL 2026*.
- Yixin Zhang, Ziyi Wang, Yiming Rong, Haoxi Wang, Jinling Jiang, Shuang Xu, Haoran Wu, Shiyu Zhou, and Bo Xu. 2026i. [World models for policy refinement in StarCraft II](#). *arXiv preprint arXiv:2602.14857*.
- Youyuan Zhang, Siyuan Li, Fangming Liu, and Jing Li. 2026j. [Toward robust personalized alignment for LLMs: Mitigating persona drift in multi-turn dialogue](#). In *Findings of the Association for Computational Linguistics: EMNLP 2026*.
- Yudi Zhang, Meng Fang, Zhenfang Chen, and Mykola Pechenizkiy. 2026k. [Self-evolving LLM agents with in-distribution optimization](#). *Preprint*, arXiv:2606.07367.
- Zeyu Zhang, Xiaohe Bo, Chen Ma, Rui Li, Xu Chen, Quanyu Dai, Jieming Zhu, Zhenhua Dong, and Ji-Rong Wen. 2024b. A survey on the memory mechanism of large language model based agents. *arXiv preprint arXiv:2404.13501*.
- Zeyu Zhang, Quanyu Dai, Luyu Chen, Zeren Jiang, Rui Li, Jieming Zhu, Xu Chen, Yi Xie, Zhenhua Dong, and Ji-Rong Wen. 2024c. [MemSim: A Bayesian simulator for evaluating memory of LLM-based personal assistants](#). In *Proceedings of NeurIPS 2024*.
- Andrew Zhao, Daniel Huang, Quentin Xu, Matthieu Lin, Yong-Jin Liu, and Gao Huang. 2023a. [ExpeL: LLM agents are experiential learners](#). In *AAAI 2024*. ArXiv:2308.10144.
- Haotian Zhao, Songlin Zhou, Yuxin Zhang, Stephen S. T. Yau, Wenyu Zhang, Lun Tian, Tianshu Zhu, Yifeng Huang, Yucheng Zeng, Jingnan Gu, Daxiang Dong, and Jianmin Wu. 2026a. [AEM: Adaptive entropy modulation for multi-turn agentic reinforcement learning](#). *Preprint*, arXiv:2605.00425.
- Qing Zhao, Haowei Li, Weijian Deng, Pengxu Wei, and Liang Lin. 2026b. [EvoSCM: Scientific belief revision through causal model evolution and experimentation](#). *Preprint*, arXiv:2609.01526.
- Siyan Zhao, Mingyi Hong, Yang Liu, Devamanyu Hazarika, and Kaixiang Lin. 2025. [Do LLMs recognize your preferences? evaluating personalized preference following in LLMs](#). In *Proceedings of ICLR 2025*.
- Yubao Zhao, Weiquan Huang, Sudong Wang, Ruochen Zhao, Chen Chen, Yao Shu, and Chengwei Qin. 2026c. [BranPO: Scalable contrastive branch sampling for long-horizon agentic reinforcement learning](#). *arXiv preprint arXiv:2602.03719*.
- Zirui Zhao, Wee Sun Lee, and David Hsu. 2023b. Large language models as commonsense knowledge for large-scale task planning. In *NeurIPS 2023*. ArXiv:2305.14078.
- Congmin Zheng, Jiachen Zhu, Zhuoying Ou, Yuxiang Chen, Kangning Zhang, Rong Shan, Zeyu Zheng, Mengyue Yang, Jianghao Lin, Yong Yu, and Weinan Zhang. 2025a. A survey of process reward models: From outcome signals to process supervisions for large language models. *arXiv preprint arXiv:2510.08049*.
- Jingsheng Zheng, Jintian Zhang, Yujie Luo, Yuren Mao, Yunjun Gao, Lun Du, Huajun Chen, and Ningyu Zhang. 2026a. [Can we predict before executing machine learning agents?](#) In *Proceedings of ACL 2026*.
- Junhao Zheng, Xidi Cai, Qiuqi Li, Duzhen Zhang, Zhongzhi Li, Yingying Zhang, Le Song, and Qianli Ma. 2025b. [LifelongAgentBench: Evaluating LLM agents as lifelong learners](#). *arXiv preprint arXiv:2505.11942*.
- Longtao Zheng, Rundong Wang, Xinrun Wang, and Bo An. 2023. [Synapse: Trajectory-as-Exemplar prompting with memory for computer control](#). In *Proceedings of ICLR 2024*.

- Yuhao Zheng, Li'an Zhong, Yi Wang, Rui Dai, Kaikui Liu, Xiangxiang Chu, Linyuan Lv, Philip Torr, and Kevin Qinghong Lin. 2026b. [Code2World: A GUI world model via renderable code generation](#). *arXiv preprint arXiv:2602.09856*.
- Yuxiang Zheng, Dayuan Fu, Xiangkun Hu, Xiaojie Cai, Lyumanshan Ye, Pengrui Lu, and Pengfei Liu. 2025c. [DeepResearcher: Scaling deep research via reinforcement learning in real-world environments](#). In *Proceedings of EMNLP 2025*.
- Yuyao Zheng, Haipeng Sun, Junwei Bao, Lemao Liu, Hongfei Jiang, Yang Song, and Dejing Dou. 2026c. [PGPO: Potential-guided policy optimization for multi-turn agentic tasks](#). *Preprint*, arXiv:2609.02236.
- Jike Zhong, Yuxiang Lai, Ming Li, Yuheng Li, Wuao Liu, Behzad Dariush, Konstantinos Psounis, and Shao-Yuan Lo. 2026a. [From shortcuts to reasoning: Robust post-training of theory of mind with reinforcement learning](#). In *Proceedings of ICML 2026*.
- Qimin Zhong, Hao Liao, Haiming Qin, Mingyang Zhou, Rui Mao, Wei Chen, and Naipeng Chao. 2026b. [Toward consistent world models with multi-token prediction and latent semantic enhancement](#). In *Proceedings of ACL 2026*.
- Wanjun Zhong, Lianghong Guo, Qiqi Gao, He Ye, and Yanlin Wang. 2023a. [MemoryBank: Enhancing large language models with long-term memory](#). In *AAAI 2024*. ArXiv:2305.10250.
- Wanjun Zhong, Lianghong Guo, Qiqi Gao, He Ye, and Yanlin Wang. 2024. [MemoryBank: Enhancing large language models with long-term memory](#). In *Proceedings of the AAAI Conference on Artificial Intelligence*.
- Zexuan Zhong, Zhengxuan Wu, Christopher D. Manning, Christopher Potts, and Danqi Chen. 2023b. [MQuAKE: Assessing knowledge editing in language models via multi-hop questions](#). In *Proceedings of EMNLP 2023*.
- Andy Zhou, Kai Yan, Michal Shlapentokh-Rothman, Haohan Wang, and Yu-Xiong Wang. 2023a. [Language agent tree search unifies reasoning acting and planning in language models](#). In *ICML 2024*. ArXiv:2310.04406.
- Huichi Zhou, Yihang Chen, Siyuan Guo, Xue Yan, Kin Hei Lee, Zihan Wang, Ka Yiu Lee, Guchun Zhang, Kun Shao, Linyi Yang, and Jun Wang. 2025a. [Memento: Fine-tuning LLM agents without fine-tuning LLMs](#). *Preprint*, arXiv:2508.16153.
- Pei Zhou, Aman Madaan, Srividya Pranavi Potharaju, Aditya Gupta, Kevin R. McKee, Ari Holtzman, Jay Pujara, Xiang Ren, Swaroop Mishra, Aida Nematzadeh, Shyam Upadhyay, and Manaal Faruqi. 2023b. [How FaR are large language models from agents with theory-of-mind?](#) In *ICLR 2024*. ArXiv:2310.03051.
- Runlong Zhou, Simon S. Du, and Beibin Li. 2024a. [Reflect-RL: Two-player online RL fine-tuning for LMs](#). In *ACL 2024*. ArXiv:2402.12621.
- Shuaiyu Zhou, Fengpeng Yue, Zengjie Hu, Yuanzhe Shen, Chenyang Zhang, feng hong, Cao Liu, and Ke Zeng. 2026. [ToolVerse: Unlocking massive environments and long-horizon tasks for agentic reinforcement learning](#). *arXiv preprint arXiv:2607.15660*.
- Shuyan Zhou, Frank F. Xu, Hao Zhu, Xuhui Zhou, Robert Lo, Abishek Sridhar, Xianyi Cheng, Tianyue Ou, Yonatan Bisk, Daniel Fried, Uri Alon, and Graham Neubig. 2023c. [WebArena: A realistic web environment for building autonomous agents](#). In *Proceedings of ICLR 2024*.
- Wangchunshu Zhou, Yuchen Eleanor Jiang, Peng Cui, Tiannan Wang, Zhenxin Xiao, Yifan Hou, Ryan Cotterell, and Mrinmaya Sachan. 2023d. [Recurrent-GPT: Interactive generation of \(arbitrarily\) long text](#). *Preprint*, arXiv:2305.13304.
- Xuhui Zhou, Jiarui Liu, Akhila Yerukola, Hyunwoo Kim, and Maarten Sap. 2025b. [Social world models](#). *arXiv preprint arXiv:2509.00559*.
- Yifei Zhou, Song Jiang, Yuandong Tian, Jason Weston, Sergey Levine, Sainbayar Sukhbaatar, and Xian Li. 2025c. [SWEET-RL: Training multi-turn LLM agents on collaborative reasoning tasks](#). *Preprint*, arXiv:2503.15478.
- Yifei Zhou, Andrea Zanette, Jiayi Pan, Sergey Levine, and Aviral Kumar. 2024b. [ArCher: Training language model agents via hierarchical multi-turn RL](#). In *ICML 2024*. ArXiv:2402.19446.
- Zijian Zhou, Ao Qu, Zhaoxuan Wu, Sunghwan Kim, Alok Prakash, Daniela Rus, Jinhua Zhao, Bryan Kian Hsiang Low, and Paul Pu Liang. 2025d. [MEM1: Learning to synergize memory and reasoning for efficient long-horizon agents](#). *Preprint*, arXiv:2506.15841.
- Kehang Zhu, Anand Shah, and David Parkes. 2026. [Engineering simplicity: Simple mechanism interfaces steer LLM agents](#). *arXiv preprint arXiv:2609.36365*.
- Wentao Zhu, Zhining Zhang, and Yizhou Wang. 2024. [Language models represent beliefs of self and others](#). In *Proceedings of ICML 2024*.
- Deyu Zou, Yongqiang Chen, Jianxiang Wang, Haochen Yang, Mufei Li, James Cheng, Pan Li, and Yu Gong. 2025. [Reducing belief deviation in reinforcement learning for active reasoning](#). In *Proceedings of ICLR 2026*.
- Yuxin Zuo, Zikai Xiao, Li Sheng, Fei Huang, Jianhong Tu, Yuxuan Liu, Tianyi Tang, Xiaomeng Hu, Yang Su, Qingfeng Lan, Yantao Liu, Qin Zhu, Yinger Zhang, Bowen Yu, Haiquan Zhao, Haiyang Xu, Jianxin Yang, Jiayang Cheng, Junyang Wang, and

14 others. 2026. [Qwen-AgentWorld: Language world models for general agents](#). *arXiv preprint arXiv:2606.24597*.

## A Paper Collection Protocol

The search is scripted and rerunnable; the configuration (including every query string and term list), raw hits, per-stage counts, and the full-text coding record for the runs reported here (identifiers 20260915\_v1 for 2025–2026 and 20260916\_v3 for the 2020–2026 window) are released with the paper.

**Queries.** Twenty-three keyword queries were defined in the first run, four to five per survey section; a second run added eleven queries on evidence acquisition and theory-of-mind belief modelling (34 in total): representation (Section 4), revision (Section 5), learning (Section 6), evidence (Section 7), and evaluation (Section 8).

**Sources.** Source A queried OpenAlex across all venues with publication date from 1 January 2025, 50 results per query. Source B queried OpenAlex restricted to the arXiv source with publication date from 1 January 2026, 50 results per query, to improve recall of preprints. The arXiv API itself was rate-limited during the run; the OpenAlex mirror of arXiv lags by a few days.

**Screening.** Records were deduplicated by identifier and then by normalised title, so that a preprint and its journal version count once. Rule-based screening on title and abstract applied one inclusion criterion for an LLM-related term (I1), one exclusion criterion for off-domain applications (E1), and one inclusion criterion requiring at least two topic terms from a fixed list (I2). Records without an abstract of at least 200 characters were excluded. Included records were ranked by a relevance score used only for ordering the manual full-text screen, never for inclusion.

**Second and third runs.** The second run (34 queries, same sources and rules) returned 834 unique records, 338 after screening, of which 96 were published in 2026; only two of those had not already been read in the first run, which we take as evidence that the first run had saturated the keyword neighbourhood. The third run extended the window to 1 January 2020 with 100 results per query and returned 4,471 unique records, 3,999 after title deduplication and 2,079 after rule screening. Keyword ranking is dominated by generic

**Table 7 | Paper-collection counts for run 20260915\_v1.**

| Stage                                   | Count         |
|-----------------------------------------|---------------|
| Queries                                 | 23            |
| Source A raw hits / unique              | 1,150 / 715   |
| Source B raw hits / unique              | 1,150 / 1,008 |
| Merged unique (identifier)              | 1,685         |
| After title-level deduplication         | 1,460         |
| Excluded: no or short abstract          | 45            |
| Excluded I1: no LLM term                | 160           |
| Excluded E1: off-domain                 | 35            |
| Excluded I2: fewer than two topic terms | 357           |
| Included after rule screening           | 863           |
| of which published in 2026              | 656           |
| Full-text screened                      | 117           |
| of which included after reading         | 106           |

high-citation papers before 2025, so the 2020–2025 foundations were enumerated as a curated list of 90 titles resolved through OpenAlex for venue and citation data (Table 17); 75 of them were read in full and 73 included.

**Citation-tree expansion.** Keyword search cannot reach papers that describe belief maintenance in other words, so the included set was expanded by snowballing over the citation graph. Each round took the papers newly included in the previous round as seeds, fetched their reference lists (backward) and the papers citing them (forward) from the Semantic Scholar Graph API, kept candidates published from 2020, removed everything already screened, and applied the same rule-based screen as the keyword runs. Because forward citation of recent preprints is dense, the rule-included candidates were sampled for full-text reading by a fixed rule (topic score  $\geq 10$ , or the word *belief* in title or abstract, or linked to at least three seeds, or a world-model paper from 2025 with score  $\geq 7$ ), and every sampled paper was read and coded with the same fields as the keyword runs. Round one was seeded by the 179 keyword-included papers (178 after the author check) and each later round by the papers the previous round added; the full-text sample shrank from 302 to 115, 83, and 26 across four rounds, and the expansion stopped after the round whose sample fell below thirty, the stopping rule fixed in advance (Table 8).

**Fifth round, targeted.** After the four rounds above, we found a method the protocol had not reached (ReBel; Tang et al., 2026b), posted after most of our seeds. We treated it as a seed

**Table 8 | Citation-tree expansion by round.**

| Stage                         | R1     | R2     | R3    | R4    |
|-------------------------------|--------|--------|-------|-------|
| Seeds                         | 178    | 165    | 55    | 45    |
| Backward references           | 9,127  | 8,601  | 2,673 | 2,406 |
| Forward citations             | 9,999  | 9,999  | 1,403 | 441   |
| Unique candidates (2020–)     | 12,872 | 13,113 | 2,536 | 1,465 |
| Already screened              | 302    | 1,284  | 605   | 465   |
| Excluded by rules             | 8,448  | 7,401  | 1,300 | 747   |
| Included after rule screening | 4,122  | 4,428  | 631   | 253   |
| Sampled for full-text reading | 302    | 115    | 83    | 26    |
| Included after reading        | 171    | 58     | 47    | 7     |
| of which peer-reviewed        | 79     | 23     | 22    | 2     |

Forward citations are capped at 9,999 per request batch by the API.

for a fifth, targeted round: the Semantic Scholar batch endpoint was rate-limited throughout, so we used ReBel’s own reference list (46 entries), an OpenAlex search on belief-consistency and belief-level supervision (85 candidates), and a Hugging Face Papers search on nine related phrases (743 candidates, 211 after the rule screen and deduplication). Full-text screening added 23 papers (3 peer-reviewed), of which four score a belief or a prediction against the agent’s own observations (two score a written belief: ReBel and ABBEL); every paper whose signal source mattered to a claim was re-read in full. One of the 23 (PABU) had passed the rule screen in the third keyword run but was not sampled. The additions are marked R5 in the released record and in Table 9.

**Sixth round.** A final targeted round added four papers found after the fifth round, two posted after the keyword and citation runs, Jev-Mem and REFLEX (Jiang et al., 2026a; Wu and Lim, 2026), and two those runs did not reach, FLAG-TRADER and Conv-FinRe (Xiong et al., 2025a; Wang et al., 2026i). They were read in full, coded with the same fields, and are counted in the totals, the sensitivity analysis, and the master table (column R6 of Table 9).

**Update run.** A final update run repeated the OpenAlex and Hugging Face Papers queries and rule screen for September 2026. It returned 1,051 unique records and 434 rule-screened candidates, of which 47 were read in full and 29 included (3 peer-reviewed): among them two surveys that formalise a Bayes-filter belief for one setting (Cai et al., 2026; Tang et al., 2026c) and three methods that credit a probabilistic belief against a known answer (Zhang et al., 2026j; Wang et al., 2026e; Hu et al., 2026a). They are counted throughout

(column U1 of Table 9), and no insight changes direction (Table 11).

**Year distribution.** Table 9 gives the included papers by year, source, and whether peer-reviewed. The share that is peer-reviewed falls with recency, from four in five before 2025 to about one in nine for 2026.

**Table 9 | Included papers by publication year and by the run that found them (keyword search, citation-tree or targeted round, or update run).**

| Year  | Keyword | R1  | R2 | R3 | R4 | R5 | R6 | U1 | Total | Peer-rev. |
|-------|---------|-----|----|----|----|----|----|----|-------|-----------|
| 2020  | 1       | 3   | 1  | 0  | 0  | 0  | 0  | 0  | 5     | 4         |
| 2021  | 2       | 2   | 1  | 0  | 0  | 0  | 0  | 0  | 5     | 3         |
| 2022  | 9       | 4   | 0  | 1  | 0  | 0  | 0  | 0  | 14    | 13        |
| 2023  | 27      | 21  | 2  | 0  | 0  | 1  | 0  | 0  | 51    | 41        |
| 2024  | 16      | 20  | 9  | 4  | 1  | 1  | 0  | 0  | 51    | 40        |
| 2025  | 18      | 69  | 20 | 20 | 2  | 5  | 1  | 0  | 135   | 51        |
| 2026  | 105     | 52  | 25 | 22 | 4  | 16 | 3  | 29 | 256   | 29        |
| Total | 178     | 171 | 58 | 47 | 7  | 23 | 4  | 29 | 517   | 181       |

**Evidence level.** Of the 517 papers included after full-text reading and the author check (178 from keyword search, 283 from citation-tree expansion, 23 from the targeted fifth round, 4 from the sixth, and 29 from the update run), 181 are peer-reviewed and 336 are preprints; every number quoted in the text is the paper’s own headline figure, taken from the abstract or the reported table.

**Full-text screening.** A paper passed full-text screening if it (a) concerns a language-model agent or a component used by one, and (b) constructs, revises, tests, learns, or evaluates an explicit state carried between decisions, or documents a failure of such a state; benchmarks and surveys that do (b) were kept and coded as such. Each paper read in full was recorded with the fields: who maintains the belief (source of the belief state), credited object, revision signal (source of the revision and test), whether belief-level metrics are reported, survey section, and role. The record is the source of Fig. 1, Tables 12–15, and Fig. 5. Screening and coding were carried out with large-language-model agents: each sampled paper was read and coded against these fields under the fixed codebook above, and every paper whose coding supports a claim in the text was re-read in full. The authors defined the framework, the inclusion criteria, and the codebook, and are responsible for the record. An author then checked every coding

by hand against the full text: of 2,590 field values on 518 papers, four were changed, one paper (a survey) was excluded and two belief-maintainer codes were corrected, giving agreement of 99.8% on inclusion, 99.6% on who maintains the belief, and 100% on credited object, revision signal, and metric. A system counts as testing a written belief against the agent’s own observations only if the model writes the belief and the signal that scores it is computed from observations the agent itself receives, with no oracle, simulator state, or teacher; the author marked this for every paper, and only ReBel and ABBEL qualify (column `own_obs_test` in the record). The score must also train the belief. Table 10 shows how the count moves when this reading is made stricter or looser.

**Table 10 | Sensitivity of the own-observation count to the definition.**

| Reading (each row adds to the one above)                                                                                                                                                                   | All | P only |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--------|
| Stricter: the next observation only (ReBel)                                                                                                                                                                | 1   | 0      |
| Ours: next or reconstructed own observation (+ ABBEL)                                                                                                                                                      | 2   | 0      |
| + written next-observation predictions trained but not carried as the belief (Wang, 2026a)                                                                                                                 | 3   | 0      |
| + written hypotheses scored by own observations at inference time, without training (Cross et al., 2024; Zhao et al., 2026b)                                                                               | 5   | 0      |
| + learned world models trained on next observations (Chae et al., 2024; Li et al., 2025f; Chen et al., 2025e; Yu et al., 2025c; Wu et al., 2025a; Zuo et al., 2026; Lu et al., 2026a; Zhang et al., 2026i) | 13  | 4      |

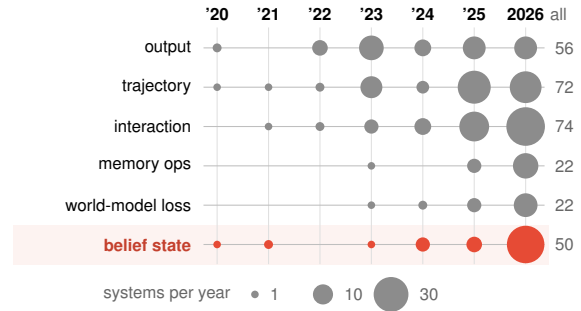
No row contains a written belief that carries uncertainty. Forecast-Dojo keeps a written probabilistic belief but scores it by event resolution for evaluation and trains on teacher trajectories, so it enters no row (Ye et al., 2026).

**Peer-reviewed subset.** Because 336 of the 517 papers are preprints, Table 11 recomputes the counts behind each insight box on the 181 peer-reviewed papers alone.

**Table 11 | Sensitivity of the five insights to preprint reliance: the counts each rests on, over all included papers and over the peer-reviewed subset alone.**

| # | Counts the insight rests on                       | All          | P only      |
|---|---------------------------------------------------|--------------|-------------|
| 1 | belief state written / filter / world model       | 70 / 27 / 55 | 21 / 8 / 19 |
| 2 | revision papers (§5)                              | 122          | 37          |
| 3 | credited: interaction / state / state by own obs. | 96 / 50 / 2  | 33 / 13 / 0 |
| 4 | acquisition papers (§7)                           | 66           | 18          |
| 5 | papers with a belief-level metric                 | 105          | 34          |

Every insight is a claim about a proportion or an absence. Insights 1, 2, 4, and 5 hold in the same direction on the peer-reviewed subset; Insight 3 rests on two preprints (ReBel, ABBEL), and its own-observation cell is empty there.



**Fig. 6 | Where credit has gone, 2020–2026.** Bubble area counts systems crediting each object per year; the right column gives totals (Table 16). Another 221 train nothing. Credit reached the belief state mostly in 2026, and only ReBel and ABBEL take its test from the agent’s own observations.

## B Additional Tables and Figures

This appendix collects the full tables and figures that the main text refers to.

## C Timeline of Learning Signals by Credited Object

**Table 12 | The belief state by who maintains it, with the revision and test each system carries.**

| Class                | Method                                               | State written by model | Append-only | Uncertainty   | Test (TEST)         | Revision (REVISE) | Domain               |
|----------------------|------------------------------------------------------|------------------------|-------------|---------------|---------------------|-------------------|----------------------|
| Context as state     | ReAct (Yao et al., 2023b)                            | ✓                      | ✓           | X             | X                   | X                 | general              |
| External memory      | MemGPT (Packer et al., 2023)                         | X                      | X           | X             | X                   | store update      | dialogue             |
| Probabilistic memory | BeliefMem (Liao et al., 2026a)                       | X                      | X           | prob.         | X                   | noisy-OR          | LoCoMo, ALFWWorld    |
|                      | D-MEM (Song and Xin, 2026)                           | X                      | X           | RPE           | X                   | gated write       | LoCoMo               |
| External filter      | Belief-State Engine (Chattopadhyay and Halder, 2026) | X                      | X           | posterior     | ✓                   | Bayes             | Tiger, security      |
|                      | Belief at Risk (Dixon, 2026b)                        | X                      | X           | posterior     | ✓                   | Bayes             | equities             |
| Model-written        | Agent-BRACE (Singh et al., 2026b)                    | ✓                      | X           | ordinal       | X                   | RL                | ALFWWorld-style      |
|                      | Linguistic forecaster (Murphy, 2026)                 | ✓                      | ✓           | numeric       | ✓                   | sequential        | ForecastBench        |
|                      | BOND (Cui and Lin, 2026)                             | ✓                      | ✓           | posterior tag | X                   | distilled         | negotiation          |
|                      | CBM (Xu et al., 2026a)                               | ✓                      | ✓           | X             | ✓(symbolic)         | RL                | closed world         |
|                      | EvoSCM (Zhao et al., 2026b)                          | ✓                      | ✓           | population    | ✓(interventions)    | rule distillation | science              |
|                      | ReBel (Tang et al., 2026b)                           | ✓                      | X           | X             | ✓(next observation) | RL                | ALFWWorld, WebShop   |
|                      | ABBEL (Lidayan et al., 2025)                         | ✓                      | X           | X             | reconstruction      | RL                | text games, ColBench |
|                      | BB-WM (Kumar et al., 2026)                           | X                      | X           | latent        | ✓                   | learned           | general              |

TEST: the belief commits to a testable prediction.

**Table 13 | Belief revision methods and benchmarks by failure mode addressed, revision trigger, and where the signal is projected.**

| Method                             | Failed stay | Failed update | Failed isolation | Failed act | Trigger             | Signal projected to |
|------------------------------------|-------------|---------------|------------------|------------|---------------------|---------------------|
| Reflexion (Shinn et al., 2023)     |             | ✓             |                  |            | task failure        | whole context       |
| BeliefShift (Myakala et al., 2026) | ✓           | ✓             |                  |            | user evidence       | (benchmark)         |
| STALE (Chao et al., 2026)          |             | ✓             |                  | ✓          | implicit conflict   | (benchmark)         |
| CBM (Xu et al., 2026a)             | ✓           | ✓             | ✓                |            | symbolic check      | belief state        |
| TOKI (Wang, 2026b)                 |             | ✓             |                  |            | contradiction       | memory rows         |
| MemOps (Hao et al., 2026)          |             | ✓             | ✓                |            | lifecycle op        | (benchmark)         |
| D-MEM (Song and Xin, 2026)         |             | ✓             | ✓                |            | reward pred. error  | memory path         |
| STOCKTAKE (Deb and Krishnan, 2026) |             | ✓             |                  | ✓          | oracle mismatch     | (benchmark)         |
| Agent-BRACE (Singh et al., 2026b)  |             | ✓             |                  | ✓          | certainty           | belief model        |
| Werewolf audit (Gao et al., 2026b) |             | ✓             |                  | ✓          | external belief log | (benchmark)         |

**Table 14 | Learning signals by credited object, with the source of the test that signs them.**

| Level       | Representative methods                                                                               | Test from                           | Granularity                     | Signed by returned $r_{t+1}$ | Delayed evidence | Partial obs. |
|-------------|------------------------------------------------------------------------------------------------------|-------------------------------------|---------------------------------|------------------------------|------------------|--------------|
| Output      | SFT, RLHF, DPO (Ouyang et al., 2022; Rafailov et al., 2023)                                          | demos, preferences                  | response                        | X                            | X                | X            |
| Trajectory  | PPO, GRPO, PRMs (Schulman et al., 2017; Shao et al., 2024; Lightman et al., 2024)                    | verifier, PRM                       | step / rollout                  | X                            | X                | X            |
| Interaction | HIPER, G*PO, ECPPO, IAPO (Peng et al., 2026a; Wang et al., 2026; Li et al., 2026; Ren et al., 2026a) | environment                         | turn / action                   | X                            | partial          | X            |
|             | Memory-R2, TAMTRL, EvoHarness-RL (Yan et al., 2026a; Wang et al., 2026b; Ning et al., 2026)          | environment, teacher                | memory operation                | X                            | partial          | ✓            |
| State       | CBM (Xu et al., 2026a)                                                                               | symbolic belief reward              | belief state                    | closed world                 | ✓                | ✓            |
|             | Agent-BRACE (Singh et al., 2026b)                                                                    | joint RL, calibration               | claim + certainty               | calibration target           | ✓                | ✓            |
|             | BOND (Cui and Lin, 2026)                                                                             | Bayesian teacher                    | posterior tag                   | teacher                      | X                | ✓            |
|             | CORE (Zhang et al., 2026)                                                                            | gold slot values                    | probabilistic slot belief       | known answer                 | X                | ✓            |
|             | GAVEL (Wang et al., 2026e)                                                                           | held-out placements                 | posterior over object locations | known answer                 | X                | ✓            |
|             | ReBel (Tang et al., 2026b)                                                                           | own next observation                | predicate belief                | ✓                            | ✓                | ✓            |
|             | ABBEL (Lidayan et al., 2025)                                                                         | reconstruction of prior observation | natural-language belief         | X(backward)                  | ✓                | ✓            |
|             | PaW (Lu et al., 2026a)                                                                               | own next observation                | prediction on the policy        | ✓                            | X                | ✓            |

**Table 15 | Benchmarks that exercise belief maintenance, by whether the evaluator can observe the latent state and what belief-level quantity is reported.**

| Benchmark                                                               | Domain                  | Horizon           | Latent state            | Observable to evaluator | Delayed evidence | Belief-level quantity reported                           |
|-------------------------------------------------------------------------|-------------------------|-------------------|-------------------------|-------------------------|------------------|----------------------------------------------------------|
| ALFWWorld (Shridhar et al., 2020)                                       | text + embodied         | episodic          | household state         | ✓                       | partial          | none (task success)                                      |
| PlanBench (Valmeekam et al., 2022)                                      | classical planning      | plan-length       | PDDL state              | ✓                       | X                | plan verification, replanning                            |
| AgentBench (Liu et al., 2023b)                                          | 8 environments          | episodic          | environment state       | ✓                       | partial          | none (task success)                                      |
| SmartPlay (Wu et al., 2023)                                             | 6 games                 | episodic          | game state              | ✓                       | partial          | none (9 capability scores)                               |
| T4D / FaR (Zhou et al., 2023b)                                          | theory of mind          | single-turn       | others' beliefs         | ✓(annotated)            | X                | belief tracking vs. action selection                     |
| False-belief tasks (Kosinski, 2023)                                     | theory of mind          | single-turn       | others' beliefs         | ✓(annotated)            | X                | false-belief accuracy                                    |
| LoCoMo (Maharana et al., 2024)                                          | long conversations      | 35 sessions       | conversation facts      | annotated               | ✓                | temporal QA (proxy)                                      |
| $\tau$ -bench / $\tau^2$ -bench (Yao et al., 2024; Barres et al., 2025) | tool-user interaction   | multi-turn        | database state          | ✓                       | ✓                | pass/k consistency; reasoning vs. communication errors   |
| MemoryAgentBench (Hu et al., 2025c)                                     | incremental multi-turn  | long-horizon      | memory contents         | annotated               | ✓                | retrieval, test-time learning, conflict resolution       |
| BeliefTrack (Xu et al., 2026a)                                          | closed-world reasoning  | multi-turn        | finite belief space     | ✓(symbolic)             | ✓                | turn-level belief accuracy; failed stay/update/isolation |
| DeltaLogic (Dhanda, 2026)                                               | logical reasoning       | 2 steps           | premise set             | ✓                       | ✓                | revision accuracy vs. initial accuracy                   |
| DTOM-Track (Nguyen et al., 2026)                                        | dialogue                | multi-turn        | interlocutor belief     | ✓(annotated)            | ✓                | prior-belief recall; change detection                    |
| BeliefShift (Myakala et al., 2026)                                      | user modelling          | multi-session     | user opinion            | annotated               | ✓                | revision accuracy; drift coherence; evidence sensitivity |
| OmniToM (Bawatneh et al., 2026)                                         | theory of mind          | narrative         | actors' beliefs         | ✓(annotated)            | X                | belief-proposition extraction; schema labels             |
| MINDGAMES (Wang et al., 2026c)                                          | social games            | repeated games    | hidden roles, opponents | ✓                       | ✓                | TrueSkill; turn-level logs                               |
| MEDLEY-BENCH (Abtahi et al., 2026)                                      | social pressure         | 2 rounds          | —                       | X                       | X                | sensitivity to manipulated consensus                     |
| STALE (Chao et al., 2026)                                               | memory validity         | multi-turn        | fact validity           | annotated               | ✓                | state resolution; premise resistance; policy adaptation  |
| MemOps (Hao et al., 2026)                                               | long conversations      | multi-session     | memory lifecycle        | annotated               | ✓                | operation-level probes                                   |
| GroupMemBench (Yang et al., 2026b)                                      | multi-party dialogue    | multi-session     | per-speaker knowledge   | annotated               | ✓                | information-update accuracy                              |
| MemGym (Xu et al., 2026c)                                               | mixed agent tasks       | long-horizon      | —                       | X                       | partial          | memory-isolated score delta                              |
| STOCKTAKE (Deb and Krishnan, 2026)                                      | inventory control       | weekly, long      | demand factors          | ✓(Bayes filter)         | ✓                | detection lag; skill vs. oracle; knowing-doing rate      |
| Flux (Alaswad et al., 2026)                                             | text game               | long-horizon      | simulator state         | ✓                       | ✓                | state-tracking error modes                               |
| YC-Bench (He et al., 2026b)                                             | startup simulation      | hundreds of turns | market, clients         | privileged outcome      | ✓                | scratchpad usage (proxy)                                 |
| RetailBench (Zhang et al., 2026c)                                       | retail POMDP            | 180 days          | demand, supply          | privileged oracle       | ✓                | evidence-acquisition behaviour (proxy)                   |
| EnterpriseArena (Han et al., 2026)                                      | lending                 | 132 months        | borrower pool           | privileged outcome      | ✓                | observation-budget allocation (proxy)                    |
| Business Arena (Pan et al., 2026a)                                      | marketplace             | long-horizon      | market                  | privileged outcome      | ✓                | action-level attribution                                 |
| Werewolf audit (Gao et al., 2026b)                                      | social deduction        | 9-player game     | roles                   | ✓                       | ✓                | action-belief consistency                                |
| CAGE-2 study (Bogdanov et al., 2026)                                    | cyber defence           | episodic          | network state           | ✓                       | partial          | state-abstraction return gain                            |
| MD5 state tracking (Pai and Xian, 2026)                                 | tool chains             | 196 calls         | registers               | ✓                       | X                | exact-state success                                      |
| FinBench (time-gated) (Ghosh and Devarakonda, 2026)                     | asset forecasting       | daily             | returns                 | realised                | ✓                | Brier; interval scores                                   |
| TradeArena (Xue, 2026)                                                  | trading                 | multi-week        | regimes                 | realised                | ✓                | representation drift (early warning)                     |
| HERCULEAN (Peng et al., 2026b)                                          | finance workflows       | multi-step        | market, books           | partial                 | ✓                | state-tracking sub-score                                 |
| FinalityBench (Sharma, 2026)                                            | financial operations    | multi-step        | canonical event log     | ✓                       | ✓                | none (effect-level loss)                                 |
| Forecast-Dojo (Ye et al., 2026)                                         | event forecasting       | dated timeline    | event outcomes          | ✓(resolution)           | ✓                | per-step Brier, log loss                                 |
| EnterpriseBench (Yang et al., 2026d)                                    | enterprise decisions    | interactive       | hidden information      | privileged              | ✓                | none (task level)                                        |
| Ledger memory benchmark (Srivastava, 2026a)                             | memory                  | multi-session     | principals' beliefs     | ✓(ledger)               | ✓                | expected belief state per principal                      |
| Werewolf accusations (Yang et al., 2026f)                               | social deduction        | single message    | player roles            | ✓                       | X                | prior suspicion and shift per accusation                 |
| Regent Chess (Joshi, 2026)                                              | board game              | game              | hidden royal piece      | ✓                       | ✓                | action-time Brier                                        |
| FinBen / InvestorBench (Xie et al., 2024; Li et al., 2024a)             | finance tasks / trading | task / episodic   | —                       | X                       | partial          | none (outcome only)                                      |

"Annotated" means human labels stand in for the latent state; "privileged" means the simulator knows it but the benchmark reports only outcomes.

**Table 16 | Systems with an explicit belief mechanism, by year.**

| System                                               | Year | State kept by | Credited    | Revision trigger                 | Metric |
|------------------------------------------------------|------|---------------|-------------|----------------------------------|--------|
| Schema-driven DST (Lee et al., 2021)                 | 2021 | written       | state       | dialogue history plus schema...  | ✓      |
| Verbalized Uncertainty (Lin et al., 2022)            | 2022 | written       | output      | fine-tuning on calibrated...     | ✓      |
| LMs Know (Kadavath et al., 2022)                     | 2022 | written       | output      | self-evaluation P and P of...    | ✓      |
| DreamerV3 (Halter et al., 2023)                      | 2023 | world model   | state       | RSSM latent posterior updated... | ○      |
| RecurrentGPT (Zhou et al., 2023d)                    | 2023 | written       | output      | LLM rewrites its own...          | ○      |
| LLM-MCTS (Zhao et al., 2023b)                        | 2023 | world model   | trajectory  | LLM-sampled world state...       | ○      |
| RAP (Hao et al., 2023)                               | 2023 | world model   | trajectory  | LLM-as-world-model predicted...  | ○      |
| Reflect-RL (Zhou et al., 2024a)                      | 2024 | written       | interaction | frozen reflection model...       | ○      |
| Hypothetical Minds (Cross et al., 2024)              | 2024 | written       | state       | hypotheses about other...        | ○      |
| Talker-Reasoner (Christakopoulou and Mourad, 2024)   | 2024 | written       | state       | Reasoner writes/updates...       | ○      |
| QUBE (Kim et al., 2024)                              | 2024 | written       | state       | question-answering over...       | ○      |
| MEM1 (Zhou et al., 2025d)                            | 2025 | written       | state       | internal state rewritten each... | ○      |
| MemAgent (Yu et al., 2025b)                          | 2025 | written       | state       | chunk-wise overwrite of...       | ○      |
| PCE (Seo et al., 2026)                               | 2026 | written       | state       | scenario likelihood scoring      | ○      |
| RB-VLA (Bagaria et al., 2026)                        | 2026 | world model   | state       | self-supervised world-model...   | ○      |
| REDEREF (Hosseini et al., 2026)                      | 2026 | prob. memory  | interaction | judge feedback updates...        | ○      |
| TAMTRL (Wang et al., 2026d)                          | 2026 | written       | trajectory  | teacher likelihood of memory...  | ○      |
| Self-Guide internal... (Wang et al., 2026g)          | 2026 | written       | trajectory  | self-generated guidance as...    | ○      |
| ReflectChain... (Luo, 2026)                          | 2026 | world model   | trajectory  | latent trajectory rehearsal...   | ○      |
| Cyclic subtask graphs (Gharzeddine and Saab, 2026)   | 2026 | written       | none        | state-analysis-and-routing...    | ○      |
| CHAL (Giovannelli and Kent, 2026)                    | 2026 | written       | state       | gradient-informed revision on... | ○      |
| Pinductor (Six et al., 2026)                         | 2026 | world model   | state       | belief-based likelihood score... | ○      |
| Belief Engine stance... (Yang et al., 2026c)         | 2026 | prob. memory  | state       | log-odds update with evidence... | ✓      |
| HIPIF (Diao et al., 2026)                            | 2026 | written       | trajectory  | subgoal completion +...          | ○      |
| ProPlay (Ma et al., 2026b)                           | 2026 | world model   | trajectory  | post-episode graph refinement... | ○      |
| POMDP validation... (Dixon, 2026c)                   | 2026 | ext. filter   | none        | Bayesian                         | ✓      |
| Internalizing the Future (Zhang et al., 2026h)       | 2026 | written       | trajectory  | foresight-conditioned RL...      | ○      |
| Hybrid-WM (Song and Cai, 2026a)                      | 2026 | world model   | state       | consistency gate                 | ✓      |
| Governance-aware POMDP... (Dixon, 2026a)             | 2026 | ext. filter   | none        | Bayesian update on evidence...   | ○      |
| BayesEvolve (Wu et al., 2026d)                       | 2026 | ext. filter   | state       | posterior refit after each...    | ✓      |
| WorldEvolver (Zhang et al., 2026g)                   | 2026 | world model   | state       | prediction-observation...        | ○      |
| WM-SAR (Song and Cai, 2026b)                         | 2026 | world model   | trajectory  | spectral error amplification     | ○      |
| BCO (Chen et al., 2026c)                             | 2026 | written       | state       | evaluation of new candidate...   | ✓      |
| EvoSCM causal belief... (Zhao et al., 2026b)         | 2026 | written       | state       | prediction-observation...        | ✓      |
| CAPTURE (Hossain et al., 2026)                       | 2026 | world model   | state       | uncertainty-triggered...         | ✓      |
| Semantic Bayesian World... (Soru, 2026)              | 2026 | ext. filter   | state       | Bayesian conditioning on...      | ○      |
| Belief-State Engine (Chattopadhyay and Hakder, 2026) | 2026 | ext. filter   | none        | Bayesian posterior               | ✓      |
| ReBel (Tang et al., 2026b)                           | 2026 | written       | state       | next observation (consistency)   | ○      |
| ABBEL (Lidayan et al., 2025)                         | 2026 | written       | state       | reconstruction of prior obs.     | ○      |
| PABU (Jiang et al., 2026c)                           | 2026 | written       | state       | teacher progress labels          | ○      |
| MARBO (Hwang et al., 2026)                           | 2026 | written       | state       | ground-truth roles               | ○      |
| CORE (Zhang et al., 2026i)                           | 2026 | prob. memory  | state       | gated slot revision              | ✓      |
| GAVEL (Wang et al., 2026e)                           | 2026 | ext. filter   | state       | online Bayesian update           | ✓      |
| BCA (Akbar et al., 2026)                             | 2026 | ext. filter   | none        | Beta update on appraised...      | ✓      |
| MoM / P-Mem (Qin and Lu, 2026)                       | 2026 | written       | none        | typed write-time operations      | ✓      |

State kept by: who maintains the belief (Section 4); Credited: the object that receives the learning signal under the criterion of Section 6, *none* if the belief is not trained; Metric: ✓ reports a belief-level quantity, ○ a proxy.

**Table 17 | Every paper included after full-text reading, ordered by year and first author.**

| System                                                       | Year | State kept by | Credited    | Revision trigger                             | Metric |
|--------------------------------------------------------------|------|---------------|-------------|----------------------------------------------|--------|
| GATA belief graphs (Adhikari et al., 2020)                   | 2020 | world model   | state       | graph updater infers and updates latent. . . | ○      |
| Text games with common sense (Dambekodi et al., 2020)        | 2020 | ext. memory   | trajectory  | commonsense inference (COMET/BERT) fill. . . | ○      |
| RAG (Lewis et al., 2020)                                     | 2020 | ext. memory   | output      | retrieved passages condition generation. . . |        |
| ToM and decision science typology (Rusch et al., 2020)       | 2020 | –             | –           | none                                         |        |
| ALFWorld (Shridhar et al., 2020)                             | 2020 | –             | output      | n/a (environment)                            |        |
| Decision Transformer (Chen et al., 2021)                     | 2021 | context       | interaction | return-to-go conditioning over past sta. . . |        |
| Do LMs have beliefs (Hase et al., 2021)                      | 2021 | context       | state       | belief-update objective (SLAG) applied. . .  | ✓      |
| BeliefBank (Kassner et al., 2021)                            | 2021 | ext. memory   | –           | weighted MaxSAT revises beliefs that cl. . . | ✓      |
| Schema-driven DST (Lee et al., 2021)                         | 2021 | written       | state       | dialogue history plus schema descriptio. . . | ✓      |
| WebGPT (Nakano et al., 2021)                                 | 2021 | context       | trajectory  | none (browsing history in context; rewa. . . |        |
| SayCan (Ahn et al., 2022)                                    | 2022 | –             | interaction | affordance value functions score skill. . .  |        |
| LMs as Agent Models (Andreas, 2022)                          | 2022 | –             | –           | n/a (position paper)                         |        |
| Towards Teachable Reasoning Syste. . . (Mishra et al., 2022) | 2022 | ext. memory   | –           | user corrections of erroneous model bel. . . | ○      |
| Inner Monologue (Huang et al., 2022b)                        | 2022 | context       | interaction | success detection, scene description, h. . . |        |
| LLMs Can Self-Improve (Huang et al., 2022a)                  | 2022 | –             | output      | self-consistency majority vote as pseud. . . |        |
| LMs (Mostly) Know (Kadavath et al., 2022)                    | 2022 | written       | output      | self-evaluation P(True) and P(IK) of ow. . . | ✓      |
| Verbalized Uncertainty (Lin et al., 2022)                    | 2022 | written       | output      | fine-tuning on calibrated verbal confid. . . | ✓      |
| HELM (Paischer et al., 2022)                                 | 2022 | ext. memory   | trajectory  | frozen LM compresses observation histor. . . |        |
| ILQL (Snell et al., 2022)                                    | 2022 | context       | trajectory  | none (offline returns from logged dialo. . . |        |
| PlanBench (Valmeekam et al., 2022)                           | 2022 | –             | output      | n/a (benchmark; IPC planning domains)        |        |
| Self-Consistency (Wang et al., 2022)                         | 2022 | –             | output      | marginalisation over sampled reasoning. . .  |        |
| Chain-of-Thought (Wei et al., 2022b)                         | 2022 | context       | output      | none (single forward pass; intermediate. . . |        |
| WebShop (Yao et al., 2022)                                   | 2022 | context       | –           | none (environment; partial observabilit. . . |        |
| ReAct (Yao et al., 2023b)                                    | 2022 | context       | –           | reasoning traces used to induce, track. . .  |        |
| Self-RAG (Asai et al., 2023)                                 | 2023 | context       | output      | self-assessed retrieval need (reflectio. . . | ○      |
| NLMs as epistemic agents (Betz et al., 2023)                 | 2023 | –             | –           | novel evidence item propagated through. . .  | ✓      |
| Self-Debug (Chen et al., 2023b)                              | 2023 | context       | output      | execution feedback + self-generated cod. . . |        |
| FireAct (Chen et al., 2023a)                                 | 2023 | context       | trajectory  | supervised fine-tuning on GPT-4 ReAct/C. . . |        |
| DreamerV3 (Hafner et al., 2023)                              | 2023 | world model   | state       | RSSM latent posterior updated from obse. . . |        |
| RAP (Hao et al., 2023)                                       | 2023 | world model   | trajectory  | LLM-as-world-model predicted next state. . . |        |
| HI-TOM (He et al., 2023)                                     | 2023 | context       | –           | none (story-based false-belief question. . . | ✓      |
| LAW (Hu and Shu, 2023)                                       | 2023 | world model   | –           | position paper: world and agent models. . .  |        |
| RCI (Kim et al., 2023a)                                      | 2023 | context       | interaction | recursive self-criticism of proposed ac. . . |        |
| FANToM (Kim et al., 2023b)                                   | 2023 | context       | –           | none (evaluation)                            | ✓      |
| LLM ToM (Kosinski) (Kosinski, 2023)                          | 2023 | –             | output      | n/a (probing)                                | ✓      |
| Semantic Entropy (Kuhn et al., 2023)                         | 2023 | –             | output      | entropy over meaning-clustered samples       | ✓      |
| RecallM (Kynoch et al., 2023)                                | 2023 | ext. memory   | memory op   | new knowledge statement triggers concep. . . | ✓      |
| TradingGPT (Li et al., 2023b)                                | 2023 | ext. memory   | interaction | layered memory decay + inter-agent deba. . . |        |
| ToM Multi-Agent Collab (Li et al., 2023a)                    | 2023 | written       | –           | prompted explicit belief-state update e. . . | ✓      |
| Let’s Verify Step by Step (Lightman et al., 2024)            | 2023 | –             | output      | per-step human process labels (PRM800K)      |        |
| AgentBench (Liu et al., 2023b)                               | 2023 | –             | output      | n/a (benchmark)                              |        |
| LLM+P (Liu et al., 2023a)                                    | 2023 | –             | output      | n/a (translate to PDDL, symbolic planne. . . |        |
| RAFA (Liu et al., 2023c)                                     | 2023 | context       | –           | switching condition: replan when predic. . . | ○      |
| Self-Refine (Madaan et al., 2023)                            | 2023 | context       | output      | self-generated natural-language feedbac. . . |        |
| RET-LLM (Modarressi et al., 2023)                            | 2023 | ext. memory   | –           | LLM emits memory-write of extracted tri. . . |        |
| DECKARD (Nottingham et al., 2023)                            | 2023 | world model   | WM loss     | LLM-hypothesized abstract world model v. . . | ○      |
| MemGPT (Packer et al., 2023)                                 | 2023 | ext. memory   | output      | self-directed memory paging via functio. . . |        |
| Generative Agents (Park et al., 2023)                        | 2023 | ext. memory   | interaction | observation stream; periodic reflection. . . |        |
| DPO (Rafailov et al., 2023)                                  | 2023 | –             | output      | none                                         |        |
| SymbolicToM (Sclar et al., 2023)                             | 2023 | ext. filter   | –           | symbolic graph update on each narrated. . .  | ✓      |
| Clever Hans N-ToM (Shapira et al., 2023)                     | 2023 | context       | –           | none (evaluation)                            | ○      |
| Sycophancy (Sharma et al., 2023)                             | 2023 | context       | output      | user pushback / stated user view (socia. . . | ○      |

Metric: ✓ belief-level metric reported, ○ proxy only.

**Table 18 | Master table, continued (2/10).**

| System                                                     | Year | State kept by | Credited    | Revision trigger                           | Metric |
|------------------------------------------------------------|------|---------------|-------------|--------------------------------------------|--------|
| Reflexion (Shinn et al., 2023)                             | 2023 | ext. memory   | trajectory  | verbal self-reflection after episode fa... |        |
| CoALA (Sumers et al., 2023)                                | 2023 | ext. memory   | –           | conceptual: learning actions update mem... |        |
| Interactive planning POMDP (Sun et al., 2023)              | 2023 | context       | –           | none                                       |        |
| LLaRP (Szot et al., 2023)                                  | 2023 | context       | trajectory  | none (egocentric observations in contex... |        |
| Voyager (Wang et al., 2023a)                               | 2023 | ext. memory   | trajectory  | environment execution feedback, errors,... |        |
| DEPS (Wang et al., 2023c)                                  | 2023 | context       | trajectory  | descriptor of execution outcome; self-e... |        |
| Math-Shepherd (Wang et al., 2023b)                         | 2023 | –             | output      | automatic per-step labels via MC comple... |        |
| SimToM (Wilf et al., 2023)                                 | 2023 | context       | –           | perspective-taking filter: context rest... | ○      |
| SmartPlay (Wu et al., 2023)                                | 2023 | –             | output      | n/a (benchmark; game environments)         |        |
| Tree of Thoughts (Yao et al., 2023a)                       | 2023 | context       | trajectory  | self-evaluation of intermediate thought... |        |
| FinMem (Yu et al., 2023)                                   | 2023 | ext. memory   | interaction | layered memory with recency/relevancy/i... |        |
| AgentTuning (Zeng et al., 2023)                            | 2023 | context       | trajectory  | hybrid instruction tuning on AgentInstr... |        |
| CoELA (Zhang et al., 2023b)                                | 2023 | ext. memory   | trajectory  | semantic memory of world/agents/tasks u... |        |
| ProAgent (Zhang et al., 2023a)                             | 2023 | written       | –           | teammate intention inferred from observ... | ○      |
| LLM-MCTS (Zhao et al., 2023b)                              | 2023 | world model   | trajectory  | LLM-sampled world state belief (object...  |        |
| ExpeL (Zhao et al., 2023a)                                 | 2023 | ext. memory   | trajectory  | cross-task insight extraction from succ... |        |
| Synapse (Zheng et al., 2023)                               | 2023 | ext. memory   | –           | none (exemplar retrieval by similarity)    |        |
| MemoryBank (Zhong et al., 2023a)                           | 2023 | ext. memory   | output      | Ebbinghaus-style forgetting with time d... |        |
| MQuAKE / MeLLO (Zhong et al., 2023b)                       | 2023 | ext. memory   | –           | edited fact stored externally; iterativ... | ✓      |
| FaR / T4D (Zhou et al., 2023b)                             | 2023 | context       | interaction | foresee-and-reflect structured promptin... |        |
| RecurrentGPT (Zhou et al., 2023d)                          | 2023 | written       | output      | LLM rewrites its own natural-language s... |        |
| LATS (Zhou et al., 2023a)                                  | 2023 | ext. memory   | trajectory  | MCTS backup of LM-valued rewards + self... |        |
| WebArena (Zhou et al., 2023c)                              | 2023 | context       | –           | none                                       | ○      |
| DigiRL (Bai et al., 2024)                                  | 2024 | context       | interaction | doubly-robust step-level advantage mixi... |        |
| Latent state estimation for UI ag... (Bishop et al., 2024) | 2024 | written       | –           | none                                       |        |
| WMA Web Agent (Chae et al., 2024)                          | 2024 | world model   | WM loss     | predicted next-observation as free-form... | ○      |
| NegotiationToM (Chan et al., 2024)                         | 2024 | –             | –           | none                                       | ✓      |
| ToMBench (Chen et al., 2024b)                              | 2024 | –             | –           | none                                       | ○      |
| Agent-FLAN (Chen et al., 2024a)                            | 2024 | context       | trajectory  | none (SFT on decomposed agent corpora;...  | ○      |
| Talker-Reasoner (Christakopoulou and Mourad, 2024)         | 2024 | written       | state       | Reasoner (System 2) writes/updates user... |        |
| Hypothetical Minds (Cross et al., 2024)                    | 2024 | written       | state       | hypotheses about other agents’ strategi... |        |
| GIF-MCTS code world models (Dainese et al., 2024)          | 2024 | world model   | WM loss     | unit-test and trajectory feedback drive... | ○      |
| InferAct (Fang et al., 2024)                               | 2024 | written       | interaction | ToM-style belief inference over the age... | ○      |
| SimpleToM (Gu et al., 2024)                                | 2024 | –             | –           | none                                       | ✓      |
| Rational belief revision in model... (Hase et al., 2024)   | 2024 | –             | –           | model edits evaluated against an ideali... | ✓      |
| From Words to Actions (He et al., 2024)                    | 2024 | context       | –           | in-context Bayesian aggregated imitatio... |        |
| Coherence norms (Hofweber et al., 2024)                    | 2024 | –             | –           | coherence norms on credence (next-token... |        |
| TimeToM (Hou et al., 2024)                                 | 2024 | written       | –           | temporal ordering of events; belief com... | ○      |
| HiAgent (Hu et al., 2024)                                  | 2024 | ext. memory   | –           | subgoal completion judged by the LLM tr... | ○      |
| MMToM-QA (Jin et al., 2024)                                | 2024 | prob. memory  | –           | Bayesian inverse planning updates poste... | ✓      |
| PercepToM (Jung et al., 2024)                              | 2024 | written       | –           | perception inference feeds a perception... | ✓      |
| QuBE (Kim et al., 2024)                                    | 2024 | written       | state       | question-answering over observations to... |        |
| VisualWebArena (Koh et al., 2024)                          | 2024 | –             | output      | n/a (benchmark)                            |        |
| SCoRe (Kumar et al., 2024)                                 | 2024 | context       | interaction | multi-turn online RL with a reward bonu... |        |
| ReadAgent (Lee et al., 2024)                               | 2024 | ext. memory   | –           | agent decides episode boundaries, compr... |        |
| Embodied Agent Interface (Li et al., 2024b)                | 2024 | world model   | –           | none (evaluation only)                     | ○      |
| MindForge (Lic et al., 2024)                               | 2024 | written       | –           | new percepts and inter-agent messages u... |        |
| LoCoMo (Maharana et al., 2024)                             | 2024 | –             | output      | n/a (benchmark; QA/event summarization...  |        |
| Agent Q (Putta et al., 2024)                               | 2024 | ext. memory   | trajectory  | MCTS with AI self-critique feedback + o... |        |
| WebRL (Qi et al., 2024)                                    | 2024 | –             | output      | outcome-supervised reward model; self-g... |        |
| r to Q* (Rafailov et al., 2024)                            | 2024 | –             | output      | none (token-level implicit reward)         |        |
| Belief state geometry (Shai et al., 2024)                  | 2024 | context       | –           | implicit Bayesian belief updating over...  | ✓      |

*Metric:* ✓ belief-level metric reported, ○ proxy only.

Table 19 | Master table, continued (3/10).

| System                                                  | Year | State kept by | Credited    | Revision trigger                           | Metric |
|---------------------------------------------------------|------|---------------|-------------|--------------------------------------------|--------|
| MuMA-ToM (Shi et al., 2024)                             | 2024 | world model   | –           | Bayesian inverse multi-agent planning o... | ✓      |
| FortUne Dial (Sicilia et al., 2024)                     | 2024 | context       | output      | fine-tuning (supervised and off-policy...  | ✓      |
| Common-ToM (Soubki et al., 2024)                        | 2024 | written       | –           | explicit belief representation supplied... | ○      |
| self-verification limits (Stechly et al., 2024)         | 2024 | context       | –           | self-critique vs sound external verifie... |        |
| AppWorld (Trivedi et al., 2024)                         | 2024 | context       | –           | none (environment; state-based unit tes... | ○      |
| SOTOPIA-pi (Wang et al., 2024a)                         | 2024 | –             | output      | LLM-rated social goal completion used f... |        |
| Agent Workflow Memory (Wang et al., 2024d)              | 2024 | ext. memory   | –           | induction of reusable workflows from su... |        |
| DistRL (Wang et al., 2024b)                             | 2024 | context       | trajectory  | none (state is screenshot history in co... |        |
| MEMORYLLM (Wang et al., 2024c)                          | 2024 | prob. memory  | state       | new text knowledge injected into latent... | ○      |
| POAD (Wen et al., 2024)                                 | 2024 | context       | interaction | token-level Bellman backup (BAD) assign... |        |
| Belief-R (Wilie et al., 2024)                           | 2024 | context       | –           | new premise that should defeat prior in... | ✓      |
| LongMemEval (Wu et al., 2024)                           | 2024 | ext. memory   | –           | knowledge-update questions test whether... | ✓      |
| AgentGym (Xi et al., 2024)                              | 2024 | context       | trajectory  | AgentEvol: iterative exploration + SFT...  |        |
| TradingAgents (Xiao et al., 2024)                       | 2024 | context       | output      | multi-agent debate (bull vs bear resear... |        |
| tau-bench (Yao et al., 2024)                            | 2024 | –             | –           | simulated-user utterances and API tool...  |        |
| LaBToM (Ying et al., 2024)                              | 2024 | ext. filter   | –           | Bayesian inverse planning over the obse... | ✓      |
| FinCon (Yu et al., 2024)                                | 2024 | ext. memory   | state       | episodic self-critique producing concep... |        |
| FinAgent (Zhang et al., 2024a)                          | 2024 | ext. memory   | interaction | low-level and high-level reflection ove... |        |
| MemSim (Zhang et al., 2024c)                            | 2024 | ext. memory   | –           | none (evaluation simulator)                | ○      |
| ArCHer (Zhou et al., 2024b)                             | 2024 | –             | interaction | hierarchical RL: utterance-level off-po... |        |
| Reflect-RL (Zhou et al., 2024a)                         | 2024 | written       | interaction | frozen reflection model produces natura... |        |
| Beliefs of self and others (Zhu et al., 2024)           | 2024 | context       | –           | none (probing and activation interventi... | ✓      |
| Bayesian geometry of attention (Agarwal et al., 2025)   | 2025 | context       | –           | implicit posterior update via FFN in HM... | ✓      |
| Markovian Thinker (Aghajohari et al., 2025)             | 2025 | written       | output      | none                                       |        |
| MemoryBench (Ai et al., 2025)                           | 2025 | ext. memory   | –           | accumulated simulated user feedback dur... | ○      |
| BASIL (Atwell et al., 2025)                             | 2025 | context       | output      | user pushback vs evidence; Bayesian-con... | ✓      |
| Vending-Bench (Backlund and Petersson, 2025)            | 2025 | ext. memory   | –           | none (evaluation only; agent has scratc... | ○      |
| Align While Search (Bae et al., 2025)                   | 2025 | ext. filter   | –           | posterior update from action-conditione... | ✓      |
| tau2-bench (Barres et al., 2025)                        | 2025 | –             | –           | user actions on shared environment obse... |        |
| LOOP (Chen et al., 2025c)                               | 2025 | context       | trajectory  | environment feedback via API observatio... |        |
| ReSearch (Chen et al., 2025d)                           | 2025 | context       | output      | search results inserted into reasoning...  |        |
| HaluMem (Chen et al., 2025b)                            | 2025 | ext. memory   | –           | memory extraction and memory updating o... | ✓      |
| SPA (Chen et al., 2025e)                                | 2025 | world model   | WM loss     | self-experience SFT supervises current...  | ○      |
| GUI-Shepherd (Chen et al., 2025a)                       | 2025 | –             | interaction | process reward model trained on 52k hum... |        |
| Agent-R1 (Cheng et al., 2025)                           | 2025 | context       | trajectory  | tool/environment feedback per interacti... |        |
| Mem0 (Chhikara et al., 2025)                            | 2025 | ext. memory   | –           | LLM-driven extraction, consolidation an... |        |
| Debate or Vote (Choi et al., 2025)                      | 2025 | context       | output      | peer arguments in multi-agent debate ro... | ○      |
| Language and Experience (Colas et al., 2025)            | 2025 | world model   | –           | Bayesian inference over executable worl... | ○      |
| ARPO (Dong et al., 2025a)                               | 2025 | context       | trajectory  | token entropy spike after tool feedback... |        |
| Active Confusion Expression (Du et al., 2025)           | 2025 | written       | output      | confusion indicators (e.g. 'tricky'/'co... | ○      |
| SoMi-ToM (Fan et al., 2025)                             | 2025 | context       | –           | new first-person multimodal observation... | ○      |
| GiGPO (Feng et al., 2025b)                              | 2025 | –             | trajectory  | episode-level group reward plus step-le... |        |
| ReTool (Feng et al., 2025a)                             | 2025 | context       | trajectory  | outcome reward only; no per-step or sta... |        |
| AgentRefine (Fu et al., 2025)                           | 2025 | context       | trajectory  | environment observation triggers correc... |        |
| Survey of self-evolving agents (Gao et al., 2025)       | 2025 | –             | –           | none                                       |        |
| AccumulatingContext (Geng et al., 2025)                 | 2025 | context       | –           | beliefs drift as context accumulates th... | ✓      |
| DIAMONDS (Ghosh et al., 2025)                           | 2025 | –             | –           | values of tracked quantities change ove... | ✓      |
| GenEnv (Guo et al., 2025)                               | 2025 | world model   | trajectory  | alpha-Curriculum Reward aligns simulato... |        |
| Martingale Score (He et al., 2025)                      | 2025 | context       | –           | new evidence across iterative reasoning... | ✓      |
| Measuring Sycophancy of Language... (Hong et al., 2025) | 2025 | context       | –           | sustained user pressure across turns       | ○      |
| MemoryAgentBench (Hu et al., 2025c)                     | 2025 | –             | –           | incremental multi-turn information inje... | ✓      |
| Memory in the Age of AI Agents (Hu et al., 2025d)       | 2025 | ext. memory   | –           | memory formation, evolution and retriev... |        |

Metric: ✓ belief-level metric reported, ○ proxy only.

Table 20 | Master table, continued (4/10).

| System                                              | Year | State kept by | Credited    | Revision trigger                           | Metric |
|-----------------------------------------------------|------|---------------|-------------|--------------------------------------------|--------|
| Agent2World (Hu et al., 2025b)                      | 2025 | world model   | trajectory  | testing-team unit tests and simulation-... | ○      |
| Text2World (Hu et al., 2025a)                       | 2025 | world model   | output      | none (one-shot world-model generation,...  | ○      |
| Bayesian Coherence Coefficient (Imran et al., 2025) | 2025 | context       | –           | in-context evidence pieces presented wi... | ✓      |
| Search-R1 (Jin et al., 2025b)                       | 2025 | context       | output      | retrieved documents interleaved with re... |        |
| SearchRLStudy (Jin et al., 2025a)                   | 2025 | context       | trajectory  | outcome reward plus format reward; inte... |        |
| Emergent World Beliefs (Kamel et al., 2025)         | 2025 | world model   | –           | none (probing of pretrained activations)   | ✓      |
| ACON (Kang et al., 2025b)                           | 2025 | written       | –           | compression guidelines refined in natur... | ○      |
| MemoryOS (Kang et al., 2025a)                       | 2025 | ext. memory   | –           | dialogue-chain FIFO from short- to mid-... | ○      |
| CoEx (Kim and Hwang, 2025)                          | 2025 | world model   | –           | subgoal experience assimilated into neu... |        |
| RefAct (Kim et al., 2025)                           | 2025 | written       | –           | each step rewrites a natural-language s... | ○      |
| LLMs get lost multi-turn (Laban et al., 2025)       | 2025 | context       | –           | new user information across turns (mode... | ○      |
| Hindsight (Latimer et al., 2025)                    | 2025 | ext. memory   | –           | reflect operation updates evolving beli... |        |
| CWM games (Lehrach et al., 2025)                    | 2025 | world model   | –           | unclear (model synthesised from rules a... | ○      |
| DeepAgent (Li et al., 2025e)                        | 2025 | ext. memory   | interaction | autonomous memory folding compresses in... |        |
| Word to World (Li et al., 2025f)                    | 2025 | world model   | WM loss     | next-state prediction under interaction    | ○      |
| Turn-PPO (Li et al., 2025c)                         | 2025 | context       | interaction | turn-level advantage estimation in turn... |        |
| SALT (Li et al., 2025b)                             | 2025 | –             | interaction | none                                       |        |
| SORL (Li et al., 2025a)                             | 2025 | context       | interaction | turn-level importance sampling; clippin... |        |
| MemOS (Li et al., 2025g)                            | 2025 | ext. memory   | –           | memory lifecycle scheduling: MemCubes w... |        |
| AgentFlow (Li et al., 2025h)                        | 2025 | ext. memory   | trajectory  | verifier output appended to evolving me... |        |
| RecToM (Li et al., 2025d)                           | 2025 | –             | –           | questions require tracking evolving int... | ○      |
| ABEL (Lidayan et al., 2025)                         | 2025 | written       | state       | belief grading: ground-truth grader or...  | ○      |
| iStar (Liu et al., 2025)                            | 2025 | context       | interaction | implicit PRM trained by trajectory-leve... |        |
| M3-Agent (Long et al., 2025)                        | 2025 | ext. memory   | trajectory  | streaming visual/auditory input continu... | ○      |
| ARPO (Lu et al., 2025)                              | 2025 | context       | trajectory  | none                                       |        |
| Agent Lightning (Luo et al., 2025b)                 | 2025 | context       | interaction | credit assignment module decomposes tra... |        |
| UltraHorizon (Luo et al., 2025a)                    | 2025 | context       | –           | none                                       | ○      |
| R-WoM (Mei et al., 2025)                            | 2025 | world model   | –           | retrieved tutorial knowledge grounds si... | ○      |
| MATPO (Mo et al., 2025)                             | 2025 | –             | trajectory  | credit assigned across planner and work... |        |
| MindGames (Moore et al., 2025)                      | 2025 | –             | –           | agent must ask about the interlocutor's... | ○      |
| LLaMAR (Nayak et al., 2024)                         | 2025 | context       | –           | none                                       |        |
| WMNav (Nie et al., 2025)                            | 2025 | world model   | –           | discrepancy between world-model plan an... | ○      |
| ReasoningBank (Ouyang et al., 2025)                 | 2025 | ext. memory   | –           | self-judged success/failure of each tas... | ○      |
| ID-RAG (Platnick et al., 2025)                      | 2025 | ext. memory   | –           | unclear (identity KG queried each decis... | ○      |
| Lookbacks belief tracking (Prakash et al., 2025)    | 2025 | context       | –           | visibility lookback updates observing c... | ○      |
| LLM Debate Overconfidence (Prasad and Nguyen, 2025) | 2025 | context       | –           | new opponent arguments each round (shou... | ✓      |
| ToolRL (Qian et al., 2025)                          | 2025 | context       | output      | reward on tool-call format and correctn... |        |
| Bayesian teaching (Qiu et al., 2026a)               | 2025 | written       | state       | distillation of a normative Bayesian mo... | ✓      |
| Zep / Graphiti (Rasmussen et al., 2025)             | 2025 | ext. memory   | –           | temporally-aware KG edge invalidation w... | ○      |
| Cognitive Memory survey (Shan et al., 2025)         | 2025 | ext. memory   | –           | none (survey; covers updating, conflict... |        |
| TTI (Thinking vs Doing) (Shen et al., 2025)         | 2025 | context       | trajectory  | additional environment observations fro... |        |
| ToMATO (Shinoda et al., 2025)                       | 2025 | context       | –           | information asymmetry between role-play... | ✓      |
| WMACT (Shu et al., 2025)                            | 2025 | context       | output      | none (outcome reward)                      |        |
| R4 (Sohn et al., 2025)                              | 2025 | world model   | –           | continuous 4D database construction fro... |        |
| R1-Searcher (Song et al., 2025)                     | 2025 | context       | trajectory  | none (search results appended to contex... |        |
| ContextFolding (Sun et al., 2025b)                  | 2025 | written       | memory op   | agent branches into a sub-trajectory an... |        |
| ZeroSearch (Sun et al., 2025a)                      | 2025 | context       | trajectory  | none (outcome reward on final answer; s... |        |
| Dynamic Cheatsheet (Suzgun et al., 2025)            | 2025 | ext. memory   | memory op   | self-curation after each query: model w... | ○      |
| TARL (Tan et al., 2025b)                            | 2025 | context       | interaction | LLM-judge turn-level adjudication          |        |
| MemBench (Tan et al., 2025a)                        | 2025 | ext. memory   | –           | none (evaluation only)                     | ○      |
| LLM cooperative game simulation (Tao et al., 2025)  | 2025 | ext. filter   | –           | Bayesian belief updating plus noisy rep... | ○      |
| NextLat belief-state latents (Teoh et al., 2025)    | 2025 | world model   | WM loss     | auxiliary next-latent prediction loss;...  | ○      |

Metric: ✓ belief-level metric reported, ○ proxy only.

**Table 21 | Master table, continued (5/10).**

| System                                                     | Year | State kept by | Credited    | Revision trigger                           | Metric |
|------------------------------------------------------------|------|---------------|-------------|--------------------------------------------|--------|
| RAGEN / StarPO (Wang et al., 2025f)                        | 2025 | context       | trajectory  | environment reward; reasoning-aware rew... |        |
| VAGEN (Wang et al., 2025d)                                 | 2025 | written       | state       | world-modeling reward on turn-level sta... | ○      |
| IGPO (Wang et al., 2025a)                                  | 2025 | context       | interaction | turn-level information gain: marginal i... | ○      |
| ENACT (Wang et al., 2025e)                                 | 2025 | world model   | –           | none (evaluation only)                     | ○      |
| EMPG (Wang et al., 2025c)                                  | 2025 | context       | interaction | step-wise policy entropy modulates grad... | ○      |
| Memento 2 stateful reflective mem... (Wang, 2025)          | 2025 | ext. memory   | memory op   | memory write = policy evaluation, read...  |        |
| SPA-RL (Wang et al., 2025b)                                | 2025 | –             | interaction | none                                       |        |
| StepSearch (Wang et al., 2025g)                            | 2025 | context       | interaction | per-step information gain and redundanc... |        |
| Fine-Grained Reward Credit (Wei et al., 2025a)             | 2025 | context       | interaction | per-turn environment rewards in turn-le... |        |
| WebAgent-R1 (Wei et al., 2025d)                            | 2025 | context       | trajectory  | none (binary task-success reward only)     |        |
| GTR-Turbo (Wei et al., 2025c)                              | 2025 | –             | trajectory  | none                                       |        |
| Evo-Memory (Wei et al., 2025b)                             | 2025 | ext. memory   | –           | memory refined after each task in a seq... | ○      |
| ReSum (Wu et al., 2025c)                                   | 2025 | written       | trajectory  | periodic tool-based summarization when...  |        |
| RLVR-World (Wu et al., 2025a)                              | 2025 | world model   | WM loss     | RL reward on decoded transition-predict... | ○      |
| CollabLLM (Wu et al., 2025b)                               | 2025 | context       | interaction | none for belief; Multiturn-aware Reward... |        |
| DEL-ToM (Wu et al., 2025d)                                 | 2025 | written       | state       | each DEL belief-update step scored by a... | ✓      |
| AgentPRM (Xi et al., 2025)                                 | 2025 | context       | interaction | TD/GAE-estimated promise and progress l... |        |
| Experience-following memory study (Xiong et al., 2025b)    | 2025 | ext. memory   | memory op   | addition gated by trajectory evaluators... | ○      |
| FLAG-TRADER (Xiong et al., 2025a)                          | 2025 | context       | interaction | policy-gradient RL on trading reward ov... |        |
| A-Mem (Xu et al., 2025e)                                   | 2025 | ext. memory   | output      | memory evolution: new notes trigger upd... |        |
| STARE-VLA (Xu et al., 2025a)                               | 2025 | –             | interaction | none                                       |        |
| PPR (Xu et al., 2025c)                                     | 2025 | context       | trajectory  | none                                       |        |
| EnigmaToM (Xu et al., 2025b)                               | 2025 | written       | –           | iterative masking over entity-state sce... | ○      |
| Memory-R1 (Yan et al., 2025)                               | 2025 | ext. memory   | memory op   | learned ADD/UPDATE/DELETE/NOOP operatio... |        |
| EvolvTrip (Yang et al., 2025b)                             | 2025 | written       | –           | narrative events append to a perspectiv... | ○      |
| ToM-agent counterfactual reflecti... (Yang et al., 2025a)  | 2025 | written       | –           | counterfactual gap between predicted an... | ○      |
| ECON (Yi et al., 2025)                                     | 2025 | ext. filter   | trajectory  | TD-loss update of a learned belief netw... |        |
| MemAgent (Yu et al., 2025b)                                | 2025 | written       | state       | chunk-wise overwrite of fixed-size memo... |        |
| Dyna-Think (Yu et al., 2025c)                              | 2025 | world model   | WM loss     | DDT stage 1 trains state prediction / s... | ✓      |
| Demystifying agentic RL (Yu et al., 2025d)                 | 2025 | context       | trajectory  | outcome reward on end-to-end tool-use t... |        |
| PersuasiveToM (Yu et al., 2025a)                           | 2025 | –             | –           | questions probe shifts of desires/belie... | ○      |
| EvolveCast (Yuan et al., 2025b)                            | 2025 | context       | –           | post-cutoff evidence presented; forecas... | ✓      |
| MARSHAL (Yuan et al., 2025a)                               | 2025 | –             | interaction | turn-level advantage estimator aligned...  |        |
| Agentic RL survey (Zhang et al., 2025d)                    | 2025 | –             | –           | n/a                                        |        |
| CriticSearch (Zhang et al., 2025i)                         | 2025 | –             | interaction | none                                       |        |
| LLMs as discounted Bayesian filte... (Zhang et al., 2025e) | 2025 | context       | –           | sequential stochastic observations unde... | ✓      |
| ACE (Zhang et al., 2025g)                                  | 2025 | written       | –           | reflector extracts lessons from executi... | ○      |
| Memory in LLMs: mechanisms/eval (Zhang et al., 2025c)      | 2025 | ext. memory   | –           | write->read->inhibit/update chain; mode... | ○      |
| ReasonRAG (Zhang et al., 2025j)                            | 2025 | context       | interaction | process-level rewards on query generati... |        |
| Ark / MedArk (Zhang et al., 2025a)                         | 2025 | context       | trajectory  | ask-vs-tell decision via self-reasoning... | ○      |
| MindPower (Zhang et al., 2025h)                            | 2025 | written       | output      | Mind-Reward rewards consistency between... | ○      |
| Beyond Markovian (Zhang et al., 2025i)                     | 2025 | context       | trajectory  | observed outcomes update a posterior ov... |        |
| PRIME (Zhang et al., 2025k)                                | 2025 | ext. memory   | –           | new user engagements update episodic me... |        |
| Sycophancy under Pressure (Zhang et al., 2025f)            | 2025 | context       | output      | adversarial user pressure in dialogue;...  | ○      |
| Overcoming Multi-step Complexity... (Zhang et al., 2025b)  | 2025 | ext. filter   | output      | stepwise Bayesian updates with small-LM... | ○      |
| PrefEval (Zhao et al., 2025)                               | 2025 | context       | –           | user preference stated explicitly or im... | ○      |
| DeepResearcher (Zheng et al., 2025c)                       | 2025 | context       | trajectory  | outcome reward on final answer from end... | ○      |
| LifelongAgentBench (Zheng et al., 2025b)                   | 2025 | ext. memory   | –           | cross-task experience replay; group sel... | ○      |
| MEM1 (Zhou et al., 2025d)                                  | 2025 | written       | state       | internal state rewritten each turn cons... |        |
| Memento (Zhou et al., 2025a)                               | 2025 | ext. memory   | interaction | environment feedback rewrites episodic...  |        |
| SWEET-RL (Zhou et al., 2025c)                              | 2025 | –             | interaction | step-level critic reward using training... |        |
| Social World Models (S3AP) (Zhou et al., 2025b)            | 2025 | written       | –           | structured state/action/mental-state re... | ○      |

*Metric:* ✓ belief-level metric reported, ○ proxy only.

Table 22 | Master table, continued (6/10).

| System                                                            | Year | State kept by | Credited    | Revision trigger                            | Metric |
|-------------------------------------------------------------------|------|---------------|-------------|---------------------------------------------|--------|
| T3 (Zou et al., 2025)                                             | 2025 | prob. memory  | trajectory  | truth-anchored potential -log b(s*) sta...  | ✓      |
| MEDLEY-BENCH (Abtahi et al., 2026)                                | 2026 | context       | —           | social disagreement / analyst summaries...  |        |
| Agent-ToM: learning to monitor ag... (Ahmed and Nafisi, 2026)     | 2026 | —             | trajectory  | deviation from task-consistent expected...  |        |
| Can LLM Agents Price Competitivel... (Ahmed et al., 2026)         | 2026 | context       | —           | market feedback and demand shocks revis...  | ○      |
| Bayesian Chronicle Agents (BCA) (Akbar et al., 2026)              | 2026 | ext. filter   | —           | Bayesian update (Beta pseudo-counts, pr...  | ✓      |
| Why world models: LLM state-track... (Alaswad et al., 2026)       | 2026 | context       | —           | none                                        |        |
| RLAW: POMDP routing + self-correc... (Ali and Tseng, 2026)        | 2026 | ext. memory   | interaction | critique score (logic/safety check)         |        |
| Bayesian orchestration (Amin, 2026)                               | 2026 | ext. filter   | state       | Bayes-rule posterior update with LLM-el...  | ○      |
| Clue possibility matrix (Ansell and Toney-Wails, 2026)            | 2026 | ext. filter   | —           | symbolic constraint update (YES/NO/MAYB...  | ✓      |
| CA-BED (Arnould et al., 2026)                                     | 2026 | prob. memory  | —           | LLM-estimated likelihoods of user answe...  | ○      |
| RoVLaP (Azran et al., 2026)                                       | 2026 | ext. filter   | —           | VLM next-token predicate probabilities...   |        |
| BAG (Baan et al., 2026)                                           | 2026 | context       | —           | user clarification in the next turn res...  | ○      |
| RB-VLA: recursive belief VLA (Bagaria et al., 2026)               | 2026 | world model   | state       | self-supervised world-model objective (...) | ○      |
| PatchWorld (Bai et al., 2026)                                     | 2026 | world model   | WM loss     | counterexample-guided repair of executa...  | ○      |
| LLM + causal model POMDP planning (Basu et al., 2026)             | 2026 | ext. filter   | —           | POMDP observation update; hybrid confid...  | ○      |
| OmniToM explicit belief modeling (Bawatneh et al., 2026)          | 2026 | —             | —           | n/a                                         | ✓      |
| MIST (Bensal et al., 2026)                                        | 2026 | ext. memory   | —           | user misconceptions written into memory...  | ✓      |
| ECCBench (Berman and Deng, 2026)                                  | 2026 | —             | —           | none                                        | ✓      |
| CAGE-2 compound agent design (Bogdanov et al., 2026)              | 2026 | ext. memory   | —           | deterministic state-tracking layer with...  |        |
| Reasoning Theater (Boppa et al., 2026)                            | 2026 | context       | output      | internal activation belief shift during...  | ○      |
| InKH financial harness (Borjigin et al., 2026)                    | 2026 | ext. memory   | —           | passive knowledge injection from user/m...  |        |
| CPS statistical control survey (Cai et al., 2026)                 | 2026 | —             | —           | none                                        |        |
| TRACE (Chang and Chang, 2026)                                     | 2026 | written       | —           | append-only revision when world-model c...  |        |
| Trivium (Chang, 2026)                                             | 2026 | world model   | WM loss     | budgeted probes and continued posterior...  | ✓      |
| STALE (Chao et al., 2026)                                         | 2026 | ext. memory   | —           | implicit conflict                           | ✓      |
| Belief-State Engine (Chattopadhyay and Halder, 2026)              | 2026 | ext. filter   | —           | Bayesian posterior                          | ✓      |
| The Horizon Gap (survey) (Chen et al., 2026b)                     | 2026 | —             | —           | n/a                                         |        |
| BCO: belief-calibrated scaffold o... (Chen et al., 2026c)         | 2026 | written       | state       | evaluation of new candidate scaffolds (...) | ✓      |
| MM-Lifelong / ReMA (Chen et al., 2026a)                           | 2026 | ext. memory   | —           | dynamic memory management iteratively u...  | ○      |
| Theory of Agent (ToA) (Cheng et al., 2026)                        | 2026 | —             | —           | none                                        |        |
| Truth stance layer (expressed dou... (Cochinescu, 2026)           | 2026 | prob. memory  | —           | rule-based revision operator (user corr...  | ✓      |
| BOND (distilling Bayesian beliefs) (Cui and Lin, 2026)            | 2026 | written       | state       | teacher posterior                           | ✓      |
| Closing the Feedback Loop (Cui et al., 2026)                      | 2026 | ext. memory   | —           | realised returns interpreted by an LLM...   |        |
| STOCKTAKE (Deb and Krishnan, 2026)                                | 2026 | —             | —           | n/a                                         | ✓      |
| MDLM text world models (Deshpande, 2026)                          | 2026 | world model   | WM loss     | world model conditions on state anchors...  | ○      |
| DeltaLogic minimal-edit revision (Dhanda, 2026)                   | 2026 | context       | output      | minimal premise edit                        | ✓      |
| HIPIF (Diao et al., 2026)                                         | 2026 | written       | trajectory  | subgoal completion (history folding) +...   |        |
| SGCD sibling-guided credit (Ding et al., 2026b)                   | 2026 | —             | output      | none                                        |        |
| APEx (Ding et al., 2026a)                                         | 2026 | ext. memory   | memory op   | reward-guided distillation of trajector...  |        |
| Belief at Risk (Dixon, 2026b)                                     | 2026 | ext. filter   | —           | Bayesian posterior over regimes             | ✓      |
| POMDP validation framework (Dixon, 2026c)                         | 2026 | ext. filter   | —           | Bayesian                                    | ✓      |
| Governance-aware POMDP for AI del... (Dixon, 2026a)               | 2026 | ext. filter   | —           | Bayesian update on evidence quality         |        |
| Mind-Studio (Dong et al., 2026)                                   | 2026 | world model   | —           | entropy-selected state-action-next-stat...  | ✓      |
| LifeSim (Duan et al., 2026)                                       | 2026 | context       | —           | user simulator driven by BDI model; ass...  | ○      |
| KbSD (Feng et al., 2026)                                          | 2026 | context       | output      | none                                        | ○      |
| MedExAgent (Gao et al., 2026a)                                    | 2026 | context       | output      | none (composite outcome reward: accurac...  |        |
| Auditing belief-conditioned Werew... (Gao et al., 2026b)          | 2026 | ext. memory   | state       | logged per-turn belief updates              | ✓      |
| CubeBench (ang Gao et al., 2025)                                  | 2026 | —             | —           | n/a (benchmark)                             | ○      |
| Cyclic subtask graphs (Gharzeddine and Saab, 2026)                | 2026 | written       | —           | state-analysis-and-routing agent applic...  |        |
| FinBench calibration benchmark (Ghosh and Devarakonda, 2026)      | 2026 | —             | output      | none                                        |        |
| CHAL: council of hierarchical age... (Giovannelli and Kent, 2026) | 2026 | written       | state       | gradient-informed revision on thesis st...  |        |
| Does Theory of Mind Improvement R... (Gong et al., 2026)          | 2026 | context       | —           | live user responses in interactive firs...  | ○      |

Metric: ✓ belief-level metric reported, ○ proxy only.

**Table 23 | Master table, continued (7/10).**

| System                                                         | Year | State kept by | Credited    | Revision trigger                             | Metric |
|----------------------------------------------------------------|------|---------------|-------------|----------------------------------------------|--------|
| ENIGMA (Gritsaenko, 2026)                                      | 2026 | prob. memory  | –           | deterministic rule-based belief crystal. . . | ✓      |
| HiDiffTIR (Guo et al., 2026c)                                  | 2026 | context       | interaction | group tool-call success rate                 |        |
| PivoARL (Guo et al., 2026b)                                    | 2026 | context       | interaction | structured reflection identifies pivota. . . |        |
| CAVE (Guo et al., 2026a)                                       | 2026 | context       | interaction | process reward decomposed into belief u. . . | ○      |
| EnterpriseArena (Han et al., 2026)                             | 2026 | context       | –           | none                                         |        |
| EVAF (Han, 2026)                                               | 2026 | ext. memory   | memory op   | surprise- and valence-gated consolidati. . . | ○      |
| Structured memory over-personaliz. . . (Hannoon et al., 2026)  | 2026 | ext. memory   | –           | none (inference-time restructuring of m. . . | ○      |
| MemOps (Hao et al., 2026)                                      | 2026 | ext. memory   | –           | lifecycle ops (remember/forget/update/r. . . | ✓      |
| YC-Bench (He et al., 2026b)                                    | 2026 | written       | –           | scratchpad persistence across context t. . . |        |
| TRACE turn credit for jailbreaking (He et al., 2026d)          | 2026 | context       | interaction | leave-one-turn-out semantic masking          |        |
| HGPO (He et al., 2026c)                                        | 2026 | –             | interaction | none                                         |        |
| CoLearn (He et al., 2026a)                                     | 2026 | prob. memory  | –           | Bayesian update (soft-evidence BKT; LLM. . . | ✓      |
| CAPTURE: preference drift vs memo. . . (Hossain et al., 2026)  | 2026 | world model   | state       | uncertainty-triggered clarification + c. . . | ✓      |
| REDEREF: training-free probabilis. . . (Hosseini et al., 2026) | 2026 | prob. memory  | interaction | judge feedback updates Thompson-samplin. . . |        |
| WMF-AM working memory probe (Hou et al., 2026)                 | 2026 | context       | –           | K sequential operations within one quer. . . | ✓      |
| MATTRL test-time multi-agent RL (Hu et al., 2026b)             | 2026 | ext. memory   | interaction | turn-level credit for experience pool c. . . |        |
| BRAID (Hu et al., 2026c)                                       | 2026 | –             | interaction | none                                         |        |
| Infinite-Parameter LLM (Hu et al., 2026a)                      | 2026 | ext. filter   | state       | known answer (gold passage relevance la. . . | ✓      |
| SDPO (Hübotter et al., 2026)                                   | 2026 | context       | output      | textual environment feedback (errors, j. . . |        |
| ToMAgent lookahead ToM (Hwang et al., 2025)                    | 2026 | written       | state       | dialogue lookahead rollouts score (ment. . . | ○      |
| MARBO (Hwang et al., 2026)                                     | 2026 | written       | state       | cross-entropy against ground-truth role. . . | ○      |
| Persuasion Propagation (Jeong et al., 2026)                    | 2026 | context       | –           | none                                         | ○      |
| Planted latent variable (Jerpelea, 2026)                       | 2026 | context       | –           | Bayesian posterior over the planted lat. . . | ✓      |
| EOG belief propagation (Jha et al., 2026)                      | 2026 | ext. filter   | –           | new evidence labelled by the LLM (cause. . . | ○      |
| Hindsight Memory-PRM (Jia et al., 2026)                        | 2026 | ext. memory   | memory op   | hindsight audit trail (retrieval hits,. . .  | ○      |
| SCRIBE (Jiang and Ferraro, 2026)                               | 2026 | context       | interaction | skill-prototype rubric reward on subgoa. . . |        |
| HAGE (Jiang et al., 2026b)                                     | 2026 | ext. memory   | memory op   | RL on downstream task jointly updates e. . . |        |
| PABU (Jiang et al., 2026c)                                     | 2026 | written       | state       | SFT on LLM-synthesised progress and ret. . . |        |
| Jev-Mem (Jiang et al., 2026a)                                  | 2026 | ext. memory   | –           | System-One controller types, links and. . .  |        |
| Regent Chess replication (Joshi, 2026)                         | 2026 | written       | –           | none (elicited belief scored only)           | ✓      |
| SuS strategy-aware surprise (Kashirskiy and Makarov, 2026)     | 2026 | context       | output      | surprise residual (pre-post prediction. . .  |        |
| Context Gathering Decision Process (Kausik et al., 2026)       | 2026 | ext. filter   | –           | n/a                                          |        |
| InfoPO (Kong et al., 2026)                                     | 2026 | context       | interaction | information-gain reward: credit to turn. . . | ○      |
| Belief-Based World Model (Kumar et al., 2026)                  | 2026 | world model   | –           | world-model belief                           |        |
| Hindsight (Latimer et al., 2026)                               | 2026 | ext. memory   | –           | retain/reflect operations; opinions car. . . | ○      |
| MINTEval (Lee et al., 2026a)                                   | 2026 | –             | –           | none (evaluation of memory under update. . . | ✓      |
| Meta-Harness (Lee et al., 2026b)                               | 2026 | ext. memory   | memory op   | outer-loop search over harness code usi. . . |        |
| Belief geometries via SAEs (Levinson, 2026)                    | 2026 | context       | –           | none                                         | ○      |
| Text World Models (review) (Li et al., 2026d)                  | 2026 | world model   | –           | n/a                                          |        |
| ECPO (Li et al., 2026e)                                        | 2026 | –             | trajectory  | n/a                                          |        |
| SERL selective hindsight distilla. . . (Li et al., 2026c)      | 2026 | context       | interaction | per-step environment feedback (error me. . . |        |
| VICT (Li et al., 2026b)                                        | 2026 | –             | interaction | verifier atom proof edges                    |        |
| InfoDelphi (Li et al., 2026f)                                  | 2026 | context       | –           | private evidence surfaced through peer. . .  | ✓      |
| SE-Search (Li et al., 2026a)                                   | 2026 | ext. memory   | memory op   | memory purification at each Think-Searc. . . |        |
| RTPO (Li et al., 2026g)                                        | 2026 | –             | interaction | reverse-turn tree updates align each tu. . . |        |
| ELPO (Liang et al., 2026)                                      | 2026 | –             | interaction | first irrecoverable step located by bin. . . |        |
| Belief Memory (BeliefMem) (Liao et al., 2026a)                 | 2026 | ext. memory   | –           | Noisy-OR update                              |        |
| TCPO turn-level credit (Liao et al., 2026b)                    | 2026 | context       | interaction | verifier score-to-credit conversion; hi. . . |        |
| UQ for LLM Agents taxonomy (Lin et al., 2026c)                 | 2026 | –             | –           | n/a                                          | ✓      |
| SkillC (Lin et al., 2026a)                                     | 2026 | context       | trajectory  | skill-vs-no-skill reward gap                 |        |
| ReAdapt (Lin et al., 2026b)                                    | 2026 | written       | –           | LLM Adapt step after each tool observat. . . |        |
| MADE: belief-driven dual-agent de. . . (Liu et al., 2026e)     | 2026 | ext. memory   | interaction | execution failure -> root-cause invalid. . . |        |

Metric: ✓ belief-level metric reported, ○ proxy only.

**Table 24 | Master table, continued (8/10).**

| System                                                        | Year | State kept by | Credited    | Revision trigger                            | Metric |
|---------------------------------------------------------------|------|---------------|-------------|---------------------------------------------|--------|
| PiCA pivot-based credit (Liu et al., 2026b)                   | 2026 | context       | trajectory  | pivot-step potential-based shaping          |        |
| SHADOW (Liu et al., 2026g)                                    | 2026 | context       | interaction | dynamics-aware state grouping and local...  |        |
| INTENT (Liu et al., 2026c)                                    | 2026 | world model   | –           | inference-time: intention-aware hierarc...  | ○      |
| SimpleMem (Liu et al., 2026d)                                 | 2026 | ext. memory   | –           | online semantic synthesis merges relate...  | ○      |
| WorldMemArena (Liu et al., 2026a)                             | 2026 | ext. memory   | –           | gold memory updates and distractors ann...  | ✓      |
| PRISM (Liu et al., 2026h)                                     | 2026 | written       | –           | new observations and VLM semantic reaso...  |        |
| MMPO (Liu et al., 2026i)                                      | 2026 | written       | state       | belief-entropy proxy (own answer entrop...  |        |
| REE diagnostic for ICL in games (Liu et al., 2026f)           | 2026 | context       | –           | none (interaction history in context)       |        |
| AutoHarness (Lou et al., 2026)                                | 2026 | ext. filter   | –           | environment feedback (illegal-move erro...  | ○      |
| HISR (Lu et al., 2026c)                                       | 2026 | context       | interaction | segment-level process reward modulated...   |        |
| UI-Copilot (Lu et al., 2026b)                                 | 2026 | ext. memory   | interaction | memory decoupling separates persistent...   |        |
| PaW (Lu et al., 2026a)                                        | 2026 | world model   | WM loss     | clipped-MAE auxiliary loss on next-obse...  | ○      |
| From Storage to Experience (memor... (Luo et al., 2026c)      | 2026 | ext. memory   | –           | reflection                                  |        |
| Graph of States (Luo et al., 2026d)                           | 2026 | ext. memory   | –           | symbolic check (causal graph constraint...  |        |
| ReflectiChain world-model supply... (Luo, 2026)               | 2026 | world model   | trajectory  | latent trajectory rehearsals; reflection... |        |
| ReflectiChain (Luo et al., 2026a)                             | 2026 | world model   | –           | retrospective reflection over the execu...  | ✓      |
| PUMA user-state modeling (Luo et al., 2026b)                  | 2026 | prob. memory  | WM loss     | new user utterance updates variational...   | ✓      |
| ProPlay (Ma et al., 2026b)                                    | 2026 | world model   | trajectory  | post-episode graph refinement from envi...  |        |
| Fine-Mem (Ma et al., 2026a)                                   | 2026 | ext. memory   | memory op   | fine-grained feedback alignment on memo...  |        |
| LongHorizon-Harness (Ma et al., 2026c)                        | 2026 | ext. memory   | –           | none                                        |        |
| CoMAM (Mao et al., 2026b)                                     | 2026 | ext. memory   | memory op   | none                                        |        |
| TreeMem (Mao et al., 2026a)                                   | 2026 | ext. memory   | memory op   | none                                        |        |
| Few-shot in-context world represe... (Matsutani et al., 2026) | 2026 | context       | –           | none (in-context inference)                 | ✓      |
| Physics of long-horizon planning (Men et al., 2026)           | 2026 | world model   | WM loss     | simulator ground-truth state transition...  | ○      |
| EquiMem (Meng et al., 2026)                                   | 2026 | ext. memory   | –           | each memory update scored against equil...  | ○      |
| PCCC / CVT-RL (Meng, 2026)                                    | 2026 | written       | state       | verifier-estimated prefix-observable la...  | ✓      |
| Episodic-Semantic Memory (Milosevic, 2026)                    | 2026 | ext. memory   | memory op   | consolidation of episodic window into s...  | ○      |
| MT-GRPO Iterative Reward Calibrat... (Modecrua et al., 2026)  | 2026 | context       | interaction | none (per-turn reward calibrated from r...  |        |
| Bayesian Linguistic Forecaster (Murphy, 2026)                 | 2026 | written       | –           | sequential update in tool loop              | ✓      |
| BeliefShift (Myakala et al., 2026)                            | 2026 | –             | –           | evidence-driven revision                    | ✓      |
| When Stale Constraints Go Uncheck... (Nakayashiki, 2026)      | 2026 | ext. memory   | –           | budgeted verification                       |        |
| DToM-Track: dynamic ToM as tempor... (Nguyen et al., 2026)    | 2026 | context       | –           | none                                        | ✓      |
| Light-Omni (Nie et al., 2026)                                 | 2026 | ext. memory   | –           | hierarchical merging consolidates episo...  |        |
| EvoHarness-RL learned harness sta... (Ning et al., 2026)      | 2026 | ext. memory   | interaction | learned read/update/consolidate policy...   |        |
| SSMG-Nav (Niu et al., 2026)                                   | 2026 | ext. memory   | –           | new observations consolidated into a pe...  | ○      |
| SkillIOS (Ouyang et al., 2026)                                | 2026 | ext. memory   | memory op   | trainable curator updates SkillRepo fro...  |        |
| MD5 state tracking (Pai and Xian, 2026)                       | 2026 | context       | –           | majority vote over arithmetic               |        |
| Business Arena marketplace benchm... (Pan et al., 2026a)      | 2026 | –             | interaction | none                                        |        |
| ContextPilot (Pan et al., 2026b)                              | 2026 | context       | memory op   | context and entropy variation identify...   |        |
| Bayes-consistent orchestration (Papamarkou et al., 2026)      | 2026 | ext. filter   | –           | Bayesian posterior update over task-rel...  |        |
| Kumiho (Park, 2026)                                           | 2026 | ext. memory   | –           | AGM-style revision on graph memory (imm...  | ○      |
| HiPER: hierarchical RL with expli... (Peng et al., 2026a)     | 2026 | context       | trajectory  | none (reward-based)                         |        |
| Explore More Drift Less (Pu et al., 2026)                     | 2026 | –             | output      | n/a                                         |        |
| STAPO trajectory-neglect steps (Qi et al., 2026)              | 2026 | context       | trajectory  | normalized-entropy outlier steps            |        |
| Agents fail to use world models (Qian et al., 2026)           | 2026 | world model   | –           | agent decides when to invoke external s...  | ○      |
| MoM / P-Mem (Qin and Lu, 2026)                                | 2026 | written       | –           | LLM typed write-time operations (accept...  | ✓      |
| SWIRL (Qiu et al., 2026b)                                     | 2026 | world model   | WM loss     | alternating GRPO updates of forward wor...  | ○      |
| MatchTIR (Qu et al., 2026)                                    | 2026 | context       | interaction | bipartite matching of predicted vs grou...  |        |
| Debugging code world models (Rahmani, 2026)                   | 2026 | world model   | WM loss     | predicted runtime state after each exec...  | ✓      |
| Oblivion (Rana et al., 2026)                                  | 2026 | ext. memory   | memory op   | read gated by agent uncertainty and buf...  |        |
| LLM explainer over Active Inferen... (Raval et al., 2026)     | 2026 | ext. filter   | –           | variational (Active Inference) postero...   | ✓      |
| IAPO: influence-aware credit for... (Ren et al., 2026a)       | 2026 | context       | interaction | none (influence-dependency graph from r...  |        |

Metric: ✓ belief-level metric reported, ○ proxy only.

**Table 25 | Master table, continued (9/10).**

| System                                                      | Year | State kept by | Credited    | Revision trigger                           | Metric |
|-------------------------------------------------------------|------|---------------|-------------|--------------------------------------------|--------|
| CSSP (Ren et al., 2026b)                                    | 2026 | ext. filter   | state       | new telemetry events update a learned s... | ○      |
| AI Scientists Reasoning (Ríos-García et al., 2026)          | 2026 | context       | –           | refutation-driven revision (rarely obse... | ✓      |
| BayesBench (Samanta et al., 2026)                           | 2026 | context       | –           | per-turn evidence arrival; compared aga... | ✓      |
| PCE (Seo et al., 2026)                                      | 2026 | written       | state       | scenario likelihood scoring                |        |
| FinalityBench (Sharma, 2026)                                | 2026 | context       | –           | none (benchmark; agent probes/waits)       |        |
| SPADER (Shi et al., 2026)                                   | 2026 | context       | interaction | none                                       |        |
| Agent-BRACE (Singh et al., 2026b)                           | 2026 | written       | state       | verbalized certainty                       | ✓      |
| Nous: when belief-based memory he... (Singh, 2026)          | 2026 | prob. memory  | state       | surprise (information-theoretic) + reli... | ✓      |
| MMA-RAG*T (Singh et al., 2026a)                             | 2026 | written       | –           | new observations at each pipeline check... | ○      |
| Pinductor: LLM-prior POMDP world... (Six et al., 2026)      | 2026 | world model   | state       | belief-based likelihood score of candid... |        |
| Parallel WebBench (Sogani et al., 2026)                     | 2026 | context       | trajectory  | GRPO on task outcome; trace diagnostics... | ○      |
| D-MEM (Song and Xin, 2026)                                  | 2026 | ext. memory   | –           | reward prediction error gating             |        |
| Hybrid-WM (Song and Cai, 2026a)                             | 2026 | world model   | state       | consistency gate: disagreement between...  | ✓      |
| WM-SAR (Song and Cai, 2026b)                                | 2026 | world model   | trajectory  | spectral error amplification               |        |
| Semantic Bayesian World Models (Soru, 2026)                 | 2026 | ext. filter   | state       | Bayesian conditioning on observations;...  |        |
| CMI (Srivastava, 2026b)                                     | 2026 | ext. memory   | state       | counterfactual answer-score intervention   |        |
| memory-bench construction (Srivastava, 2026a)               | 2026 | ext. memory   | –           | hidden ledger of facts defines expected... | ✓      |
| External affective state dynamics (Subaharan, 2026)         | 2026 | ext. memory   | –           | exponential smoothing or momentum updat... |        |
| When Memory Updates but Behavior... (Sun and He, 2026)      | 2026 | ext. memory   | –           | state dependency repair                    |        |
| VerMem (Sun et al., 2026b)                                  | 2026 | ext. memory   | memory op   | local verifier scores each memory trans... | ○      |
| AEWM / EditAct (Sun et al., 2026a)                          | 2026 | world model   | interaction | teacher or judge (annotation-agent labe... |        |
| HCAPO hindsight critic (Tan et al., 2026)                   | 2026 | context       | trajectory  | post-hoc LLM critic refinement             |        |
| MSCE (Tang et al., 2026a)                                   | 2026 | ext. memory   | trajectory  | reflection-weighted value backfilling p... | ○      |
| ReBel (Tang et al., 2026b)                                  | 2026 | written       | state       | belief-consistency reward: each predica... | ○      |
| Proactive Service Agents survey (Tang et al., 2026c)        | 2026 | –             | –           | Bayesian belief update (formal framewor... |        |
| STAGE (Tian et al., 2026)                                   | 2026 | written       | state       | new scene evidence triggers recursive u... | ✓      |
| Memora (Uddin et al., 2026)                                 | 2026 | ext. memory   | –           | user-stated add/update/delete of prefer... | ✓      |
| Rank-Bounded Memory (Verbovoy, 2026)                        | 2026 | ext. memory   | –           | storage schema invariants (ownership pa... | ✓      |
| TOKI (Wang, 2026b)                                          | 2026 | ext. memory   | –           | bitemporal operators (4 heuristics)        |        |
| CAPO: critic-guided action-aligne... (Wang et al., 2026a)   | 2026 | context       | interaction | none (critic value at action boundary)     |        |
| SLEA-RL: step-level experience au... (Wang and Jiang, 2026) | 2026 | ext. memory   | trajectory  | observation change triggers step-level...  |        |
| G2PO (Wang et al., 2026j)                                   | 2026 | context       | trajectory  | globally standardized TD error over sta... |        |
| Self-Guide internal reward co-evo... (Wang et al., 2026g)   | 2026 | written       | trajectory  | self-generated guidance as internal rew... |        |
| TAMTRL (Wang et al., 2026d)                                 | 2026 | written       | trajectory  | teacher likelihood of memory update        |        |
| T2PO (Wang et al., 2026b)                                   | 2026 | context       | interaction | token-level entropy/confidence plateau     |        |
| CrEST (Wang et al., 2026k)                                  | 2026 | context       | interaction | verified turn reward with self-teacher...  |        |
| MINDGAMES live arena (Wang et al., 2026c)                   | 2026 | –             | –           | n/a                                        |        |
| APPO (Wang et al., 2026h)                                   | 2026 | –             | output      | none                                       |        |
| LENS (Wang et al., 2026f)                                   | 2026 | prob. memory  | –           | LLM relevance-oracle verdicts update a...  | ○      |
| Mind2Dialogue (Wang et al., 2026l)                          | 2026 | written       | output      | simulator updates a shared evolving use... | ○      |
| Dark Room (Wang, 2026a)                                     | 2026 | written       | state       | rule-verified next-observation predicti... | ○      |
| Conv-FinRe (Wang et al., 2026i)                             | 2026 | –             | –           | n/a (benchmark; investor risk preferenc... | ○      |
| GAVEL (Wang et al., 2026e)                                  | 2026 | ext. filter   | state       | known answer (RSN prior over object roo... | ✓      |
| Agentic Reasoning survey (Wei et al., 2026)                 | 2026 | –             | –           | n/a                                        |        |
| AC1 self-model loop replication (Wilson, 2026)              | 2026 | ext. filter   | –           | Bayesian accumulator over own interacti... | ✓      |
| Bayesian-Agent (Wu et al., 2026b)                           | 2026 | ext. filter   | –           | verified trajectories                      |        |
| BayesEvolve: explicit belief for... (Wu et al., 2026d)      | 2026 | ext. filter   | state       | posterior refit after each evaluated ca... | ✓      |
| HumanLM state alignment (Wu et al., 2026a)                  | 2026 | written       | state       | RL aligns generated natural-language la... | ○      |
| RL recipe long-horizon tool agents (Wu et al., 2026c)       | 2026 | context       | trajectory  | staged vs dense reward shaping on Trave... |        |
| REFLEX (Wu and Lim, 2026)                                   | 2026 | context       | –           | confidence threshold escalates bounded...  |        |
| SkillRL (Xia et al., 2026)                                  | 2026 | ext. memory   | trajectory  | experience distilled into hierarchical...  |        |
| WebWorld (Xiao et al., 2026b)                               | 2026 | world model   | WM loss     | trained on 1M+ open-web interactions; u... | ○      |

Metric: ✓ belief-level metric reported, ○ proxy only.

Table 26 | Master table, continued (10/10).

| System                                                     | Year | State kept by | Credited    | Revision trigger                            | Metric |
|------------------------------------------------------------|------|---------------|-------------|---------------------------------------------|--------|
| AeroBelief (Xiao et al., 2026a)                            | 2026 | prob. memory  | –           | rule-based accumulation of VLM match co...  |        |
| TIPS (Xie et al., 2026)                                    | 2026 | context       | interaction | teacher answer-likelihood gain              |        |
| LLM Agents for Forecasting (surve... (Xu et al., 2026d)    | 2026 | –             | –           | n/a                                         |        |
| Contextual Belief Management (Bel... (Xu et al., 2026a)    | 2026 | written       | state       | symbolic verification                       | ✓      |
| NeSyFS: neuro-symbolic fast-slow... (Xu and Fekri, 2026)   | 2026 | ext. memory   | interaction | repeated fast-action failure triggers s...  |        |
| GPM: governed persistent memory (Xu, 2026)                 | 2026 | ext. memory   | –           | symbolic check (source binding, retract...  |        |
| MemGym (Xu et al., 2026c)                                  | 2026 | ext. memory   | –           | none                                        |        |
| EnvACE world rehearsal (Xu et al., 2026e)                  | 2026 | world model   | trajectory  | task-success RL reward jointly optimize...  |        |
| META (Xu et al., 2026b)                                    | 2026 | ext. memory   | –           | trade outcome + LLM reflection; prune/c...  |        |
| Representation signatures in trad... (Xue, 2026)           | 2026 | context       | output      | structured risk feedback                    |        |
| MemStrata temporal validity (Yadav, 2026)                  | 2026 | ext. memory   | –           | contradiction of a stored (subject, rela... | ✓      |
| Memory-R2 / LoGo-GRPO (Yan et al., 2026a)                  | 2026 | ext. memory   | state       | local rerollouts from shared intermedia...  |        |
| DenoiseFlow (Yan et al., 2026b)                            | 2026 | context       | trajectory  | semantic entropy over samples               |        |
| Belief Engine stance dynamics (Yang et al., 2026c)         | 2026 | prob. memory  | state       | log-odds update with evidence uptake an...  | ✓      |
| GroupMemBench multi-party memory (Yang et al., 2026b)      | 2026 | ext. memory   | –           | none                                        |        |
| Graph-based agent memory survey (Yang et al., 2026a)       | 2026 | ext. memory   | –           | memory evolution stage (update/consolid...  |        |
| EvoTool (Yang et al., 2026e)                               | 2026 | context       | interaction | none                                        |        |
| world models as intermediary (Yang, 2026)                  | 2026 | world model   | WM loss     | none (position paper)                       |        |
| EnterpriseBench (Yang et al., 2026d)                       | 2026 | context       | –           | none (benchmark)                            |        |
| Werewolf accusation belief-shift (Yang et al., 2026f)      | 2026 | written       | –           | LLM in-context update after accusation...   | ✓      |
| Forecast-Dojo (Ye et al., 2026)                            | 2026 | written       | trajectory  | LM rewrites JSON belief notebook (proba...  | ✓      |
| AgeMem (Yu et al., 2026b)                                  | 2026 | ext. memory   | memory op   | agent-initiated store/retrieve/update/s...  | ○      |
| LOCARD structured belief state (Yu and Knottenbelt, 2026)  | 2026 | written       | –           | strategic core flips a boolean SOP-mile...  |        |
| InfIAgent (Yu et al., 2026a)                               | 2026 | written       | –           | none                                        |        |
| SAVeR: self-audited verified reas... (Yuan et al., 2026a)  | 2026 | context       | state       | adversarial audit locating logical/evid...  |        |
| RPMS (Yuan et al., 2026b)                                  | 2026 | written       | –           | lightweight belief state gates memory a...  | ○      |
| Critic Experience Bank (Zeng et al., 2026)                 | 2026 | ext. memory   | –           | hindsight LLM majority vote conditioned...  | ✓      |
| MICA: intertemporal credit for em... (Zhang et al., 2026d) | 2026 | ext. memory   | interaction | none (potential over structured user st...  |        |
| Q-Evolve (Zhang et al., 2026k)                             | 2026 | context       | trajectory  | advantage from in-distribution IQL crit...  |        |
| Internalizing the Future (Zhang et al., 2026h)             | 2026 | written       | trajectory  | foresight-conditioned RL calibrating pl...  |        |
| RetailBench long-horizon retail P... (Zhang et al., 2026c) | 2026 | context       | –           | none                                        |        |
| WorldEvolver (Zhang et al., 2026g)                         | 2026 | world model   | state       | prediction-observation mismatch             |        |
| Theory of Space (Zhang et al., 2026f)                      | 2026 | written       | –           | new partial observations during active...   | ✓      |
| MemSkill (Zhang et al., 2026b)                             | 2026 | ext. memory   | memory op   | controller selects extract/consolidate/...  | ○      |
| OASES (Zhang et al., 2026a)                                | 2026 | context       | state       | none                                        |        |
| HiMem (Zhang et al., 2026e)                                | 2026 | ext. memory   | –           | conflict detected between retrieved not...  | ○      |
| StarWM (Zhang et al., 2026i)                               | 2026 | world model   | WM loss     | offline next-observation prediction fro...  | ○      |
| CORE (+PERSIST) (Zhang et al., 2026j)                      | 2026 | prob. memory  | state       | known answer (gold slot values: state-f...  | ✓      |
| AEM adaptive entropy modulation (Zhao et al., 2026a)       | 2026 | context       | trajectory  | response-level surprisal x advantage        |        |
| EvoSCM causal belief revision (Zhao et al., 2026b)         | 2026 | written       | state       | prediction-observation discrepancy dist...  | ✓      |
| BranPO (Zhao et al., 2026c)                                | 2026 | –             | interaction | none                                        |        |
| PGPO (Zheng et al., 2026c)                                 | 2026 | context       | interaction | potential difference between consecutiv...  |        |
| FOREAGENT predict-before-execute (Zheng et al., 2026a)     | 2026 | world model   | –           | predict-then-verify: prediction replace...  | ○      |
| Code2World (Zheng et al., 2026b)                           | 2026 | world model   | WM loss     | render-aware RL rewards rendered next-s...  | ○      |
| Thinking-RFT for ToM (Zhong et al., 2026a)                 | 2026 | –             | output      | verifiable-reward RL on shortcut-free T...  | ○      |
| LSE-MTP (Zhong et al., 2026b)                              | 2026 | context       | WM loss     | multi-token prediction anchored to grou...  | ○      |
| ToolVerse (Zhou et al., 2026)                              | 2026 | context       | interaction | turn-aware relative advantage from outc...  |        |
| Simple mechanism interfaces (Zhu et al., 2026)             | 2026 | context       | –           | none                                        |        |
| Qwen-AgentWorld (Zuo et al., 2026)                         | 2026 | world model   | WM loss     | next-state prediction reasoning trained...  | ✓      |

Metric: ✓ belief-level metric reported, ○ proxy only.